SIMULATION OF CLINICAL SKILLS:
A PRACTICAL GUIDE

Janja Tarčuković, Erika Šuper-Petrinjac (editors) et al.
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PUBLISHER
University of Rijeka, Faculty of Medicine

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PRINT
Printshop Name d.o.o

ISBN
978-953-8341-46-5

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LIST OF ABBREVATIONS

ABCDE	Airway, Breathing, Circulation, Disability, Exposure and Environment
ABG	Arterial blood gases
ACS	Acute coronary syndrome
ACEi	Angiotensin-converting enzyme inhibitor
ADH	Antidiuretic hormone
AF	Atrial fibrillation and flutter
ALI	Acute lung injury
ALS	Advanced Life Support
ARB	Angiotensin II receptor antagonist
ARDS	Acute respiratory distress syndrome
AV	Atrioventricular
AVPU	Alert, Voice, Pain, Unresponsive
BE	Base excess
BNP	B-type natriuretic peptide
BP	Blood pressure
CBC	Complete blood count
CHF	Chronic heart failure
CMV	Controlled mechanical ventilation
CNS	Central nervous system
CO	Cardiac output
COPD	Chronic obstructive pulmonary disease
CPAP	Continuous positive airway pressure
CPP	Cerebral perfusion pressure
CPR	Cardiopulmonary resuscitation
CRP	C-reactive protein
CRT	Capillary refill time
CSF	Cerebrospinal fluid
CT	Computed tomography
CVC	Central venous catheter
DO_2	Oxygen delivery
ECF	Extracellular fluid
ECG	Electrocardiogram
EEG	Electroencephalogram
ERC	European Resuscitation Council
ETC	European Trauma Course
ETT	Endotracheal tube
$EtCO_2$	End-expiratory carbon dioxide
EWS	Early Warning Score
FAST	The Focused Assessment with Sonography in Trauma ultrasound protocol
FiO_2	Fraction of inspired oxygen
G	Gauge
GCS	Glasgow Coma Scale
Hb	Haemoglobin
HES	Hydroxyethyl starch
IBW	Ideal body weight
ICP	Intracranial pressure
IM	Intramuscular
IO	Intraosseous
ITLS	International Trauma Life Support
IV	Intravenous
LAD	The anterior descending branch of the left coronary artery
LBBB	Left bundle branch block
LCx	The circumflex branch of the left coronary artery
LMA	Laryngeal mask
LVEDP	Left ventricular end-diastolic pressure
MAP	Mean arterial pressure
MOI	Mechanism of injury
MRI	Magnetic resonance imaging
NGT	Nasogastric tube
NIV	Noninvasive ventilation
NSAIDs	Non-steroidal anti-inflammatory drugs
NSTE-ACS	Non-ST-segment elevation acute coronary syndrome
NSTEMI	Non-ST-segment-elevation myocardial infarction
$PaCO_2$	Carbon dioxide partial pressure in arterial blood
PaO_2	Oxygen partial pressure in arterial blood
PAO_2	Oxygen partial pressure in the alveoli
PCI	Percutaneous coronary intervention
PE	Pulmonary embolism
PEA	Pulseless electrical activity
PEEP	Positive end-expiratory pressure
PEF	Peak expiratory flow
PS	Pressure support
PSVT	Paroxysmal supraventricular tachycardia
PTX	Pneumothorax
pVT	Pulseless ventricular tachycardia
RCA	The right coronary artery
SAMPLE	Symptoms, Allergies, Medications, Past medical history, Last oral intake, Events leading to the incident
SaO_2	Arterial blood oxygen saturation
SGA	Supraglotttic airway device
SIADH	Syndrome of inappropriate antidiuretic hormone secretion
SID	Strong ion difference
SIMV	Synchronised intermittent mandatory ventilation
SOFA	Sequential Organ Failure Assessment score
SpO_2	Peripheral blood oxygen saturation
STE-ACS	ST-segment elevation acute coronary syndrome
STEMI	ST-segment-elevation myocardial infarction
SVT	Supraventricular tachycardia
TBI	Traumatic brain injury
TBSA	Total body surface area
VF	Ventricular fibrillation
VT	Ventricular tachycardia
WOB	Work of breathing
WPW	Wolff-Parkinson-White syndrome
4H	hypoxia, hypovolaemia, hyperkalaemia, hypokalaemia and metabolic disorders, hypothermia (reversible causes of death)
4T	tension pneumothorax, tamponade, thrombosis, toxic agents (reversible causes of death)

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This book stands in the space between deterioration and decision — where knowledge becomes the path from chaos to care.

Part 1: Deteriorating patient

13—56

Katarina Božić, Matej Bura

1.1. INITIAL ASSESSMENT OF ACUTELY ILL PATIENTS (ABCDE AND SAMPLE)

An acutely ill patient is one with an immediate threat of deterioration of vital functions, such as the cardiovascular, respiratory, and neurological systems, with a risk for subsequent cardiac arrest. This can result from their initial illness or a combination of chronic comorbidities and an acute state. Handling these patients requires standardised, systematic, and simple initial assessment and management. Simplified and focused examinations bring enough information about the patient's current status and need for therapeutic interventions. The accidental or deliberate omission of one or more vital parameters may delay recognition of the problem and potentially further worsen the patient's condition.

With a structured examination, in most situations, we obtain enough information to form a differential diagnosis and plan for the initial management of the acutely ill patient. One of the tools for standardised and systematic initial patient assessment is the **ABCDE algorithm.** It consists of five categories, where each letter represents one category during the assessment. The assessment is performed in alphabetical order. If a deviation/problem is encountered during the inspection, immediate therapeutic intervention is performed. In other words, if there is a problem on "B", the recommendation is to solve that problem, stabilise it if possible, and only then move on to "C". Life—threatening conditions in patients are most often manifested through changes in cardiocirculatory, respiratory and neurological systems; therefore, they have the most important place in the rapid assessment.

"A" stands for **airway** and evaluates problems with **airway obstruction**. Obstruction can be partial or complete and caused by various reasons such as decreased tone of soft tissues in the upper airway, food, blood, gastric content, oedema of the soft tissues of the tongue, pharynx, and larynx, laryngospasm, bronchospasm, and others. Complete airway obstruction can quickly cause cardiac arrest. Sometimes after complete airway obstruction, a see—saw phenomenon can be seen in the form of paradoxical inverse movement of the chest cavity and abdomen. If the respiratory drive is still present, the diaphragm and intercostal muscles contract and prepare for air to enter. If there is no inspired air, a reverse movement of the chest and abdomen can be seen (instead of the regular simultaneous abdomen and chest movement).

Partial obstruction is a common problem in clinical practice. In that case, sound phenomena can usually be heard as snoring, gargling, and inspiratory stridor. Depending on which of these are detected, one can estimate an approximate location of obstruction (upper or lower respiratory tract). A common problem is partial airway obstruction caused by acute deterioration in the level of consciousness. It is expected that patients with a Glasgow Coma Score (GCS) of 8 or fewer points don't have protective reflexes, so protection of the airway against gastric reflux and regurgitation of content with endotracheal tube is required in that case!

HOW TO KEEP THE AIRWAY OPEN?

Opening the airway is one of the key points of patient treatment whatsoever. If airway obstruction is visible, it is necessary to act immediately. One of the basic approaches is the head—tilt chin—lift manoeuvre in its original form, which can be supplemented with adjutant manoeuvres. The upper hand is put on the patient's forehead, and the lower hand pushes the bony part of the mandible until resistance is felt. If a fracture of the cervical vertebrae is suspected (for example, fall from height, traffic accident, or jump into a shallow pool or beach), the manoeuvre should be modified. Instead of head tilt, fixation of the skull is made with pressure on zygomatic arches on both sides, followed by regular chin lift. If there is suspicion of spinal fracture combined with visible trauma of the facial bones, the head is fixated in the area of temporal bone bilaterally with regular chin lift movement.

Once the airway is opened, it should be kept that way, which is achieved with oro— or nasopharyngeal airway or supraglottic (SGA) or infraglottic devices. Usually, we start with the lowest level device (e.g., nasopharyngeal airway) and escalate according to the patient's state. Commonly used are the oropharyngeal airway and nasopharyngeal airway.

The oropharyngeal airway is inserted through the mouth. Its concave side "slides" along the hard palate. Once the top of the

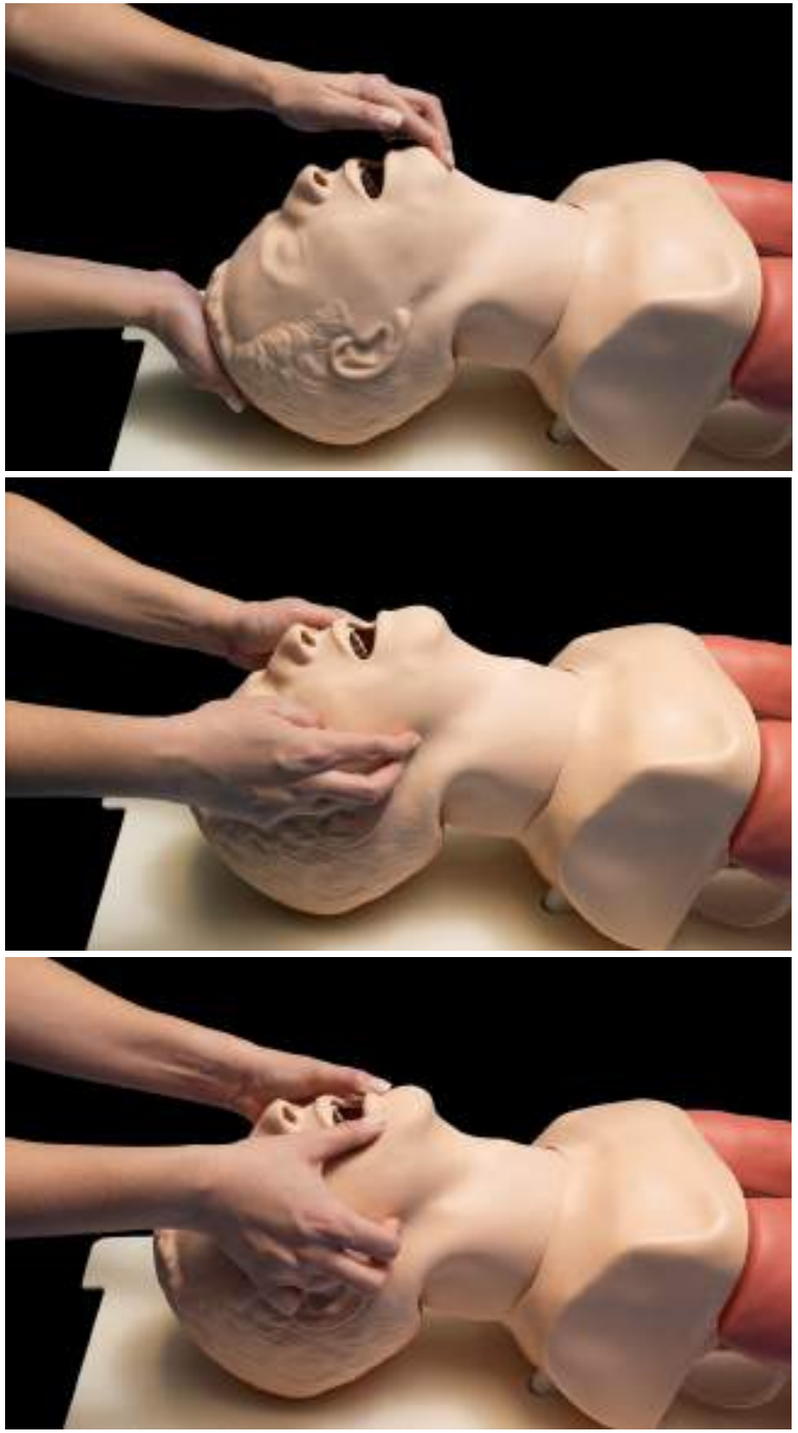

FIGURE 1. *Head—tilt chin—lift manoeuvre*

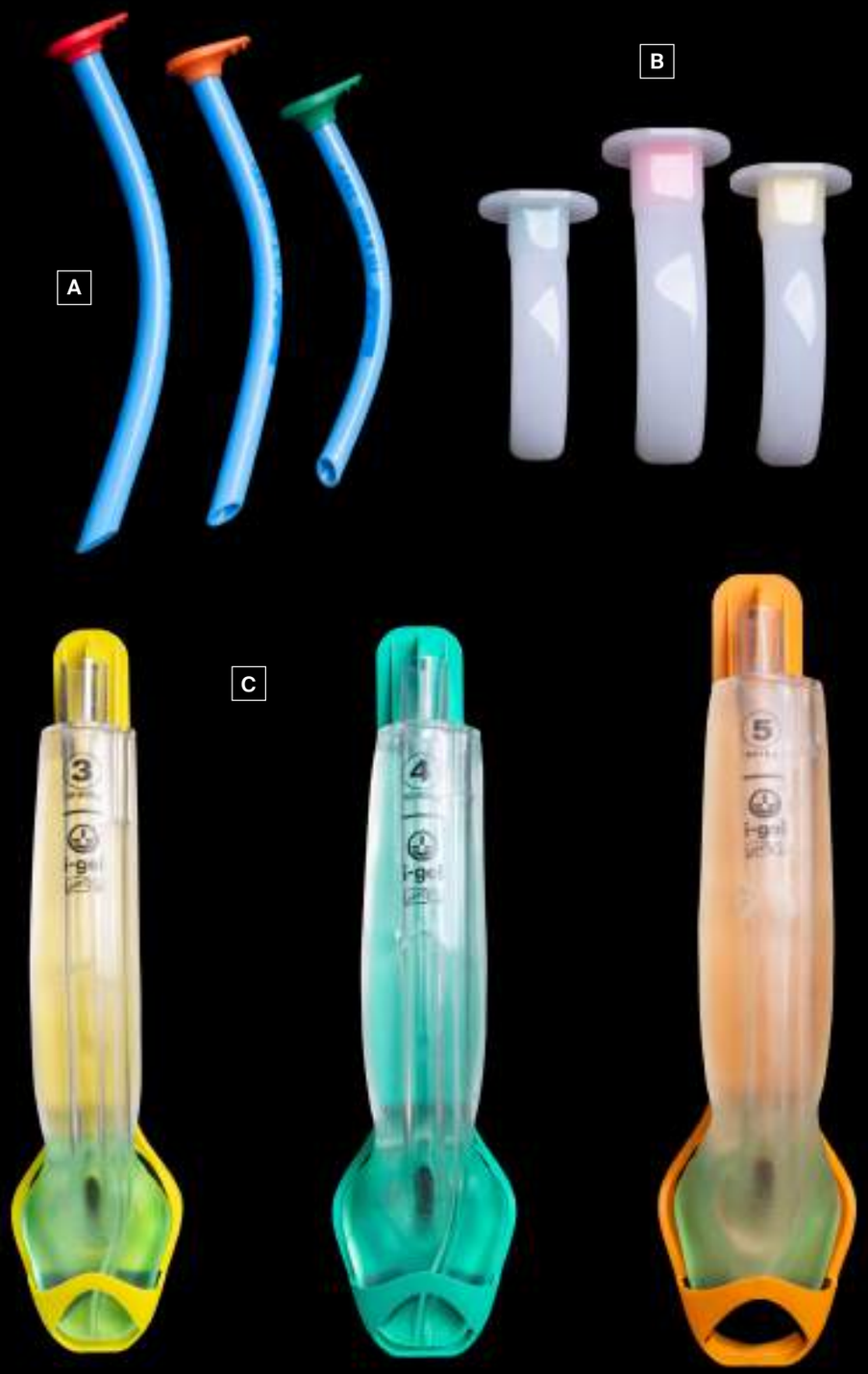

FIGURE 2. *Supraglottic devices of different sizes: A) nasopharyngeal airways, B) oropharyngeal airways, C) laryngeal masks.*

airway reaches the soft palate, a change of resistance is felt, and the tube should be rotated by 180° and sunken for the air to freely pass from outside through the oral cavity and pharynx into the larynx. For the airway tube to function appropriately, the right size should be placed in the patient's oral cavity. The appropriate size is estimated by placing it in a position in which it stands in the oral cavity, placing the lower edge at the mandibular angle. Tube of proper size should have the upper edge at the level of incisors. If the proper size is unavailable, a bigger size is more appropriate. As with every other tube, there can occur practical problems. Firstly, the tube is near the oropharyngeal aperture and base of the tongue, which are areas with significant vagal innervation. This can provoke a vomiting reflex if one is still present in the patient (in that case, the airway must be removed immediately). Secondly, the trauma of the oral cavity and trismus could make insertion difficult, maybe even impossible.

The **nasopharyngeal airway** is inserted into the pharynx through the nasal cavity. It is cut diagonally on its lower edge, and a shorter margin should be placed medially when inserting. Rotating movements are done by prospering through the lower nasal cavity perpendicularly to the patient's head. The appropriate size of the airway is estimated by placing the lower edge on the earlobe, and the upper edge should be at the nostril opening. Some of the commercially available nasopharyngeal tubes come with adjustable flange to mark and secure the airway in optimal depth (for a particular patient) after the insertion. The nasopharyngeal airway is better tolerated than the oropharyngeal in patients that still have protective reflexes (pharyngeal, laryngeal, cough reflex). There can be difficulties placing the airway if there is a nose fracture present. On top of that, if there is a suspected skull base fracture (Brillen hematoma, otorrhea, rhinorrhoea, mastoid hematoma), it is contraindicated to place nasopharyngeal airway because potentially fractured lamina cribrosa of ethmoidal bone can be additionally damaged by placing the device.

One of the essential supraglottic devices is a **laryngeal mask** (LMA). The most commonly used version is i—gel, which got its value by having thermoplastic elastomers on its lower edge that ensure better placement on the laryngeal aditus. This provides a better seal and less probability of aspiration of gastric content into the lungs. Insertion is made by holding LMA in the hand like a pencil, with the progression of the mask through the oral cavity. It is useful to push the soft tissues of the tongue aside to place it more easily. Its upper edge has a connector that enables bag valve mask placement and assists ventilation of the patient if needed. ALS algorithm for cardiac arrest states that LMA is an adequate airway protection device for providers who are not highly skilled in intubation. However, research shows that, although it does provide some protection, it does not completely eliminate the risk of aspiration of gastric content in the case of regurgitation. Moreover, it can even provoke vomiting if the vagal reflexes are present or the ventilation is performed under higher

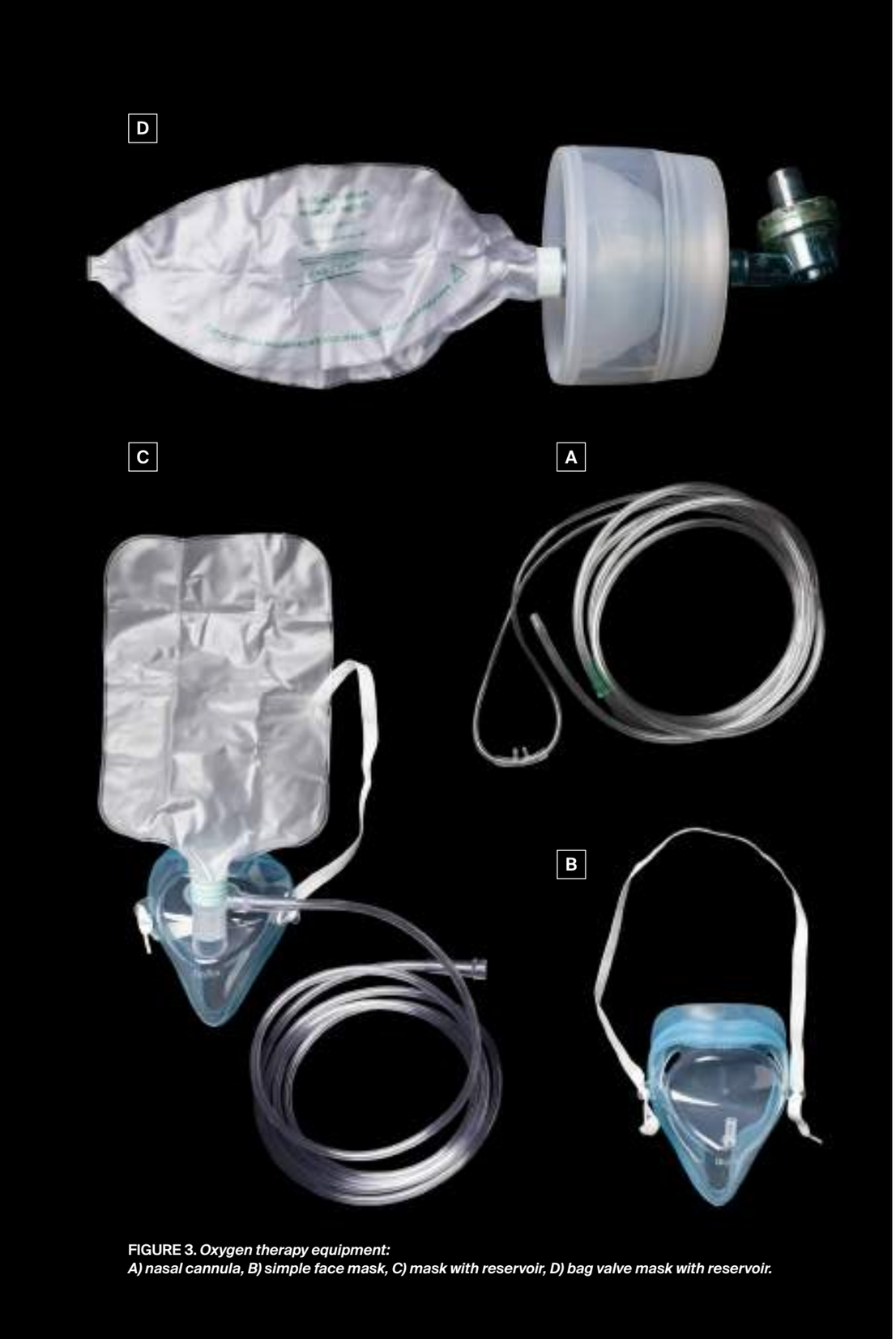

FIGURE 3. *Oxygen therapy equipment:*
A) nasal cannula, B) simple face mask, C) mask with reservoir, D) bag valve mask with reservoir.

pressure, with a leak in the oesophagus with subsequent gastric distention. Endotracheal intubation remains the gold standard for airway protection.

KEY POINTS:
- Manoeuvres: head-tilt chin-lift + modifications
- Devices: oropharyngeal airway, nasopharyngeal airway, laryngeal mask, endotracheal tube).

The second category is "B", which stands for the **breathing** parameters. Firstly, the respiratory rate is assessed. It is measured by observing breathing mechanics in parallel with placing a hand on the lower part of the patient's chest for 30 seconds. Normal respiratory rate is between 10 and 20 breaths per minute (eupnoea). Bradypnoeic patient (< 10/min) has, despite the appropriate tidal volume, inappropriate exchange of oxygen and carbon dioxide in the lungs, so the patient needs assisted ventilation (for example, if the patient breaths 6/min, 6 additional ones should be added to have appropriate frequency and, therefore, adequate minute volume). An apnoeic patient requires constant assisted ventilation. Tachypnoeic patient (> 20/min) has significantly higher energy expenditure on work of breathing, which eventually leads to muscle fatigue and use of accessory respiratory muscles (intercostal muscles, muscles of the neck). Normal breathing mechanics also implies a symmetric movement of both sides of the thorax synchronously with the abdomen. In the case of asymmetric movement, further examination should be performed to distinguish potential pathology. Auscultation of the lungs can bring a lot of information that can help in the clinical examination: inspiratory stridor can point to a problem in the upper respiratory tract, and inspiratory and expiratory crackles can point to a problem in the lower respiratory tract. If lung auscultation shows abnormalities, percussion of the thorax can distinguish whether air is trapped in the thorax (pneumothorax — hyperresonance) or there is effusion (dull, flat sound). Lung ultrasound and radiogram are additional points in examination that can help in further diagnosis and are commonly used in these cases. In the case of tension pneumothorax (PTX), urgent decompression of the compressed lung is necessary. Furthermore, pulse oximetry is a vital parameter in ABCDE assessment. It is a non—invasive measurement of a patient's peripheral blood oxygen saturation. In most patients, normal saturation is between 94 and 98%, while 88—92% is tolerated in patients with chronic obstructive pulmonary disease (COPD). If there is a lack of oxygen, it can be compensated with a nasal cannula, simple face mask, or face mask with a reservoir.

The mixture of air that is normally inhaled (fraction of inspired oxygen — FiO_2) contains 21% of oxygen. The fraction of inspired oxygen can be enriched through nasal cannulas and masks. The nasal cannula can supplement oxygen through 1—4 L/min flow, thus increasing the inspiratory fraction to 40%. Face mask supplements 4—10 L/min flow (FiO_2 35—60%). Face mask with reservoir

supplements 10–15 L/min (FiO_2 60–95%). Which device to use depends entirely on the patient's clinical state, which will be further explained in the chapter about acute respiratory insufficiency.

KEY POINTS:
- respiratory rate, symmetry of chest movement, lung auscultation, pulse oximetry
- supplementation of oxygen: nasal cannula, simple face mask, face mask with reservoir
- assisted ventilation

"C" stands for **circulation**. Cardiovascular disturbances are often a sign of general deterioration in a patient. There are several parameters included in the "C" assessment: characteristics of a patient's pulse, blood pressure, skin colour, capillary refill time, and diuresis. The individual parameter might not sometimes give much information, while the combined analysis of all the parameters and their trends often paints a good general picture of the patient's haemodynamic profile and risk for deterioration and cardiac arrest. Pulse is characterised by its rate, rhythm, and strength. Normal heart rate ranges from 60 to 100 beats per minute, while a rate below 60/min is called bradycardia, and above 100/min is called tachycardia. Bradycardic pulse can sometimes be found in young, healthy individuals without cardiac pathology. A typical example of a completely arrhythmic pulse can be found in patients with atrial fibrillation, a commonly found pathological rhythm in elderly patients. Heart rhythm and its pathology are explained in more detail in chapter 2.2. Pulse is regularly equally palpable centrally (e.g., carotid artery) and peripherally (radial artery). If the pulse strength is weaker on the radial artery, the estimated systolic blood pressure is less than 80 mmHg (hypotensive state). The focus in the assessment of skin is on its warmth and wetness. Normal skin is dry and warm, while in the case of haemodynamic disturbances, it is often clammy, sweaty, and cold. Pale skin can point to anaemia and/or centralisation of blood flow. Cyanotic skin can indicate hypoxia, and red and swollen skin can indicate anaphylaxis. Capillary refill time is a parameter that indicates the adequacy of peripheral blood flow; as the skin is the organ with the lowest level of autoregulation, it will also be the first organ from which the blood will be centralised in the case of reduced cardiac output, e.g., in hypovolaemia. Measurement is performed by applying pressure on the skin near bony tissue (typically sternum or forehead). If it takes longer than 2 seconds for a distal capillary bed to regain its colour, blood flow through the skin and other peripheral tissues is insufficient. The longer the capillary refill time, the stronger the centralisation (a typical example is an actively bleeding patient).

Centralisation of blood flow is an adaptive response to haemodynamic disturbance. It consists of vasoconstriction in the "peripheral" organ system (skin, muscles) in order to preserve adequate blood flow through vital organs (brain, heart, lungs). Clinically, centralisation is seen in characteristic manifestations: pale and clammy skin, capillary refill time (CRT) longer than 2 seconds, and tachycardia. Blood pressure measurement in these cases might show hypotension with systolic blood pressure below 90 mmHg.

Besides clinical examination, there are several interventions that have to be performed during the assessment of the cardiovascular system. An acutely ill patient should be put on continuous electrocardiographic (ECG) monitoring. Basic monitoring includes at least three Einthoven leads: the red one is put on the upper right side of the thorax, the yellow one is put on the upper left part of the thorax, and the green electrode is put on the lower left part of the thorax. After the patient is put on ECG monitoring, the cardiac rhythm strip should be analysed. Depending on the patient's medical history and clinical examination, a 12–lead ECG should be recorded. Placing intravenous access is one of the basic interventions in acutely ill patients. Once IV access is placed, blood samples can be analysed, and intravenous medications can be administered.

KEY POINTS:
- basic examination: pulse (rate, rhythm, strength), skin (colour, wetness, warmth), blood pressure, capillary refill time, diuresis
- ECG monitoring, IV access.

Short neurological assessment is performed during the "D" category (**disability**). State of consciousness, pupillary response, and blood glucose levels are regularly monitored. Basic quantitative assessment of the state of consciousness is checked through the AVPU score. "A" stands for an alert patient, "V" represents a patient who reacts to verbal stimuli (corresponds to somnolence). The patient in the "P" category does not react to verbal stimuli but reacts to painful stimuli (corresponds to stupor). "U" stands for patient unresponsive to any stimuli, which corresponds to coma. AVPU roughly shows categories measured more thoroughly by the GCS, a scoring system that should be considered if the patient has a primary neurological disorder. Pupils are typically equally wide and respond to light with constriction bilaterally. Asymmetry of pupils can imply increased intracranial pressure. In acutely ill patients, bilaterally dilated pupils that don't respond to light stimuli imply global cerebral circulation disorder. Hypoglycaemia is one of the most common causes of acutely deteriorated neurological status, which makes glucose measurement an essential part of the neurological assessment.

KEY POINTS:
- AVPU scale
- pupillary response, blood glucose measurement.

The last category in the assessment includes **exposure and environment** ("E"). It includes the examination of other organ systems

	EXAMINATION	INTERVENTION	GOAL
A	• airway noises • position of head • foreign body • fluid, secretions • oedema	• open • suction • secure • O_2	patent airway
B	• look – listen – feel approach • raspiratory rate and effort • breath and added sounds • subcutaneous emphysema • symmetry of chest movement • tracheal deviation • jugular vein distention • cyanosis	• O_2 according to SpO_2 • pneumothorax therapy • inhalation therapy • ventilation	sufficient oxygenation and ventilation
	SpO_2 - $EtCO_2$ – USG – X-ray - CT		
C	• heart rate • blood pressure • capillary refill timee • bleeding • skin colour • blood samples • diuresis	• I.V./I.O. access • control of bleeding • massive haemorrhage protocol • fluids • drugs • transfusion	stabilisation of circulation
	ECG – USG – CT – X–ray		
D	• AVPU / GCS • reactivity and symetry of pupils • blood glucose level • basic neurological examination • posture • toxicological examination	• glucose • antidotes	evaluation of neurological state
E	• head to toe examination • medical history • temperature • injuries • oedemas • scars • signs of drug abuse • skin changes • signs of infection/sepsis	• identified cause therapy • thermomanagement • trauma treatment • insertion of NGT, IUC	revealing other symptoms and thermomanagement

FIGURE 4. *ABCDE examination, an overview of possible interventions, and goals (retrieved from Peran et al. ABCDE cognitive aid tool in patient assessment – development and validation in a multicenter pilot simulation study. BMC Emergency Medicine 2020; 20: 95.*

previously not included in the examination and a detailed examination of the skin of the whole body in search of further clinical signs that can be connected to already found pathology. Body temperature should also be measured. Depending on the symptoms observed, the examination can include an abdomen examination, rectal examination, neurological orientation examination, FAST (The Focused Assessment with Sonography in Trauma) ultrasound examination, and others.

> KEY POINTS:
> – temperature
> – FAST ultrasound, last menstrual cycle.

MEDICAL HISTORY

In order to help with differential diagnosis and decide on the plan for initial management of the acutely ill patient, target medical history should be taken from the patient or their caregiver. A short medical history can be sampled through six basic categories that form the acronym SAMPLE (Table 1).

Symptoms are very often connected to the circumstances that led to them. Multiple comorbidities of the patient and numerous chronic medications can direct the therapeutic approach and examination (e.g., due to polyneuropathy in acute coronary syndrome, people with diabetes do not necessarily have to have chest pain and usually present with non–specific symptoms, which leads to the conclusion that a 12–lead ECG must be recorded). The time passed since the last meal indicates stomach fullness with food, which impacts the strategy for potential endotracheal intubation. The stomach is considered empty 6 hours after the consumption of solid food, 4 hours after the consumption of liquid food, and 2 hours after the consumption of water. In women of reproductive age, indicative information in the medical history can be the time of the last menstruation (important in

TABLE 1. *Basic components of the SAMPLE history*

S	SYMPTOMS	duration, intensity, character (e.g.,pain)
A	ALLERGIES	primarily to medications
M	MEDICATIONS	therapy for chronic diseases
P	PAST MEDICAL HISTORY	in correlation with the previous section (M)
L	LAST ORAL INTAKE	the time elapsed since the last meal or drink
E	EVENTS LEADING TO THE INCIDENT	circumstances in which symptoms happened

the differential diagnosis of abdominal pain). In case it is not possible to obtain the data from the patient, taking a history from a caregiver or next of kin plays a significant role. Acutely ill patients may deteriorate during the examination, therefore, it is important to take as much anamnestic data as possible while it is possible. In patients able to communicate, it is essential to initially inquire about basic data and then continue the examination according to the ABCDE approach so that if the condition deteriorates, there is some anamnestic data available.

APPENDIX 1: RECOGNITION OF AN ACUTELY ILL PATIENT IN HOSPITAL CONDITIONS

Abnormal physiology is common in general wards, yet research shows that observation and measurement of vital parameters occur less than recommended. Assessment of simple vital signs may help with the prevention of cardiopulmonary arrest. One of the ways to increase the success of early identification of critical patients in the wards is to use the Early Warning Score (EWS) protocol. EWS enables repeated assessment of the patient's vital parameters in defined time intervals. The assessment is mainly carried out by a medical technician. If the patient scores 3—5, they are observed once every 4 hours. If the sum is greater than 6 or more, it is necessary to alert the ward doctor, who then assesses the need for more frequent observation (for example, once every hour). Patients with a score of 9 or more require continuous monitoring, which is most often carried out in intensive care units, emergency room observation rooms, and other appropriate places. The most important is the recognition of major changes for specific parameters in time intervals. If the patient's condition deteriorates, a decision is made on further treatment and the need to refer the critical patient to treatment services.

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TABLE 2. *Example of a scoring system of monitoring and recognition of patient deterioration*

	POINTS						
	3	2	1	0	1	2	3
AVPU				A			V, P, U
RASPIRATORY RATE (/min)	≤8		9	10-20		21-24	≥ 25
INSPIRED OXYGEN				air			oxygen therapy
OXYGEN SATURATION (%)	≤91	92-93	94-95	≥ 96			
PULSE (/min)		≤40	41-50	51-90	91-110	111-130	>131
SYSTOLIC BP (mmHg)	≤90	90-100	100-110	110-250	>250		
TEMPERATURE (°C)	≤35.0		35.0-36.0	36.1-38.0	38.1-39.0	≥ 39.1	

1. 2. ESSENTIALS FOR ECG INTERPRETATION IN MEDICAL EMERGENCIES

An electrocardiogram (ECG) is a recording of the heart's electrical activity. Depending on the indication, it can be continuously monitored on an ECG monitor or recorded on graph paper in the form of a 12-lead ECG. The recording and interpretation of ECG is an indispensable part of the assessment of an acutely ill patient, and it is a basic skill that every doctor must acquire. It is important to point out that the ECG shows the electrical activity of the heart but not the mechanical activity of the heart (heart contraction). Hence, it is necessary to palpate the pulse for the assessment if electrical activity ultimately produces a mechanical one.

1. 2. 1. THE CONDUCTION SYSTEM OF THE HEART AND THE NORMAL CARDIAC CYCLE ON THE ECG

In order for the heart to perform its basic role as a pump, two steps are necessary:

1. Generation and conduction of the electrical signal through the heart.
2. Cardiac contraction and relaxation.

The first step requires the so-called conductive heart fibres and the second the contractile heart fibres. The conductive heart fibres represent the part of the heart specialised for initiating and efficiently conducting the electrical impulse to the cells that are capable of

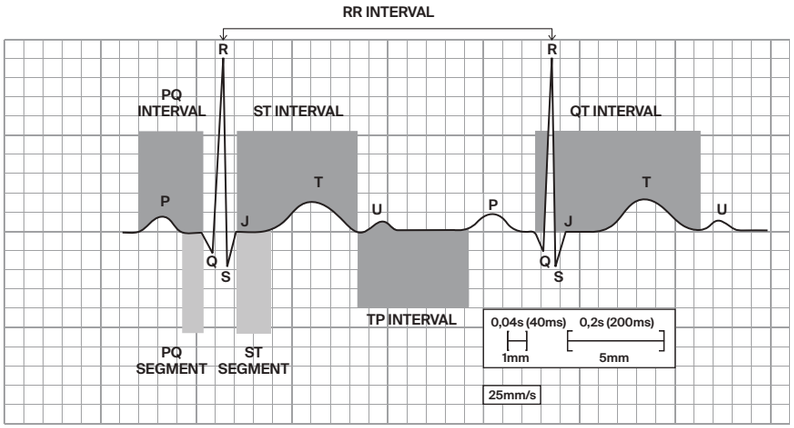


FIGURE 1. *Normal cardiac cycle on ECG (courtesy of Josip Eljuga, MD)*

contraction. In a healthy patient, the electrical signal (in the form of an action potential) originates in the SA node ("cardiac pacemaker"), from where it spreads through the atria to the AV node. The signal is briefly "slowed down" while passing through the AV node and then quickly transmitted through the bundle of His, its right and left branches, and the network of Purkinje fibres towards the contractile ventricular fibres. The mentioned slowing of the signal in the AV node allows the atria to contract first during the cardiac cycle and expel the blood towards the ventricles, after which the filled ventricles contract. In order to better understand the ECG recording, it is necessary to note that each action potential consists of depolarisation and repolarisation, whereby depolarisation stimulates cardiac contraction (systole), while repolarisation enables cardiac relaxation (and filling of chambers with blood during diastole).

"Translation" of the mentioned process into one cardiac cycle on the ECG (Figure 1): atrial depolarisation is represented by the P wave, conduction through the atria and the AV node is represented by the PR interval (from the beginning of the P wave to the beginning of the QRS complex; normally 0.12–0.20 s), where deceleration through the AV node is the PR segment (from the end of P to the beginning of the QRS), ventricular depolarisation is represented by the QRS complex (normally up to 0.12 s), while ventricular repolarisation is represented by the T wave.

1. 2. 2. ECG MONITORING

ECG monitoring is used for continuous monitoring of the heart's electrical activity, which enables the assessment of rhythm and rate in real time. Using ECG monitoring, it is not possible to analyse the ST junction and the duration of individual segments (PR, QT, QRS). If such an analysis is necessary, it is indicated to record a 12-lead ECG.

Indications for patient monitoring are rhythms during cardiac arrest and other significant arrhythmias, chest pain, cardiac decompensation, collapse/syncope, shock/hypotension, palpitations, and any other condition in which the patient is acutely endangered (e.g., sepsis, polytrauma, etc.).

TYPES OF MONITORING ARE:

a) **3-lead monitoring:** it is recorded using electrodes attached according to the "traffic light principle" (red electrode above the right clavicle, yellow above the left clavicle, green in the projection of the apex of the heart). This enables the monitoring of leads I, II, and III.

b) **Monitoring using self-adhesive electrodes or defibrillator pads:** it is recorded by placing one electrode/pad under the right clavicle and the other in the middle axillary line to the left (anteroposterior placement is also possible). This is also the method of choice in the arrest and post-resuscitation period. It enables only monitoring of the lead II.

In practice, we use the following six questions for quick ECG analysis in medical emergencies:

1. ***Is there any electrical activity?*** If there is no electrical activity, the rhythm is asystole. If electrical activity is present but disorganised ("bizarre", without identifiable QRS complexes), the rhythm is ventricular fibrillation (VF). Asystole and VF are rhythms incompatible with life because they cannot produce a pulse and cardiopulmonary resuscitation (CPR) should be started immediately. If we see those while monitoring a living patient (patient with a pulse), it is probably artefacts and/or wrong position/connection of the electrodes. If organised electrical activity is seen, the rhythm is further analysed according to questions 2–6. It is important to point out the case in which electrical activity is seen, but the patient has no pulse, which is called pulseless electrical activity (PEA). It does not require further analysis of the rhythm, and it is extremely important to start CPR without delay.
2. ***What is the ventricular (QRS) rate?*** The normal heart rate is 60–100/min. A rate slower than 60/min is called bradycardia, and a rate faster than 100/min is called tachycardia. The rate reading is usually presented on the monitor, but it can also be calculated from a longer printout on graph paper (explained later).
3. ***Is the QRS rhythm regular or irregular?*** We compare the lengths of the R-R intervals, and we can also use sound signals from the monitor. A regular rhythm with intermittent irregularities (i.e., premature complexes) most often indicates extrasystoles (they can be of atrial or ventricular origin). If the rhythm is completely irregular ("irregularly irregular") and all

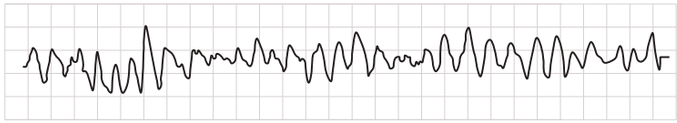
QRS complexes are morphologically equal, it is most likely atrial fibrillation (AF).

4. *Is the QRS complex normal or prolonged?* QRS complexes normally last up to 0.12 s (up to three "small squares" on the printout). A normal/narrow QRS complex indicates that the rhythm is "supraventricular" (above the bifurcation of the bundle of His). An extended QRS (prolonged duration) indicates that the rhythm originates from the ventricle (e.g., ventricular tachycardia) or might be supraventricular but transmitted with aberrant conduction (branch block, accessory pathway conduction, etc.).
5. *Is atrial activity present?* It is necessary to look for P waves: sinus P waves are most often clearly visible in the lead II as upright waves. The absence of atrial activity, i.e., its disorganisation, indicates AF. Regular atrial activity in the form of fast "saw-tooth" waves indicates atrial undulation. When possible, interpretation of atrial activity should be facilitated (especially in fast rhythms) by recording a 12-lead ECG and analysing all leads (usually, atrial activity is easiest to determine in V1).
6. *How does atrial activity relate to ventricular activity?* Does a QRS follow each P wave, and is there a P wave before each QRS? What is the length of the PR interval?

The most precise answer is obtained by analysing a longer printout. Conduction between atria and ventricles is normal if PR < 0.20 s. Each P wave is followed by a QRS complex, and the P wave is located before each QRS complex. First-degree AV block is characterised by prolongation of PR (> 0.20 s). Second-degree AV block Mobitz Type I (Wenckebach AV block) is characterised by a progressive prolongation of the PR interval until there is no QRS, after which the cycle starts from the beginning. Second-degree AV block Mobitz Type II is characterised by a constant PR interval with occasional individual QRS absences. In the third-degree AV block (complete AV block, "AV dissociation"), there is no conduction through the AV node, so the frequencies of the atria and ventricles are independent of each other (the ratio of P and QRS waves is irregular and unrelated).

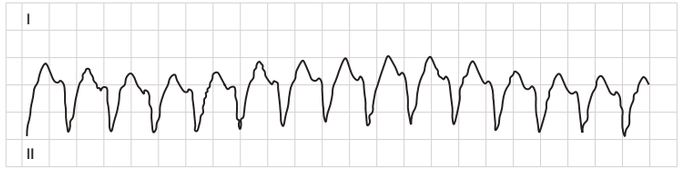
RHYTHM ANALYSIS (EXAMPLES):

a) **Ventricular fibrillation (VF)**: VF is incompatible with life, and the patient has no pulse. Since it is a "shockable" resuscitation rhythm, the priority in CPR is defibrillation as early as possible.



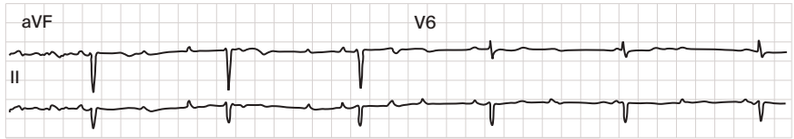
b) **Atrial fibrillation (AF)** with an average ventricular rate of around 130/min: P waves are not visible, and absolute ventricular arrhythmia is present. If we see this recording on the monitor, and the patient has no pulse, this rhythm is called **PEA**, and it is necessary to start CPR without delay. PEA can look like any other rhythm (except for the remaining "resuscitation rhythms", which can be asystole, ventricular fibrillation, and pulseless ventricular tachycardia).

c) **Broad-complex tachycardia** — in this case, **ventricular tachycardia (VT)** with a rate of about 150/min. VT is a rhythm compatible with life, and the patient's pulse should be palpated. If there is no pulse, it is a pulseless VT, so CPR with defibrillation as early as possible is a priority. In case the patient has a pulse, we follow the tachycardia algorithm.



d) **Atrial undulation** with a ventricular rate of about 80/min.

e) **Third-degree AV block** with a ventricular rate of 32/min. The ventricles have "their own rhythm", which is slower and independent of the atria. There is no QRS after each P wave, no P wave before each QRS, and the PR interval, if present, is variable.



1.2.3.INTERPRETATION OF 12-LEAD ECG

A more detailed analysis of the heart's electrical activity is not possible without a 12-lead ECG, and there are numerous causes of changes in the ECG (ischaemia, pericarditis, arrhythmias, pulmonary embolism, electrolyte disorders, drugs, implanted pacemaker, etc.). A standard 12-lead ECG is usually recorded by placing 10 electrodes, which results in a recording of six peripheral (I, II, III, aVR, aVL, aVF) and six precordial leads (V1–V6). Peripheral leads are projected in the vertical plane and the precordial leads are projected in the horizontal plane which enables a three-dimensional analysis (see Figure 7). In addition to the standard ones, recording can be supplemented by placing additional leads (Figure 2), for example, posterior leads (to confirm posterior myocardial infarction) or right leads (to confirm right ventricular infarction).

THE BASIC ANALYSIS CONSISTS OF FOUR PARTS:

1. **RHYTHM** – we determine it using the "six questions for rhythm analysis" discussed earlier. Sinus rhythm (SR) (Figure 3) is a normal rhythm and is the result of regular activity of the SA node and normal conduction of the electrical signal to the contractile heart muscles (P waves are upright in leads I and II, each QRS is preceded by P waves, the PR interval is constant). There are many irregular rhythms (FA, VT, heart blocks, arrhythmias with bundle branch blocks, and many others), but we will not go into a more detailed analysis here (Figure 4).
2. **RATE** – we can use the "300 method". We do this by finding the R wave that coincides with the thick line on the ECG recording and, depending on which following thick line the R wave falls on, we calculate the rate. The first line represents 300, the second 150, the third 100, the fourth 75, the fifth 60, and the sixth 50/min. This rule only applies to regular rhythms. For irregular and slow rhythms, take 25 cm of recording paper (represents 10 s), count the QRS complexes, and multiply the number by 6 to get a heart rate expressed as beats/min.
3. **ELECTRICAL AXIS** – represents the direction of depolarisation of the ventricles and is normally between –30° to +90°. Axis deviation can indicate pathology (left axis deviation is from –30° to –90°, right axis deviation is from +90° to +180°, and extreme axis deviation is from +180° to +270°). The simplest method is to determine the axis by quadrants. We observe the QRS complexes in the leads I and aVF and determine whether they are predominantly positive or negative. A positive QRS in the lead I means that the electrical axis is going to the left, and a negative QRS means that the axis is going to the right. Likewise, a positive QRS in aVF means that the axis is going down, and a negative QRS means that the axis is going up. By combining these values in the coordinate system, we determine the electrical axis of the heart. In the case of a positive QRS in lead I and

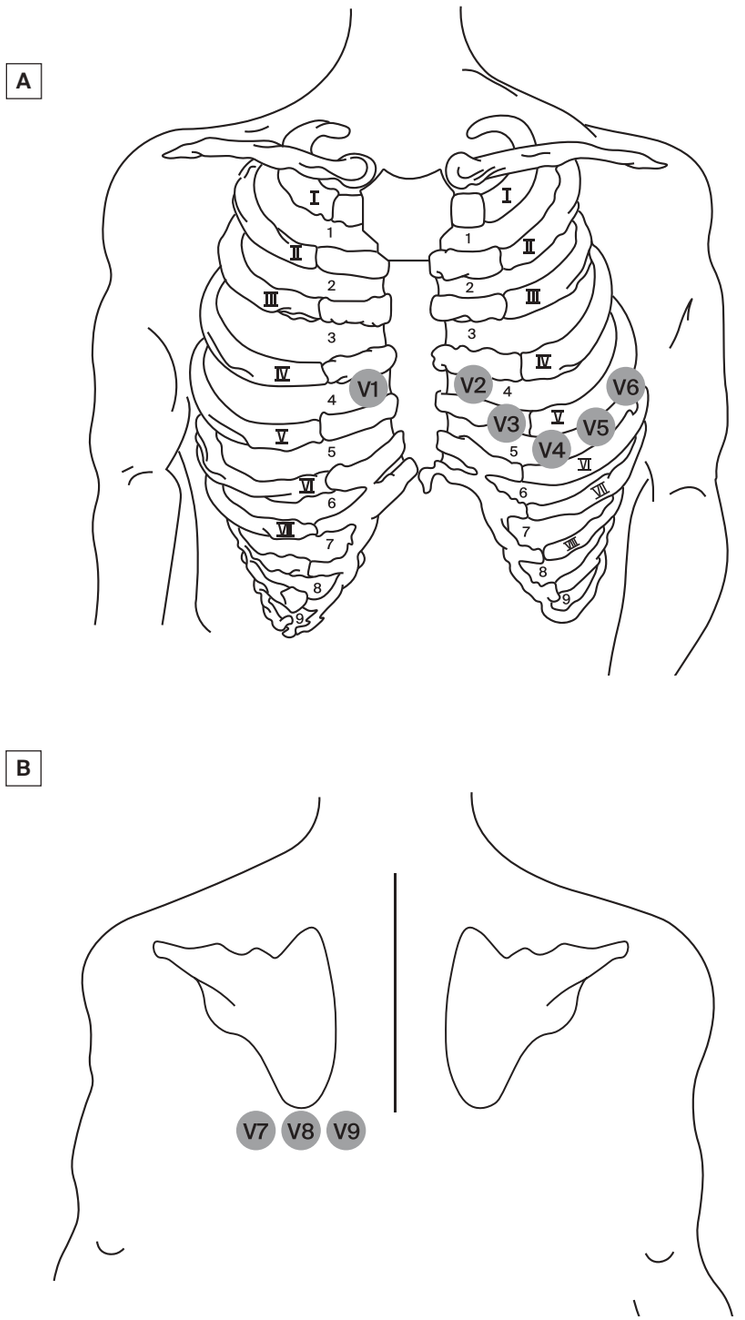


FIGURE 2. *A) STANDARD PRECORDIAL LEADS (the right leads, V3R–V6R, would be recorded by switching electrodes V3–V6 to the right side of the chest at the same level) B) POSTERIOR LEADS – recorded from the posterior of the chest at the same level as V6 (V7 in the left posterior axillary line, V8 on top of the left scapula, V9 to the left paravertebrally) (courtesy of Josip Eljuga, MD)*

a negative QRS in lead aVF, to confirm whether it is a left axis deviation or a physiological axis, look at lead II — a negative QRS in the same lead indicates axis deviation.

4. **ANALYSIS OF THE ST SEGMENT AND T WAVE** — normally, the ST segment is in the isoelectric line (the J point — the beginning of the ST segment is compared with the TP interval, which represents the isoelectric line) (Figure 1), and the T waves are positive (except in aVR and V1). Changes that suggest myocardial ischaemia are inverted (negative) T wave (Figure 6) and ST-segment depression. ST-segment elevation can be a sign of acute myocardial infarction and some other conditions (pericarditis, vasospastic angina, benign early repolarisation, left bundle branch block, etc.). It is necessary to look at all 12 leads and analyse these segments, and it is also useful to monitor the dynamics of these changes over time by periodically repeating the ECG recording.

1. 2. 4. ST-SEGMENT-ELEVATION MYOCARDIAL INFARCTION (STEMI) ON ECG

ECG criteria for the diagnosis of STEMI are ST elevation in two or more contiguous leads. Contiguous leads are leads that "look at" the same myocardial wall, i.e., the vascularisation area of the same coronary artery (Figure 7 and Figure 8), with the elevation having to be ≥ 1 mm (0.1 mV, "one small square"), except in V2 and V3 where more strict criteria apply, e.g., ≥ 2.5 mm for men < 40 years, ≥ 2.0 mm for men > 40 years, ≥ 1.5 mm for women). We determine the location of the infarction according to the leads in which the ST elevation is present. In patients with inferior STEMI (ST elevation in leads II, III, aVF), it is recommended to record the right leads (V3R—V6R) for possible confirmation of right ventricular infarction (Figure 9B).

ST elevation is often "paired" with a reciprocal ST depression in the "opposite" leads (Figure 9A, Figure 9B, and Figure 9C). Thus, ST depression in V1—V3 may suggest a "mirror image" of posterior STEMI, so it is necessary to record the posterior leads (V7—V9). ST elevation ≥ 0.5 mm is sufficient to confirm posterior STEMI (Figure 8). New-onset left bundle branch block (LBBB) in the clinical presentation of acute coronary syndrome (ACS) is considered equivalent to STEMI. ST depression ≥ 1 mm in eight or more leads followed by ST elevation in aVR and/or V1 suggests obstruction of the left coronary artery ("left main") or severe "triple-vessel coronary artery disease/ischaemia" (Figure 10).

Figure 7 represents the largest coronary arteries: the left anterior descending artery (LAD), the circumflex branch of the left coronary artery (LCx), and the right coronary artery (RCA). Contiguous leads "look at" the same wall, i.e., the vascularisation area of the same coronary artery, and the location of the infarction is determined according to the leads in which the ST elevation is present (notice how the

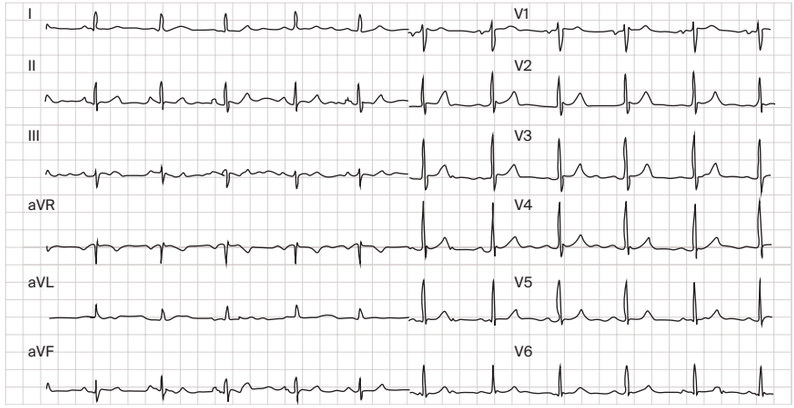


FIGURE 4. *Normal ECG: sinus rhythm, rate 60/min, normal electrical axis, without ischaemic changes*

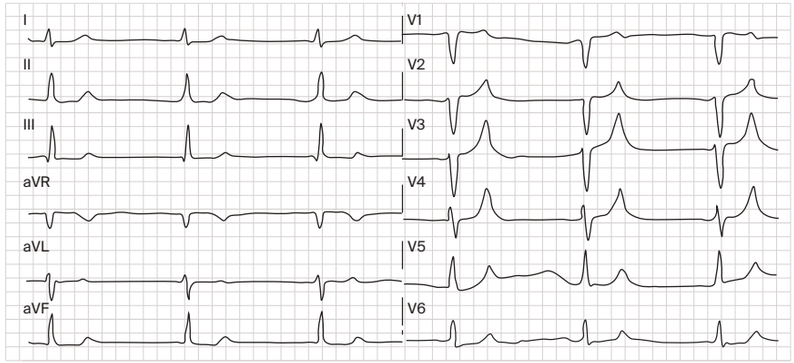


FIGURE 5. *Rhythm disorder in a chronic kidney disease patient with potassium levels over 8 mmol/L. The ECG shows absent P waves ("atrial standstill") with an additional broad QRS complex rhythm with a rate of 30/min. Hyperkalaemia can also cause other arrhythmias (asystole, VF, VT, AV blocks, bundle branch blocks, etc.).*

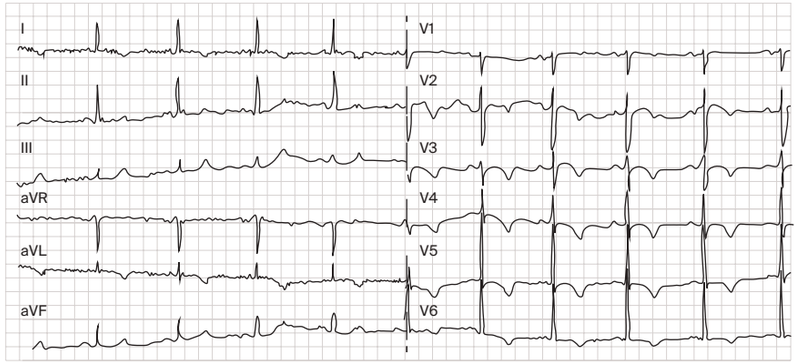


FIGURE 6. *ECG performed in a patient with chest pain. The ECG shows sinus rhythm, rate 60/min, normal electrical axis, negative T waves in leads I, aVL, V2–V6. Further assessment verified non-ST-segment-elevation myocardial infarction (NSTEMI); percutaneous coronary intervention (PCI) was performed with placement of stents in the anterior descending and circumflex branches of the left coronary artery (LAD and LCx).*

peripheral leads are projected in the vertical plane, and the precordial leads In the horizontal plane, which enables 3D analysis). We distinguish the following:

- Septal STEMI (LAD occlusion): ST elevation in V1–V2.
- Anterior STEMI (LAD occlusion): ST elevation in V3–V4.
- Lateral STEMI (LCx occlusion): ST elevation in V5, V6, I, aVL.
- Posterior STEMI (RCA or LCx occlusion): ST elevation in V7–V9, reciprocal ST depression in V1–V3.
- Inferior STEMI (RCA occlusion): ST elevation in II, III, aVF, reciprocal ST elevation in lateral leads (right leads V3R–V6R should also be recorded to confirm/exclude right ventricular infarction!)
- Combinations of the previously mentioned are also possible, e.g., anteroseptolateral STEMI (also called “extensive anterior infarction”, ST elevation in V1–V6, I+aVL) and inferoposterior STEMI (ST elevation in II, III, avF, and V7–V9, reciprocal ST depression in V1–V3).

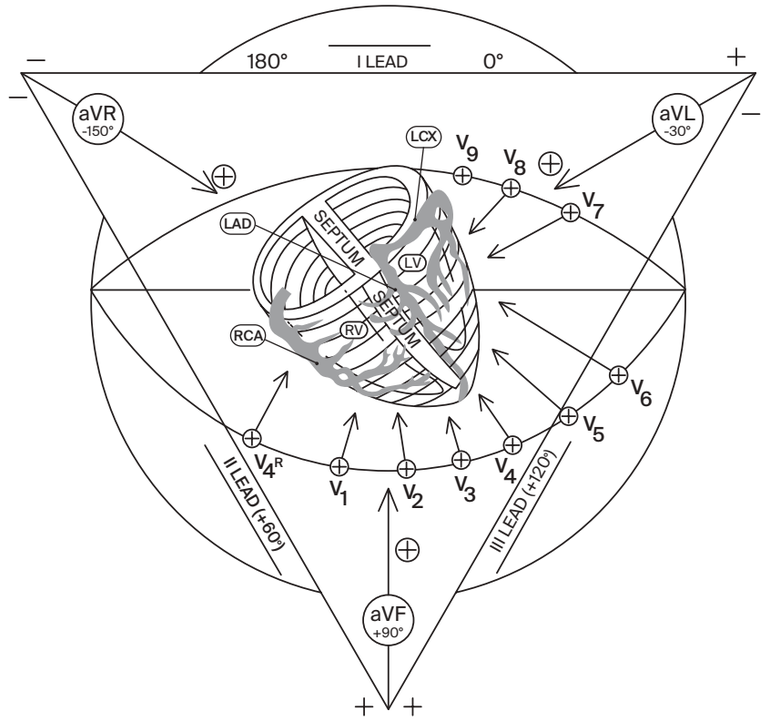


FIGURE 7. *Simplified representation of the heart and leads (courtesy of Josip Eljuga, MD)*

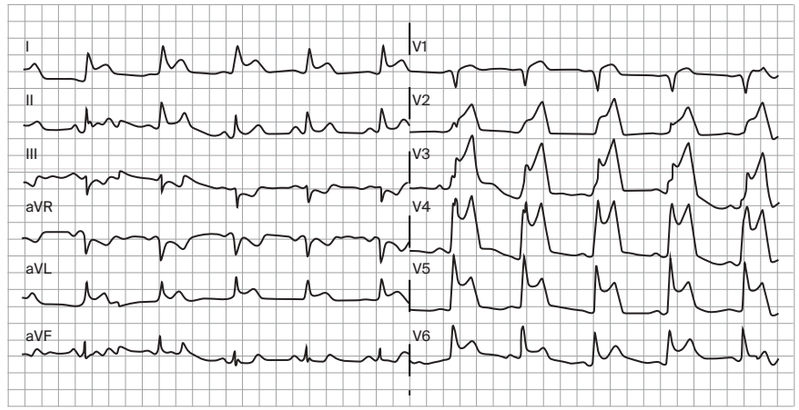
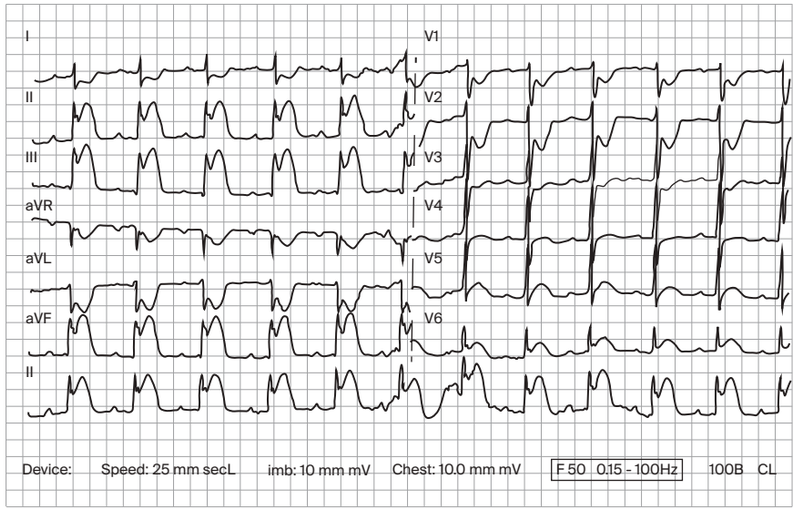
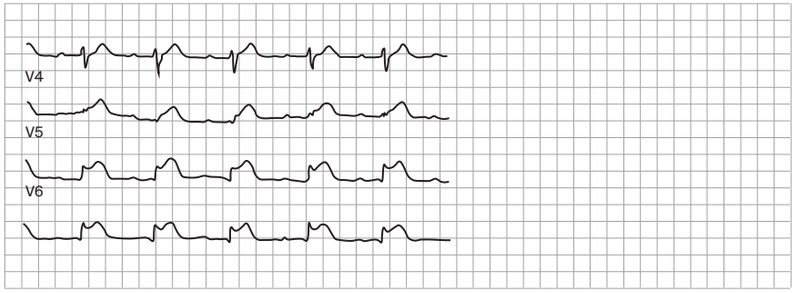


FIGURE 8. *Anteroseptolateral STEMI — ST elevations in I, aVL, V1–V6. The patient underwent primary PCI with stent placement in the anterior descending branch of the left coronary artery (LAD).*

a)



b)



c)

FIGURE 9. *a) Inferoposterior STEMI of the right ventricle (the result of right coronary artery occlusion), b) Due to ST elevation in the inferior leads (II, III, aVF), the right leads were recorded, which confirmed right ventricular infarction, c) ST depression in V1–V3 suggests a reciprocal change, so the recording of the posterior leads confirmed the infarction of the posterior wall; Noticeable reciprocal ST depressions in I, aVL, and aVR and ST elevation in V6.*

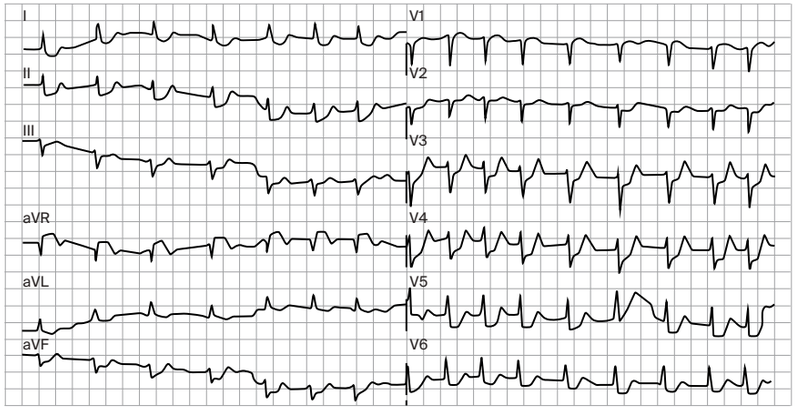


FIGURE 10. *Triple vessel coronary artery disease. Diffuse ST depression with ST elevation in aVR and V1. The rhythm shows atrial fibrillation.*

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1. 3. INTRAVASCULAR AND INTRAOSSEOUS ACCESS

1. 3. 1. INTRAVASCULAR ACCESS

Adequate intravascular access is one of the main preconditions of adequate care for acutely ill patients. It ensures fast and direct administration of medication and fluid therapy, including blood products and other intravenous fluids. Intravascular access can be established via a peripheral vein, a central vein, or intraosseous access. In practice, one or more peripheral venous accesses are most often placed. In cases when that is not possible, alternatives for peripheral intravascular access include intraosseous access (in out-of-hospital conditions) and central venous access (in hospital conditions). This chapter focuses on techniques for establishing intravenous (IV) and intraosseous (IO) access, competencies that every medical doctor should have after graduation. As with every other skill, it is important to point out that the technique requires practice and experience.

Intravenous access is established by placing a cannula in the peripheral vein. There are several sizes of cannulas available, each size having its specific colour. The flow through the cannula depends on its size, i.e., diameter, which is measured in Gauges (G) and written on the cannula packaging. The larger the Gauge, the smaller the diameter of the catheter and the slower the infusion flow. In an adult acutely ill patient, **the standard is to place at least two large-bore IV cannulas with a diameter greater than 18 G**, colloquially “green, white, or grey.”

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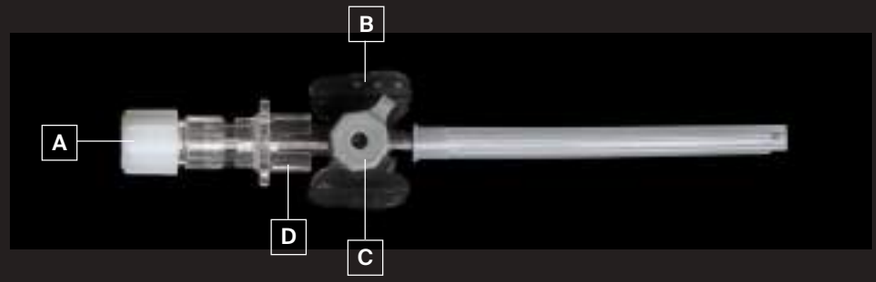

FIGURE 1. *1) intravenous cannulas of different sizes, 2) IV cannula in standard commercial set, A) Luer lock plug with flashback chamber and needle grip (which is removed after the puncture of the vein and placement of the cannula), B) wings that are used for manipulation of the cannula during placement and its fixation after placement, C) Blushing, valve and injection port cap, D) Luer connector*

If possible, the procedure needs to be explained in detail to the patient before the beginning. Compliance and, therefore, the success of the procedure is higher in that case. It is also necessary to check the equipment with which the procedure is performed (for example, whether the infusion system is completely filled with infusion fluid). The procedure begins with palpation of the desired venepuncture site; standard positions are the branches of the dorsal venous arch of the basilic vein, the basilic vein located cubically, the beginning of the cephalic vein located on the hand, and the cephalic vein in the upper arm. There are facilitating manoeuvres that can be used: tying the limb cranially to the venepuncture site (tourniquet), instructing the patient to lower their hand below the level of the heart and squeeze the hand into a fist several times, rubbing with an alcohol antiseptic solution, after which the vein is more clearly visible, and choosing a vein that has a straight course in the length of several centimetres. It is often more important to palpate the vein well rather than to see it well. Placing an IV cannula is performed in several steps: the skin 3 to 4 cm distal to the venepuncture site is stretched (especially in elderly patients, in whom placement may be significantly more difficult due to reduced skin turgor) and then disinfected. The patient is warned to expect a puncture. The vein is punctured at an angle from 10° and 30° from the surface of the patient's skin. After obtaining blood return in the flashback chamber, the cannula is placed another millimetre into the lumen of the vein and then laid horizontally next to the patient's skin. The tourniquet is released, and the cannula should then be "swiped" over the needle into the vein while simultaneously pulling the needle out. The distal part of the cannula is then connected to the infusion system, and flow is started. The patient may feel a sensation of coldness, but pain proximal to the site of the catheter placement, as well as swelling of the surrounding tissue may indicate a blown vein due to improper needle insertion and/or a fragile vein with an escape of the infusion paravascularly. In that case, the IV cannula should be removed and the procedure repeated on another, more proximally located vein. Once the cannula has been successfully placed, secure its position on the skin using the stickers.

KEY POINTS:
- The technique must be practised.
- In acutely ill patients, at least two wider-bore accesses are the basis of patient care.

1. 3. 2. INTRAOSSEOUS ACCESS

Circulatory shock and cardiac arrest significantly disturb the patient's haemodynamics, and setting up intravenous access to such a patient sometimes presents a major challenge. Furthermore, the procedure can be difficult in certain groups of patients, such as children (due to subcutaneous fatty tissue that makes it difficult to palpate the vein

FIGURE 3. *Equipment for establishing intraosseous access*

and place the cannula), the elderly (they often have fragile atherosclerotic blood vessels), and intravenous drug addicts. According to the guidelines for the management of acutely ill patients, two failed attempts to establish intravenous access are an indication for the establishment of intraosseous access.

There are typical places for establishing intraosseous access in adults:

- Proximal part of the humerus: the patient's arm is adjusted in adduction and internal rotation, and the hand is fixed under the lower back. The greater tubercle and the surgical neck of the humerus are palpated. The ideal insertion site is on the greater tubercle, 1 cm above the surgical neck of the humerus.
- Proximal part of the tibia: the patient's leg is placed in extension. The ideal insertion point is 2 cm distal and 2 cm medial from the tibial tuberosity, on the bone surface.
- Distal part of the tibia: the ideal place of insertion is 3 cm proximal to the most prominent part of the medial malleolus. Palpation of the anterior and posterior borders of the tibia is necessary in order to perform the insertion on a flat part of the bone.

In children, there are some special considerations when choosing a place of insertion. One should take into account the epiphyseal cartilage and bone growth zones. The distal femur can be considered an alternative to the previously mentioned sites.

Contraindications for intraosseous access placement are bone fracture, osteomyelitis, infection of the skin and soft tissues over the bone, the patient with an artificial joint or bone in that place, intraosseous access already placed on that bone in the last 48 hours, and the inability to determine typical palpation sites.

The equipment consists of the following: a drill or a gun with a single shot on which the needle is (magnetically) inserted, a system for connecting to a classic set for infusion solutions, and a cuff for applying the infusion solution under pressure. Once the correct position for insertion is selected and aseptic preparation of the skin at the site of the planned insertion is made, the drill with the attached needle is oriented perpendicularly to the patient's bone. Piercing the skin and subcutaneous tissue is done with a needle, without drilling, and after contact with the periosteum, the drilling can start. The goal is to pass through the compact part of the bone and reach the spongious tissue of bone, which can be felt as a loss of resistance. Once the cancellous bone is reached, the drilling should be stopped. The stylet is removed from the catheter, and the remaining part is fixed to the skin. Aspiration of the content (bone marrow) confirms the adequate position. Prior to fluid administration, breakage of the fibrin mesh in the spongious part of the bone should be done, usually with the application of a fluid bolus (volume of 5–10 mL). The procedure itself is painful, therefore, if the patient reacts to pain, it is necessary to apply 2–4 mL of 2% lidocaine before the bolus to anesthetise the septal area. After applying the bolus, it is possible to connect the intraosseous access

to the infusion system, but the infusion solution must be administered under pressure higher than atmospheric. This is most often done by using a cuff/pump. All solutions and drugs that are regularly administered intravascularly, except for cytostatics, can be administered via intraosseous access. Possible complications are infection, compartment syndrome or fluid extravasation. It should be removed from the patient 48 hours after placement or sooner.

KEY POINTS:

- Two failed attempts of intravenous access are an indication for intraosseous access in acutely ill patients.
- Typical placement positions: proximal humerus, proximal tibia, distal tibia.
- Fluid bolus to break fibrin mesh – analgaesia.

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1. 4. ADVANCED LIFE SUPPORT (ALS) ALGORITHM

The advanced life support (ALS) algorithm is a standardised approach for the management of patients in cardiac arrest. It is extremely useful because it allows starting the treatment without losing time for discussion, thus increasing the team's efficiency in resuscitation.

Heart rhythms associated with cardiac arrest can be divided into two groups:

1. shockable **rhythms**, which include pulseless ventricular tachycardia (pVT) and ventricular fibrillation (VF) and
2. **non-shockable rhythms**, which include pulseless electrical activity (PEA) and asystole (Figure 1).

The main difference in the management of the two mentioned groups is that patients with pVT/VF need to be defibrillated as early as possible, i.e., receive an electrical shock to the heart. Other cardiopulmonary resuscitation (CPR) actions, such as heart compressions, airway management and ventilation, intravenous/intraosseal access, administration of adrenaline, and identification and management of reversible causes, are common to both groups. Other additional procedures may be indicated if cardiac arrest is caused by special circumstances. It is essential to note that procedures that improve survival after cardiac arrest include prompt and effective bystander CPR, high-quality chest compressions with minimal interruptions and early defibrillation.

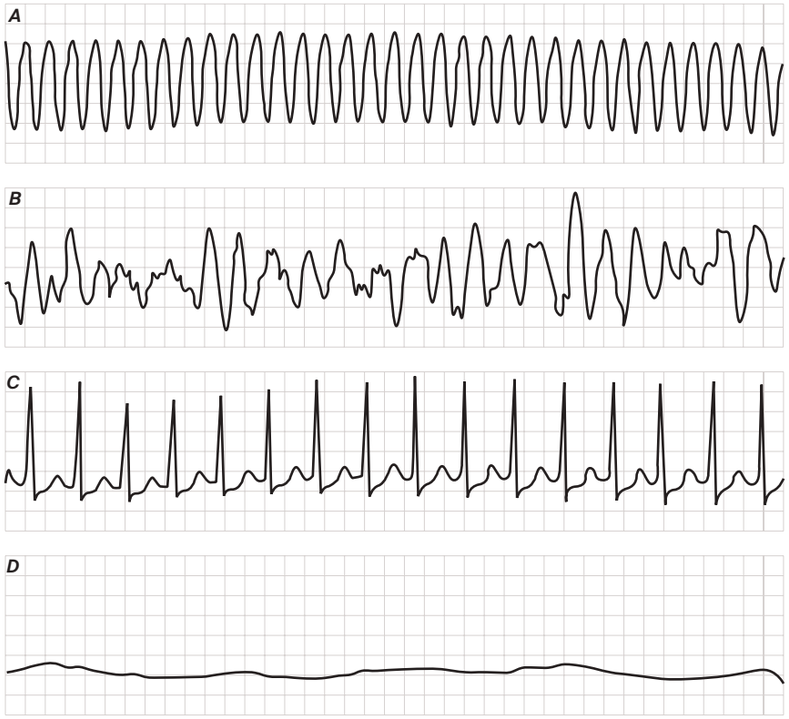

FIGURE 1. *Examples of cardiac arrest rhythms: A: ventricular tachycardia, B: ventricular fibrillation, C: PEA, D: asystole*

1.4.1. MANAGEMENT OF SHOCKABLE RHYTHMS (PVT, VF)

1. Confirm cardiac arrest by simultaneously checking the pulse and breathing for 10 seconds.
2. Call the resuscitation team.
3. Initiate chest compressions 30:2 while simultaneously placing self-adhesive defibrillation/monitoring pads — one below the right clavicle and the other at the V6 position in the midaxillary line.
4. Stop chest compressions to avoid artefacts on the monitor. Confirm VF/pVT on ECG.
5. Immediately continue chest compressions. At the same time, the person on the defibrillator should choose the appropriate energy (150–200 J biphasic for the first defibrillation and 150–360 J biphasic for the following defibrillations) and press the charge button.
6. While the defibrillator is charging, warn all present team members, except the one performing CPR, to stand back and move the oxygen source at least one meter away from the defibrillator.

7. When the defibrillator is charged, tell the person performing the massage to “step back”. When the person moves away, deliver the shock.
8. Immediately after shock delivery, proceed with CPR (30:2) without checking the rhythm or pulse, starting with chest compressions.
9. Continue CPR for 2 minutes and secure the airway with an advanced device during that time.
10. After 2 minutes, stop briefly and check the monitor.
11. If the rhythm is VF/pVT, repeat steps 6–10 and deliver a second shock. Continue CPR for 2 minutes. If the airway is secured with an advanced device (e.g., laryngeal mask or endotracheal tube), perform CPR continuously without a break for ventilation with chest compressions at the rate of 100-120/min and respiratory rate of 10/min. Think about reversible causes of death (4H and 4T). If a reversible cause is detected, begin correcting it.
12. If VF/pVT persists, repeat steps 6–8 and deliver a third shock. Immediately resume CPR and give adrenaline 1 mg IV/IO and amiodarone 300 mg IV/IO while performing CPR for the next 2 minutes.
13. If VF/pVT persists, repeat the 2-minute sequence of CPR — rhythm/pulse check — defibrillation.
14. Give another 1 mg of adrenaline IV after delivery of every other shock (e.g., every 3–5 min).

1.4.2. TREATMENT OF NON-SHOCKABLE RHYTHMS (PEA, ASYSTOLE)

1. Begin CPR 30:2.
2. As soon as the IV/IO access is established, give adrenaline 1 mg IV/IO.
3. Continue CPR 30:2 until the airway is secured (then continue with chest compressions without stopping during ventilation) for 2 minutes.
4. After 2 minutes, check the rhythm: If organised electrical activity is shown, the pulse and/or signs of life should be checked.
   a) If there is a pulse and/or signs of life, start post-resuscitation care. b) If there is no palpable pulse and/or no signs of life (PEA — pulseless electrical activity), continue CPR.
5. While performing CPR, think about reversible causes of death (4H and 4T).
6. After 2 minutes, check the rhythm and continue depending on the assessed rhythm (in the case of persistent cardiac arrest, once given, repeat adrenaline 1 mg IV every 3–5 minutes — every other 2-minute cycle of CPR):
   — If the monitor shows VF/pVT, follow the other side of the algorithm that includes defibrillation.
   — If asystole is shown on the monitor, continue CPR.

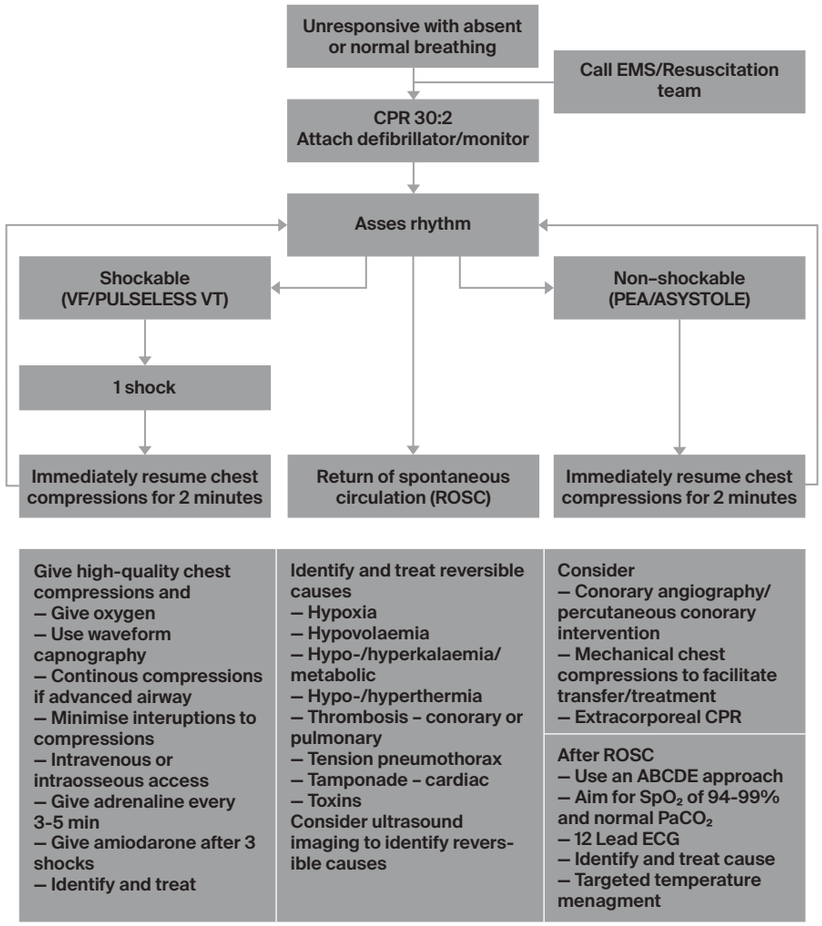

FIGURE 2. *Advanced life support algorithm (retrieved from Adult Advanced Life Support Course Manual published by the European Resuscitation Council, 2021)*

TABLE 1. *Reversible causes of death (4H and 4T)*

4H	4T
Hypoxia	Tension pneumothorax
Hypovolaemia	Tamponade
Hyperkalaemia, hypokalaemia, and metabolic disorders	Thrombosis (coronary thrombosis, pulmonary embolism)
Hypothermia	Toxic agents

During the treatment of cardiac arrest, you should keep in mind the possible reversible causes that led to it, which are divided into two groups (4H and 4T) to make them easier to remember (Table 1).

Hypoxia: reduce the risk of hypoxia by oxygenating with 100% oxygen, adequate ventilation, and securing the airway by placing an i-gel or an endotracheal tube.

Hypovolaemia: pulseless electrical activity is often caused by hypovolaemia, which can result from, for example, bleeding or sepsis. The intravascular volume should be replaced immediately with intravenous fluids and blood through a wide-lumen peripheral venous cannula (min. green IV cannula, diameter 18 G and wider).

Hyperkalaemia, hypokalaemia, hypoglycaemia, hypocalcaemia, and other metabolic disorders can be detected by laboratory tests or from medical history (e.g., kidney failure). Each of these conditions requires specific therapy.

Hypothermia: hypothermia should be suspected in all drowning victims and patients found outdoors. When measuring temperature, use a thermometer with a low-temperature reading option. Such patients should be warmed by passive (blankets, aluminium foil) and active methods (heating the air, giving warm IV fluids, rinsing cavities with warm fluids).

Tension pneumothorax: may be the primary cause of pulseless electrical activity. The most important causes of tension pneumothorax are trauma, barotrauma in patients on positive pressure ventilation, and asthma. The treatment is rapid thoracostomy or needle thoracocentesis.

Tamponade: can be diagnosed by placing an ultrasound probe below the xiphoid. If tamponade is the cause of cardiac arrest, it is necessary to perform an emergency thoracotomy or pericardiocentesis as soon as possible.

Drug poisoning: can be detected later by laboratory tests. If available, use appropriate antidotes.

The most common cause of **thromboembolic** or mechanical **obstruction** is massive pulmonary embolism (PE) or **coronary artery thrombosis**. If it is suspected that the cardiac arrest is caused by a pulmonary embolism, consider immediate administration of thrombolytics. Furthermore, the suspicion of a thromboembolic incident of the coronary arteries can be established from the patient's history (chest pain) or by pre-arrest 12-lead ECG (with documented ischaemic changes).

SOURCES

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PART 1: DETERIORATING PATIENT

Always do the ABCDE approach for initial assessment. DO NOT skip any part. By the end of the assessment the goals are — patent airway, adequate oxygenation and ventilation, haemodynamic stability, correction of neurological state (when possible, otherwise — just note the current state) and to identify possible causes.

Include SAMPLE whenever possible (ask the family, witnesses, etc.).

Use 6-question approach for ECG rhythm analysis in medical emergencies.

Basic interpretation of 12-lead ECG includes analysis of the rhythm, rate, electrical axis and ST segment and T wave changes.

CPR — if you do not provide high quality chest compressions/rescue breaths, all else is in vain.

Follow the ALS algorithm, identify the rhythm as soon as possible and treat accordingly. High quality CPR is equally important in shockable and non-shockable rhythms because it keeps the brain alive.

Shockable rhythms — defibrillate IMMEDIATELY, get to know the defibrillator and needed energies at your workplace prior to approaching a patient. Add adrenaline/amiodarone in appropriate doses after 3rd defibrillation.

Non-shockable rhythms — adrenaline 1 mg iv in a fast bolus.

Reassess the rhythm every two minutes. Repeat adrenaline every 3-5 minutes after first application.

Secure the AIRWAY (usually with i-gel) and establish IV/IO access after the first rhythm recognition, not excuses for delaying the rhythm recognition.

Treat reversible causes — 4H/4T.

Post-resuscitation care — ABCDE again. Try to normalize every parameter you can.

Part 2: Cardiac Emergencies

57—78

2.1. INITIAL ASSESSMENT AND MANAGMENT OF A PATIENT WITH ACUTE CORONARY SYNDROME

2.1.1. INTRODUCTION

Acute coronary syndrome (ACS) is a clinical condition caused by sudden, critical myocardial ischaemia. It most often occurs due to atherosclerotic plaque rupture or erosion and the consequent clot formation in the coronary artery, leading to partial or complete luminal obstruction. This results in acute derangement in myocardial perfusion, which, depending on the severity and duration, leads to cardiomyocyte necrosis and myocardial infarction. The ACS is one of the leading causes of death in developed countries and is considered one of the greatest emergencies in modern cardiology and medicine in general. Therefore, it is crucial to promptly recognise a patient with ACS and start treatment as early as possible to reduce the risk of complications and mortality.

2.1.2. DIAGNOSIS

Since this clinical condition is assigned the highest level of emergency, making the diagnosis as early as possible is crucial. It is essential to recognise the clinical signs and analyse the ECG, and then make a working diagnosis of the acute coronary syndrome without waiting for laboratory results to confirm or exclude cardiomyocyte injury.

CLINICAL PRESENTATION

In most patients, ACS is manifested by the sudden onset of chest discomfort that lasts more than 20 minutes. It usually occurs in the space behind the sternum as a burning, discomfort, pressure, or dull pain. Recurrent, shorter episodes of chest pain are also possible. The pain can spread to the neck, arms (most often to the left shoulder and arm), lower jaw, or back. In the case of inferior wall myocardial infarction, epigastric pain can occur with or without spreading to the retrosternal area. The pain often occurs at rest, without being provoked, typically early in the morning, and it usually awakens the patient from sleep. The onset of symptoms can also be triggered by physical exertion or emotional stress. When ischaemia affects a large part of the myocardium, it can lead to acute heart failure, the presentation of which can vary from mild signs of pulmonary congestion to acute pulmonary oedema or cardiogenic shock. Due to the increased electrical excitability of the ischaemic myocardium, various rhythm disorders are possible, from paroxysmal atrial fibrillation, ventricular extrasystoles, and short-term episodes of unstable ventricular tachycardia to persistent ventricular tachycardia accompanied by haemodynamic instability or ventricular fibrillation and consequent cardiorespiratory arrest. Conduction disorders at different levels are also possible; for example, varying degrees of AV block due to injury to the conduction system or accentuated vagal tone, usually associated with an inferior myocardial infarction. In some patients, like the elderly or people with long-standing diabetes, the clinical presentation of ACS can be atypical in the sense of undefined chest discomfort, shortness of breath, nausea, or generalised weakness.

After the targeted medical history, a 12-lead ECG should be recorded, if possible, within 10 minutes of the first contact.

CLASSIFICATION OF ACUTE CORONARY SYNDROMES

Based on the initial ECG findings, ACS is classified into two forms:

1. **ST-segment elevation ACS (STE-ACS) (> 20 minutes)** (Figure 1)
2. **Non-ST-segment elevation ACS (NSTE-ACS)** – characterised by signs of myocardial ischaemia in the form of ST-segment depression, transient ST-segment elevation, and negative T waves, but the ECG findings may be normal (Figure 2 and Figure 3).

This classification correlates with the pathophysiology of a critical coronary lesion and, at the same time, determines the treatment method. STE-ACS is usually caused by complete atherothrombotic occlusion of one of the coronary arteries and results in ST-segment-elevation myocardial infarction (STEMI). The aim with such patients is to enable reperfusion as soon as possible, i.e., to re-establish flow in the endangered myocardium area using percutaneous coronary intervention (PCI) or, very rarely nowadays, using fibrinolytic drugs. On the other side, NSTE-ACS is usually caused by partial or intermittent coronary artery occlusion, which in some patients also results in non-ST-segment-elevation myocardial infarction (NSTEMI), while in others, cardiomyocyte necrosis is absent (normal cardiac troponin levels), which is called unstable angina pectoris. In a small number of patients, NSTE-ACS can manifest as persistent myocardial ischaemia and consequent recurrent or refractory chest pain, acute heart failure, haemodynamic instability, or malignant ventricular arrhythmias. Accordingly, the approach to the treatment of NSTE-ACS depends on the estimated risk of adverse events. In certain patients, as well as in STEMI patients, emergency invasive approach and revascularisation are necessary, while in others, it is possible to start with drug therapy.

2. 1. 3. MANAGEMENT OF ACUTE CORONARY SYNDROME

Despite the aforementioned classification, ACS is actually a spectrum of clinical conditions with a common aetiology and pathogenesis and, therefore, entails numerous overlaps in approach and treatment, especially in the early prehospital phase or emergency hospital admission.

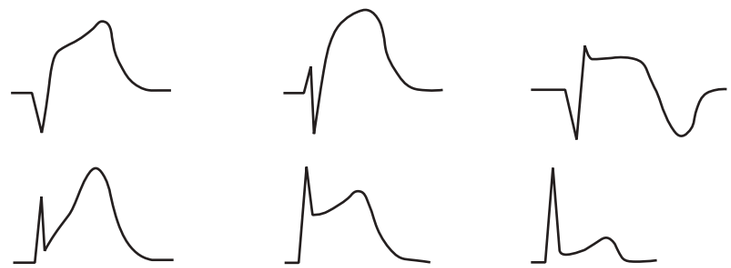

FIGURE 1.
Morphology of ST-segment elevation in patients with STE-ACS

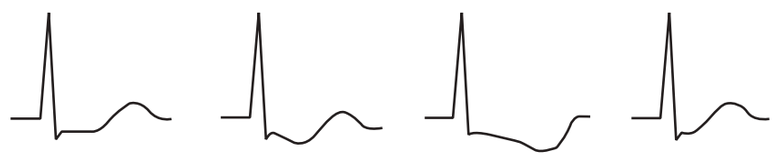

FIGURE 2.
Morphology of ST-segment depression in patients with NSTE-ACS

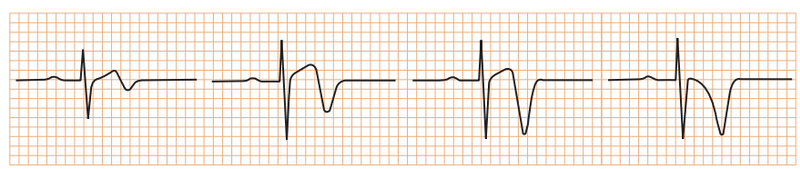

FIGURE 3.
T-wave morphologic variations in patients with NSTE-ACS

Along with a targeted medical history and 12-lead ECG, the approach to each patient with a working diagnosis of acute coronary syndrome emphasises the assessment and treatment of cardiac and circulatory disorders (C from circulation in ABCDE) and must include:

- Setting up venous access,
- Measuring vital parameters (blood pressure, heart rate, SpO_2 levels),
- Short physical examination focused on the cardiovascular and respiratory systems,
- Continuous ECG monitoring with an available defibrillator
- Taking blood samples for laboratory analysis (in emergency hospital admission),
- Suppression of pain and dyspnoea,
- Application of antiplatelet therapy.

TREATMENT OF ACUTE CORONARY SYNDROME

Suppression of pain and dyspnoea

Reducing pain is one of the important initial steps in treating ACS patients since pain is associated with activation of the sympathetic nervous system, consequent vasoconstriction and tachycardia, and thus additional oxygen needs in an already ischaemic myocardium. A dose of 0.4 mg sublingual **nitroglycerin** is most often administered, which can be repeated up to three times at intervals of 5–10 minutes. Nitrates should be avoided in patients with systolic blood pressure below 90 mmHg and in suspected right ventricular infarction.

In the case of persistent pain, it should be considered to administer intravenous **morphine** in single doses of **2–4 mg** every 5 minutes until the pain stops, i.e., until reaching the highest dose of 20 mg. For preventing nausea and vomiting, antiemetic drugs should always be used with opioids. Morphine should be administered slowly by intravenous injection to prevent respiratory depression, hypotension, or bradycardia. Besides acting as an analgaesic, morphine also has a desirable haemodynamic effect due to vasodilation of the venous and arterial system, thus reducing the myocardial oxygen demand.

In hypoxic patients, **oxygen** should be administered in case the $Sa0_2$ is below 90%, while routine oxygen administration is not recommended due to the possible harmful effect of hyperoxia on the injured myocardium. In the case of signs and symptoms of pulmonary congestion, the patient should be given an intravenous **diuretic** (furosemide 20–40 mg or the dose the patient takes at home), and depending on the severity of the accompanying respiratory insufficiency, **non-invasive** or **invasive ventilation** may be indicated.

TREATMENT OF STE-ACS

In STE-ACS patients in whom symptoms appeared within 12 hours, urgent coronary angiography and primary PCI are indicated to mechanically open the occluded coronary artery as early as possible. Therefore, when diagnosing STE-ACS on-site or in health institutions that do not have a PCI centre, the patient should urgently be transported to a centre with a primary PCI program.

Since the activation of platelets and the coagulation cascade play a key role in the pathophysiology of ACS, it is essential to give **dual antiplatelet therapy** as early as possible, i.e., a combination of **acetylsalicylic acid** and a **potent P2Y12 receptor inhibitor** in a loading dose of:

- Oral **acetylsalicylic acid 300 mg**.
- Oral P2Y12 inhibitor – **ticagrelor 180 mg** or **prasugrel 60 mg** (or clopidogrel 600 mg).

Compared to clopidogrel, ticagrelor and prasugrel lead to faster, more reliable, and stronger inhibition of platelet aggregation (onset of action 30 min vs. 2–6 h for clopidogrel), which is why clopidogrel is used only in case the two abovementioned drugs are not available or are contraindicated.

PERIPROCEDURAL AND POSTPROCEDURAL TREATMENT OF STE-ACS

During coronary angiography and PCI, patients also receive an anticoagulant drug, i.e., **unfractionated heparin (70–100 U/kg)** or, less often, **low-molecular-weight heparin (enoxaparin 0.5 mg/kg IV)**. If there are no other indications, further use of anticoagulant therapy after PCI is not indicated except for thromboprophylaxis (prevention of deep-vein thrombosis and pulmonary thromboembolism).

Intravenous administration of **GPIIb/IIIa inhibitors**, which belong to the group of antiplatelet drugs, should be considered in patients with angiographically detected extensive coronary artery thrombosis or in patients without a no-reflow or slow-flow phenomenon.

After PCI, patients continue with dual antiplatelet therapy in a maintenance dose for the next 12 months:

- **Acetylsalicylic acid 75–100 mg**
- **Ticagrelor 2x90 mg** or **prasugrel 10 mg** (reduced dose of 5 mg for BW < 60 kg or age > 75) or **clopidogrel 75 mg**

At the end of the mentioned period, the P2Y12 inhibitor is discontinued, and the patient continues to take acetylsalicylic acid permanently. Depending on the individually assessed risk of bleeding and risk of ischaemic events, the period of dual therapy can be shortened or extended in certain patients.

FIBRINOLYTIC THERAPY

If the estimated time from STEMI diagnosis to PCI reperfusion is longer than 120 minutes, it is recommended to administer intravenous fibrinolytic therapy with a **fibrin-specific agent (alteplase)**. Thanks to the developed network of primary PCI, it is rarely indicated today. Fibrinolysis is also followed by dual antiplatelet therapy in loading doses (acetylsalicylic acid 300 mg and clopidogrel 300 mg, i.e., 75 mg at age > 75), and the patient must then be transported to the

PCI centre to perform routine early coronary angiography and PCI within 2–24 hours.

TREATMENT OF NSTE-ACS

As previously mentioned, there are different categories of NSTE-ACS patients. The correct choice of treatment strategy is based on an early assessment of the risk of ischaemia and bleeding in each patient. NSTE-ACS patients are divided into three categories according to the risk of ischaemia, which also determines the degree of urgency and time of invasive cardiac assessment (Table 1):

1. Very high risk — emergency coronary angiography (< 2 hours from admission),
2. High risk — early invasive assessment (< 24 hours from admission),
3. Low risk — selective invasive assessment (only in patients in whom non-invasive assessment shows signs of inducible ischaemia).

In very high-risk patients, the treatment strategy is the same as in STEMI patients. Before referring to urgent invasive assessment, we should try to suppress the patient's pain and administer **acetylsalicylic acid 300 mg** as early as possible. In NSTE-ACS, it is no longer recommended to routinely administer P2Y12 inhibitors before coronary angiography and insight into the anatomy of the coronary arteries. Studies have shown that pretreatment with P2Y12 inhibitors does not improve outcomes compared with their use during PCI and can significantly endanger the patient and increase the risk of bleeding in the case of indication for urgent surgical revascularisation due to multivessel coronary artery disease.

In other NSTE-ACS patients, initial drug therapy includes **anti-thrombotic** and **anti-ischaemic therapy**. Antithrombotic therapy includes:

- **Acetylsalicylic acid 300 mg** initially, then in a maintenance dose of **100 mg**.
- Anticoagulant: **fondaparinux 2.5 mg SC** or **enoxaparin 2 × 1 mg/kg**.

Beta-blockers are indicated for reducing myocardial oxygen demand (if the patient is haemodynamically stable and without signs of acute heart failure). In persistent pain, we should also administer **nitrates** sublingually or intravenously. Most NSTE-ACS patients undergo invasive assessment in the aforementioned time intervals, except for patients who do not have high or very high-risk criteria, with normal troponin levels, in whom non-invasive testing excluded inducible ischaemia.

PERIPROCEDURAL AND POSTPROCEDURAL TREATMENT OF NSTE-ACS

A patient with an indication for PCI after a coronary angiography should receive a potent P2Y12 inhibitor in a loading dose – **prasugrel 60 mg** or **ticagrelor 180 mg** and additionally a bolus of **unfractionated heparin (70–100 U/kg)** or **enoxaparin (0.5 mg/kg BW)** if the patient has previously received it. If there is no other indication, further use of anticoagulant therapy after PCI is not indicated even in NSTE-ACS patients, except for thromboprophylaxis.

Recommendations regarding dosage and duration of dual antiplatelet therapy in NSTE-ACS patients are the same as in STE-ACS patients.

TABLE 1. Classification of NSTE-ACS patients depending on the risk of ischaemic events and complications

VERY HIGH RISK	HIGH RISK	LOW RISK
— haemodynamic instability / cardiogenic shock — acute heart failure — recurrent/refractory pain — malignant arrhythmias — mechanical complications of MI — ECG pattern of multivessel coronary artery disease	— NSTEMI diagnosis — emerging/dynamic ST/T segment changes — out-of-hospital cardiac arrest without ST elevation or cardiogenic shock — GRACE score > 140	— absence of very high and high-risk criteria — absence of recurrent episodes of pain

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2. 2. INITIAL ASSESSMENT AND MANAGEMENT OF A PATIENT WITH CARDIAC ARRHYTHMIA

2. 2. 1. INTRODUCTION

Cardiac rhythm disorder, or arrhythmia, refers to any abnormality in the generation and/or conduction of a cardiac impulse. Arrhythmias include a spectrum of disorders, from harmless and haemodynamically insignificant to those leading to haemodynamic collapse and cardiac arrest. We often encounter patients with various cardiac rhythm disorders in daily clinical practice, especially in outpatient emergency services and emergency hospital admissions. Therefore, it is essential to know how to approach such a patient and assess whether their condition is life-threatening and requires urgent treatment. For this to be possible, it is necessary to be familiar with the most common arrhythmias and their potential complications.

2. 2. 2. CLASSIFICATION OF ARRHYTHMIAS

Cardiac rhythm and conduction disorders are usually classified according to heart rate into tachyarrhythmias (> 100/min) and bradyarrhythmias (< 60/min).

Depending on the place of origin, tachyarrhythmias are divided into supraventricular and ventricular, and in everyday practice, a more practical classification is into narrow (QRS ≤ 120 ms) and broad (QRS > 120 ms) QRS complex tachycardias. In the case of broad QRS complex tachycardia, cardiac rhythm is most often of ventricular origin, but it can also be a supraventricular disorder (for example, SVT

with bundle branch block or aberrancy, SVT in preexcitation syndrome). Ventricular tachycardia always represents a serious diagnosis and a greater risk of haemodynamic collapse and/or transition to ventricular fibrillation with consequent cardiac arrest.

Bradyarrhythmias occur as a result of disorders in the generation or conduction of impulses at the level of the sinus node or lower parts of the conduction system (AV node, bundle of His). They can be manifested by various disorders, from sinus bradycardia, sinoatrial block, sinus arrest, bradycardia-tachycardia syndrome, or varying degrees of atrioventricular block. After the initial assessment of the patient's condition and, if necessary, emergency treatment, we must distinguish between a reversible disorder (due to the effects of drugs, electrolyte disorders, hypothyroidism, hypothermia, or myocardial infarction) or an irreversible condition resulting from an intrinsic conduction system disorder. It is also important to recognise physiological bradycardia caused by excess vagal tone or in athletes when a sinus rhythm of 40–50/min at rest or up to 30/min during sleep is considered a normal finding.

The haemodynamic effect of a cardiac rhythm disorder in each patient will depend on the type of arrhythmia, its frequency, duration, and the presence of structural heart disease. Thus, a person with a healthy myocardium will tolerate extreme heart rate well (> 150/min, < 40/min), while in people with heart failure or coronary disease, even significantly smaller heart rate deviations will cause symptoms and unfavourable haemodynamic effects.

2.2.3. INITIAL APPROACH TO THE PATIENT WITH A CARDIAC ARRHYTHMIA

In patients with cardiac arrhythmia, two procedures are essential:

- assessment of the clinical condition,
- recognition of arrhythmia and being aware of its potential haemodynamic effects.

Assessment of the clinical condition implies the detection of adverse signs or symptoms, the presence of which indicates a serious rhythm disorder that haemodynamically threatens the patient and requires immediate treatment. Adverse signs include:

1. Hypotension (systolic blood pressure < 90 mmHg) – if it is accompanied by symptoms of hypoperfusion of the target organs (pale skin, cold sweat, confusion, inappropriate urine output) we are talking about shock.
2. Heart failure — signs of pulmonary congestion that can vary from the auscultatory finding showing late inspiratory crackles over the lung bases to the presentation of acute pulmonary oedema.
3. Myocardial ischaemia — can present with symptoms such as chest pain and/or ECG changes (ST depression, negative T waves).
4. Syncope — a short-term, temporary loss of consciousness with complete recovery that occurs due to a sudden drop in cardiac output and thus brain hypoperfusion.

A very efficient and rapid method for the initial assessment and detection of adverse signs in a patient is the **ABCDE** (airway, breathing, circulation, disability, exposure) approach and patient care. Special emphasis is placed on the intravenous route, recording a 12-lead ECG, and continuous ECG monitoring, blood pressure, and SpO_2.

2.2.4. TREATMENT

Initial treatment in the outpatient emergency department or emergency hospital admission will depend primarily on the patient's haemodynamic stability and, of course, the type of arrhythmia. Treatment options are drugs, electroshock delivery in the form of synchronised cardioversion or defibrillation for tachycardia disorders, and external electrostimulation for bradycardic rhythm disorders. In some patients, we can use vagal manoeuvres. It is also important to remove possible triggers (electrolyte disorders, acid-base disturbances, hypoxia, medications).

TREATMENT OF TACHYCARDIA IN HAEMODYNAMICALLY UNSTABLE PATIENTS

In patients who are haemodynamically unstable due to tachycardia, it is necessary to perform synchronised cardioversion, which involves shock delivery during the R wave to prevent delivery during the T wave and consequent ventricular fibrillation. The patient needs to be sedated beforehand, usually with propofol in a 1 mg/kg dose. The initial energy of the biphasic defibrillator is 120–150 J for broad QRS complex tachycardias and atrial fibrillation and 70–120 J for regular narrow QRS complex tachycardias, including atrial flutter. In the case of failure, increasing the energy level in the next shock is recommended, with a maximum of three attempts. If conversion does not occur after the third shock, administer amiodarone 300 mg IV over 10–20 minutes, then repeat synchronised cardioversion.

TREATMENT OF TACHYCARDIA IN HAEMODYNAMICALLY STABLE PATIENTS

If the patient has no adverse symptoms, treatment will depend on the type of arrhythmia. Therefore, the 12-lead ECG should be analysed in detail, and a differential diagnosis should be made to detect the disorder. In that sense, the classification of tachycardias according to the QRS complex broadness and the RR interval regularity is useful.

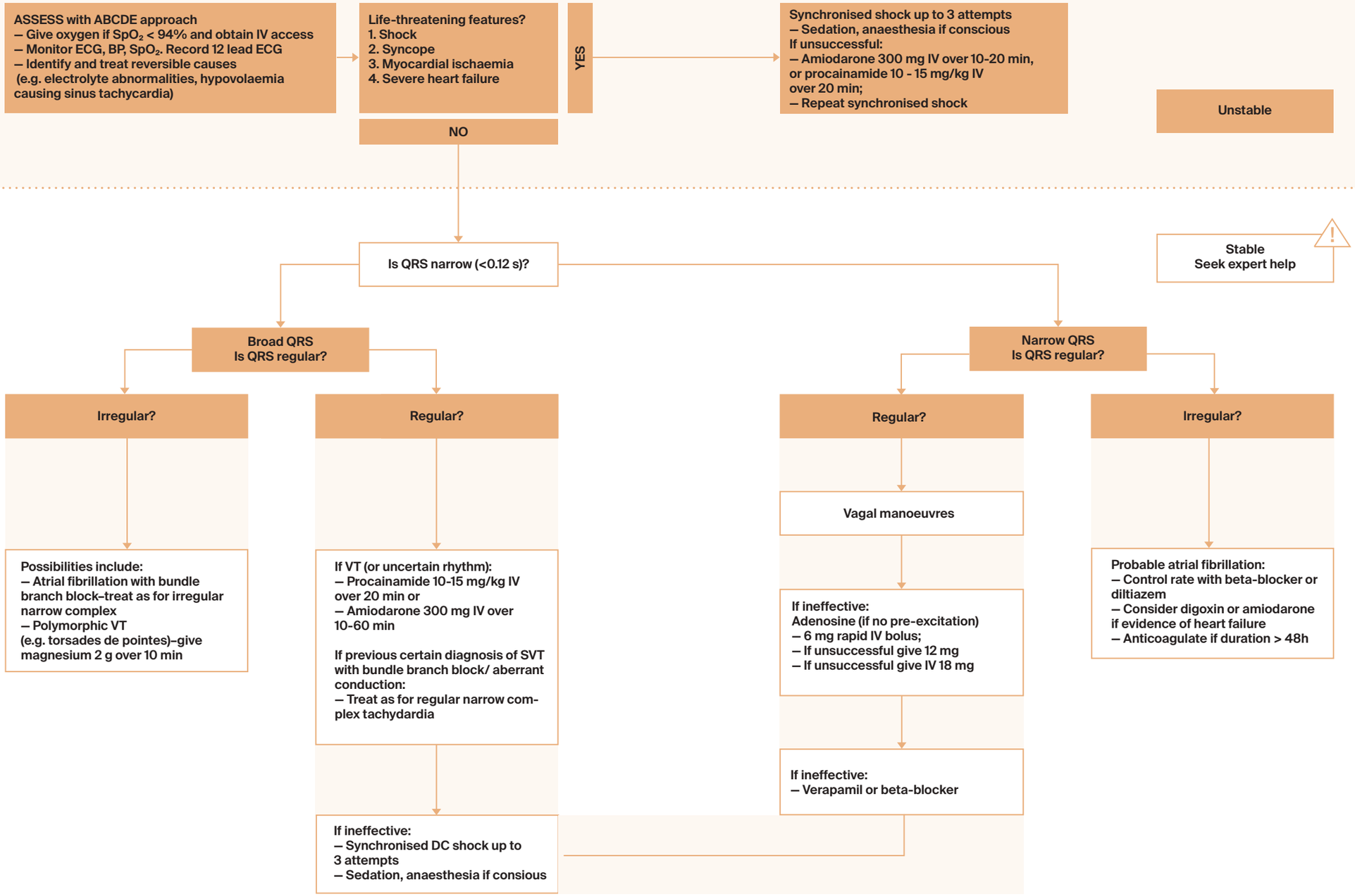


FIGURE 1. *Tachycardia algorithm (retrieved from www.crpguidelines.eu, under permission from the European Resuscitation Council, 2021)*

REGULAR BROAD QRS COMPLEX TACHYCARDIA

Regular broad QRS complex tachycardia can be classified as ventricular tachycardia (80%) or supraventricular tachycardia (SVT) with bundle branch block (15%) or with anterograde conduction over the accessory pathway (5%). Any regular broad QRS complex tachycardia is considered ventricular tachycardia until proven otherwise. The procedure is the following:

- Administer **amiodarone 300 mg** IV over 20–60 minutes or procainamide 10-15 mg/kg IV over 20 minutes (currently not available in our country).
- Previous certain diagnosis of SVT with bundle branch block – treat as for the narrow complex tachycardia.

IRREGULAR BROAD QRS COMPLEX TACHYCARDIA

In stable patients, it is usually atrial fibrillation with bundle branch block, while other options are atrial fibrillation with preexcitation (Wolff-Parkinson-White-WPW-syndrome) and polymorphic ventricular tachycardia, which are more often accompanied by haemodynamic instability.

In the case of atrial fibrillation with bundle branch block, the approach to the patient is described in detail in the section on narrow QRS complex tachycardias.

Atrial fibrillation with pre-excitation represents an emergency and, most often, the need for urgent **synchronised cardioversion** due to the high risk of transition to ventricular fibrillation. Another option is the use of antiarrhythmic drugs, of which **procainamide** has proven to be the safest. Among the drugs available in our country, **propafenone** IV in a dose of 1.5–2 mg/kg can be considered. It is contraindicated to use drugs that slow down the AV conduction (beta-blockers, calcium channel blockers, adenosine) and thus stimulate conduction through the accessory pathway, which can lead to a further increase in heart rate and occurrence of ventricular fibrillation.

Polymorphic VT with a prolonged QT interval in the basic ECG (*torsade de pointes*) also usually requires urgent treatment because of the high incidence of transition to ventricular fibrillation. In such patients, it is necessary to administer **magnesium sulphate** 2 g IV, correct electrolyte disorders, and stop using drugs that prolong the QT interval. If the arrhythmia persists, the option is overdrive pacing, for which it is necessary to place temporary transvenous pacing.

REGULAR NARROW QRS COMPLEX TACHYCARDIA

Paroxysmal supraventricular tachycardias (AVRT, AVNRT), atrial tachycardia, or atrial flutter with regular AV conduction are most often presented as regular narrow QRS complex tachycardias.

It is initially recommended to apply a **vagal manoeuvre** (pressure on the carotid sinus or the Valsalva manoeuvre by blowing into a 10 mL syringe), which can terminate PSVT in 54% of cases if performed correctly. On the other hand, in cases of atrial flutter or atrial tachycardia, by slowing AV conduction, the vagal manoeuvre can unmask the flutter waves or P waves. Sometimes it can stop atrial tachycardia, depending on the mechanism of its origin. If the vagal manoeuvre does not stop the tachycardia, **adenosine** should be administered in a rapid IV bolus dose of 6 mg, then 12 mg and 18 mg if the previous administration was unsuccessful.

If adenosine does not stop the tachycardia nor unmask the atrial flutter, **metoprolol** 2.5–5 mg IV can be administered up to a maximum of 15 mg or **verapamil** 2.5–10 mg IV. If the pharmacological treatment fails, we can perform **synchronised cardioversion**, which is rarely necessary for PSVT.

In the case of atrial flutter, the treatment strategy is the same as in patients with atrial fibrillation, including the prevention of thromboembolic events.

IRREGULAR NARROW QRS COMPLEX TACHYCARDIA

It most often presents as atrial fibrillation or, less often, atrial flutter with variable AV conduction. It is essential to know when the arrhythmia occurred and to assess the risk of thromboembolic complications. If arrhythmia lasts > 48 h or the time of onset is unknown, a heart rate control strategy is indicated using **beta-blockers (bisoprolol** 1.25–10 mg orally, metoprolol 2.5–5 mg IV up to a total dose of 15 mg or 25–200 mg orally) or very rarely by using **calcium channel blockers (verapamil** 2.5–10 mg IV). The target ventricular rate is < 110/min at rest, and if it is not achieved by one of the mentioned drugs, we can consider **digoxin** (0.2–0.4 mg IV up to a total saturation dose of 1 mg in 24 h; oral maintenance dose 0.1 mg). In such cases, conversion to sinus rhythm without prior anticoagulation for at least 3 weeks is contraindicated due to the risk of thromboembolic complications. If the patient has been regularly taking oral anticoagulant therapy, it is possible to proceed with the cardioversion procedure.

If AF occurs within 48 hours, pharmacological cardioversion can be attempted; if this is unsuccessful, synchronised electrocardioversion can be attempted. Choosing antiarrhythmic drugs depends on the presence of structural heart disease. In patients without structural heart disease, the option is **propafenone** 1.5–2 mg/kg IV for 10 minutes, and in patients with known heart disease, the option is **amiodarone** 5–7 mg/kg for 1–2 hours. Depending on the estimated long-term risk of thromboembolism (CHA2DS2-VASc score), some patients will need permanent oral anticoagulant therapy.

TREATMENT OF BRADYARRHYTHMIA IN HAEMODYNAMICALLY UNSTABLE PATIENTS

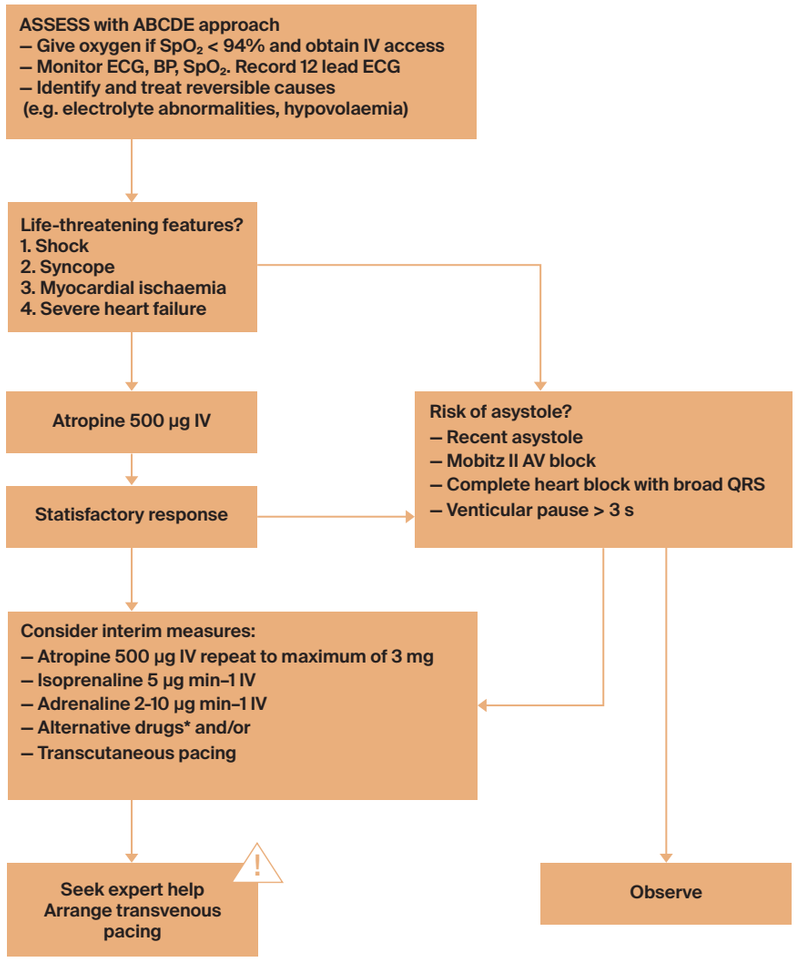
In patients with haemodynamic instability due to bradyarrhythmia, we often encounter third-degree AV block, second-degree Mobitz Type II AV block or AV block 2:1, and sinus arrest. Treatment options

include the use of atropine and cardiac electrostimulation. **Atropine** is administered in a dose of 0.5 mg IV, which can be repeated at intervals of 3–5 min up to a total of 3 mg. The effect of atropine will depend on the mechanism and localisation of conduction disorders. Thus, atropine will improve conduction in patients with increased vagal tone or an AV node disorder. However, if the disorder is localised below the AV node, it will worsen the impulse conduction and the patient's condition. For this reason, the better option is to apply **transcutaneous external cardiac electrostimulation** and urgently call the cardiology service for further treatment.

BRADYARRHYTHMIA IN HAEMODYNAMICALLY STABLE PATIENTS

Immediate treatment is unnecessary in patients who do not have adverse symptoms. We should try to find reversible causes and remove them. It is also important to know which disorders inevitably require further cardiological assessment and treatment due to possible progression to more severe forms and, thus, an increased mortality risk. The following disorders are most often encountered:

- First-degree AV block and second-degree Mobitz Type I AV block – mostly benign and do not require active treatment.
- Second-degree Mobitz Type II AV block and third-degree AV block – high risk of progression and mortality and represent an indication for the implantation of a permanent pacemaker, except in the case of a clear reversible cause.
- Sinus bradycardia – indication for the implantation of a permanent pacemaker only when it is symptomatic and after reversible causes are excluded (sick sinus syndrome).



*Alternatives include:
– Aminophylline
– Dopamine
– Glucagon (if bradycardia is caused by beta–blocker or calcium channel blocker)
– Glycopyrrolate (may be used instead of atropine)

FIGURE 2. *Bradycardia algorithm (retrieved from www.crpguidelines.eu, under permission from the European Resuscitation Council, 2021)*

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Stick to the ABCDEs!

Not all patients have the exact same symptoms during acute coronary syndrome (ACS).

Read the 12-lead EKG properly. Make sure to differentiate between STE-ACS and NSTE-ACS because the therapy is different.
For all ACS patients — administer oral acetylsalicylic acid 300 mg (even First Aid providers are advised to give it to a person with chest pain), morphine for pain (nausea might happen!), give sublingual nitroglycerine (beware of contraindications) and oxygen when SpO_2 is below 90%.

STE-ACS — if within 12 hours — primary PCI is indicated as soon as possible, along with oral P2Y12 inhibitors (prasugrel/ticagrelor).

NSTE-ACE — assess the risk, treat accordingly. Do not administer P2Y12 inhibitors prior to coronary angiography.

When treating cardiac arrhythmias – look for life-threatening features:
— Shock
— Syncope
— Myocardial ischaemia
— Severe heart failure

Tachycardias with a life-threatening feature present are treated with synchronised cardioversion at appropriate energy (up to 3 attempts), followed by amiodarone infusion (300 mg IV over 10-20 min).

Bradycardias with a life-threatening feature present are treated with atropine 0,5 mg IV (up to 3 mg). If not responding — switch to transcutaneous pacing (alternative medicines might be given, but the authors prefer this approach).

In the absence of life-threatening features, one might still have to treat bradycardias. Check for risk of asystole (recent asystole, Mobitz II AV block, complete heart block with broad QRS, ventricular pause >3s).

Part 3: Respiratory Emergencies

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3. 1. ACUTE RESPIRATORY FAILURE

3. 1. 1. INTRODUCTION

Acute respiratory failure happens as a primary failure in breathing or as a secondary failure following other disorders in the body. It is characterised by the inability of the lungs to meet the body's requirements for oxygen and/or carbon dioxide removal.

Respiratory control is a complex activity. Automated rhythmic breathing is controlled by the brain stem through multiple neuronal groups. It can be interrupted by coughing and vomiting and is susceptible to voluntary control (e.g., speech). Central chemoreceptors in the medulla oblongata are sensitive to hydrogen ion concentration (H+). An increase in the arterial carbon dioxide partial pressure ($PaCO_2$) increases the PCO_2 in the cerebrospinal fluid (CSF), which causes a drop in pH and drives breathing. Peripheral chemoreceptors (in the carotid bodies) are sensitive to hypoxia, but only by sensing a drop in the arterial oxygen partial pressure (PaO_2) and not so much the total oxygen content. Acidaemia, fever, and hypoperfusion are also important ventilation drives.

OXYGEN TRANSPORT IN THE BLOOD

Oxygen is transported via haemoglobin (oxyhaemoglobin), and a smaller fraction is dissolved in the blood.

Oxygen delivery (DO_2) is described by the following equation:

$$DO_2 = CO \times ((1.39 \times [Hb] \times SaO_2) + (PaO_2 \times 0.03))$$

DO_2 — oxygen delivery (mL/min)
CO — cardiac output (L/min)
1.39 mL/g — oxygen-binding capacity of haemoglobin
[Hb] — haemoglobin concentration (g/L)
SaO_2 — blood oxygen saturation (haemoglobin), expressed as a fraction (e.g., 0.97 instead of 97%)
PaO_2 — oxygen partial pressure in arterial blood (mmHg)
0.03 — amount of dissolved oxygen in the blood (mL/L/mmHg) (37 °C)

Under normal circumstances, oxygen delivery is abundant compared to oxygen consumption. When something pathological occurs, there is an imbalance between oxygen delivery and oxygen consumption. This is not exclusive to breathing difficulties. When a patient bleeds, the haemoglobin content decreases, which also alters the oxygen delivery capacity.

OXYGEN-HAEMOGLOBIN DISSOCIATION CURVE

The curve shown above describes the relationship between oxygen saturation and partial pressure. It is not linear because the haemoglobin-oxygen affinity increases in an oxygen-rich environment. Haemoglobin is a tetramer, and each molecule binds four oxygen molecules. Each oxygen molecule binds independently. The binding of the first oxygen molecule increases the affinity for the next one (and so on). The curve shifts left or right depending on certain conditions (as shown in the figure above). A leftward shift (higher affinity/low tissue release) occurs with hypothermia, alkalosis, and carbon monoxide poisoning. A rightward shift (lower affinity/high tissue release) occurs in fever and acidosis. In practical terms, it means that more oxygen is released in the tissues during fever (which is good when the body is fighting an infection).

OK, ENOUGH ABOUT OXYGEN, WHAT ABOUT CO_2?

Unlike oxygen transport, 98% of which is bound to (oxy)haemoglobin and 1–2% of which is dissolved in the blood, carbon dioxide transport is slightly different. About 10% is dissolved in the blood, 10–20% is conjugated as carbamates (on deoxyhaemoglobin), and the majority (70–90%) is transported as a bicarbonate ion after the following reaction occurs in the erythrocyte.

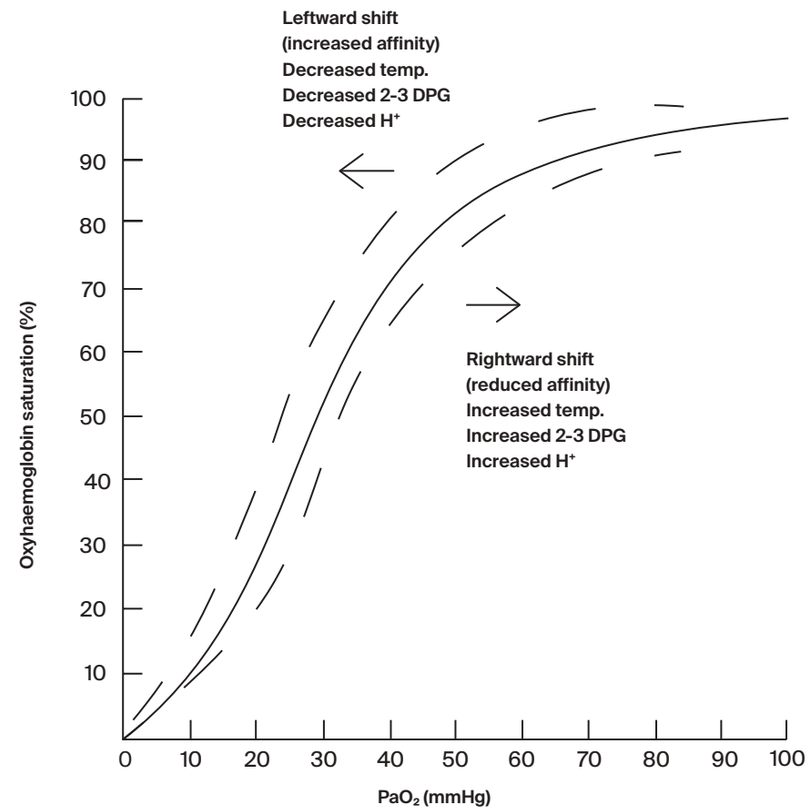

FIGURE 1. *Oxygen — haemoglobin dissociation curve. Note the nonlinear relationship between PaO_2 and SaO_2. P_{50} is the PaO_2 at which the SaO_2 is 50%. The P_{50} value changes when the curve shifts.*

Reduced oxygen delivery and/or increased consumption are the main causes of tissue hypoxia.

$$CO_2 + H_2O \rightleftharpoons H_2CO_3 \rightleftharpoons HCO_3^- + H^+$$

Deoxyhaemoglobin has approximately 3.5 times greater affinity for CO_2. Why is that important? In chronic obstructive pulmonary disease (COPD), patients have trouble breathing the air out of their lungs effectively, which increases the amount of CO_2 content in the body. Usually, it binds with deoxyhaemoglobin. When such a patient receives oxygen therapy, this often leads to a further increase in $PaCO_2$ levels, which can be fatal (why and how is explained in a later chapter of this manual).

If we sum this all up and try to organise it in a better and clinically useful way, we can define four types of respiratory failure:

TYPE 1. CLASSIC HYPOXAEMIC RESPIRATORY FAILUR
PaO_2 < 60 mmHg (< 8 kPa)

CAUSES:
- alveolar oxygenation failure (low FIO_2, alveolar collapse, alveoli filled with fluid or blood, infiltration...)
- disrupted transition of oxygen from the alveoli to the blood (interstitial lung disease, pulmonary vascular disease)
- reduced blood oxygen transport (obstructed flow, shunting, low Hb concentration, defective Hb molecule).

TYPE 2. HYPERCAPNIC RESPIRATORY FAILURE
$PaCO_2$ > 45 mmHg (> 6kPa)

Defined as an inability to adequately remove CO_2 because of reduced alveolar minute ventilation. This increase also leads to hypoxaemia because of the substitution of O_2 with CO_2. Hypoxia in this case is usually easily corrected with oxygen therapy.

TYPE 3. PERIOPERATIVE RESPIRATORY FAILURE
A specific type of failure due to the circumstances in which it occurs is caused by abnormal movements of the chest and abdomen due to surgery/anaesthesia (residual drug effects) or injury. Both hypoxia and hypercarbia occur.

Treatment: prevention with an adequate level of anaesthesia and/or analgaesia, reducing unnecessary sedation, elevated head level, etc.

TYPE 4. HIGH-DEMAND RESPIRATORY FAILURE
The inability of healthy (or not primarily affected) lungs to maintain increased ventilation in the case of a systemic hypermetabolic state. Sepsis is one such condition. If the primary cause is left untreated or there is no therapeutic response due to the nature of the state, respiratory muscle fatigue sometimes requires mechanical ventilation.

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3. 2. OXYGEN THERAPY AND PULSE OXIMETRY

Oxygen is one of the most commonly used drugs in modern medicine, and until recently, it was considered harmless and no special dosing recommendations were needed. However, today, we are aware of many adverse effects of oxygen, which, like any other drug, has a recommended dosage regimen, administration route, side effects, indications, and contraindications.

We all breathe oxygen at slightly less than 21% (F_IO_2 0.21). When climbing to high altitudes, the percentage of oxygen remains the same, but the partial pressure of gases decreases.

Administering oxygen increases oxygen concentration in the inspired air. The oxygen dose is the F_IO_2, which is determined clinically (administration route and flow rate). It can be administered to a spontaneously breathing person via nasal cannula, simple oxygen mask, and non-rebreather mask. The approximate F_IO_2 is presented in the table below. Venturi masks are specially designed to deliver fixed F_IO_2 (colour-coded).

Indications for administering oxygen are compromised airway patency and/or low SaO_2 level (below 94% in previously healthy individuals, below 88% in COPD patients). Oxygen acts as a vasoconstrictor of coronary arteries. In a patient with chest pain, oxygen administration is indicated when the SaO_2 level drops below 90%. Oxygen produces ROS (reactive oxygen species), which demonstrates tissue toxicity. As an exemption, in haemorrhagic shock, we can administer high-flow oxygen therapy even with normal SaO_2

TABLE 1. *Approximate oxygen dose, measured as inspired oxygen fraction (FIO2), depending on the device for oxygen delivery and suggested set flow rate*

Device for oxygen delivery	Oxygen flow rate	F_IO_2 delivered
Nasal cannula	1-4 L/min	≈0,21–0,30
Simple oxygen face mask	4–10 L/min	≈0,35–0,60
Non-rebreather mask	10–15 L/min	≈0,85 (can be up to 1)
Venturi face mask	Colour coded, fixed flow masks, exact F_IO_2 written on the mask	

levels. Loss of haemoglobin in bleeding significantly reduces the oxygen-carrying capacity of the blood and that can lead to severe tissue hypoxia, even with normal saturation levels of the (remaining) haemoglobin.

Pulse oximetry is a non-invasive method of continuous measurement of haemoglobin oxygen saturation. The probe is placed at the fingertip, and the screen shows the SpO_2 percentage, oximetry curve, and numerical heart rate. It measures light absorption from the blood at two different wavelengths – 660 nm and 940 nm. One fraction of light passes through the tissue and another one is absorbed in the tissue. It uses two wavelengths to distinguish between oxyhaemoglobin and deoxyhaemoglobin molecules.

LIMITATIONS

- It does not provide information about ventilation, only about oxygenation.
- In the case of carbon monoxide poisoning, a falsely high reading occurs due to similar wavelengths of oxyhaemoglobin and carboxyhaemoglobin.

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3. 3. ARTERIAL BLOOD GASES (ABGs)

Partial pressure describes the number of molecules in a mixture of gases better than the percentage. For example, if we double the total pressure of a mixture, partial pressures go double, and the percentage stays the same. The atmospheric air we breathe is at 1 atm (100 kPa or 750 mmHg). The nitrogen partial pressure is 78 kPa (and 78%) and the oxygen partial pressure is 21 kPa (21%). When we breathe a 40% oxygen mixture, the oxygen partial pressure is 40 kPa. When a gas is dissolved in a liquid (e.g., blood), partial pressure in the liquid is the same as the gas that is in contact with the fluid. That is the Pmixture of a specific gas, e.g., PaO_2 (partial pressure of oxygen in arterial blood).

For arterial blood gas analysis and determining acid-base status, the following four key parameters are measured:

1. **pH** — hydrogen ion concentration in the body is extremely small (e.g., sodium concentration is 3 million times bigger), therefore, we use the pH scale (1–14) to determine hydrogen ion concentration. **The pH of a normal blood sample is 7.35–7.45.** The relationship between hydrogen ions and pH is inversely proportional. High pH means a drop in hydrogen ion levels. The increase in blood pH is called alkalaemia. Low pH means an increase in hydrogen ion levels. The drop in blood pH is called acidaemia. Small changes in pH mirror big changes in hydrogen ion concentration. Body enzymes function properly

only in a very narrow pH range. The human body makes a significant amount of hydrogen ions daily and then it buffers them to prevent acidaemia. The most important intracellular buffers are proteins and phosphates. The most important extracellular buffers are plasma proteins and bicarbonates (HCO_3-). Even if the body pH is normal, an acid-base disturbance may still be present, although in a compensated form.

2. **$PaCO_2$ — carbon dioxide is a by-product of cellular metabolism.** To get to the lungs and out of the body, it is transported in the blood, either dissolved in the plasma or bound to the haemoglobin and proteins. **Normal values are 4.7–6.0 kPa** (35–45 mmHg).

$$CO_2 + H_2O = H^+ + HCO_3^-$$

It acts as an acid. An increase in CO_2 will increase the hydrogen ion concentration and promote acidaemia. Reduction of alveolar ventilation is the leading cause of the increase in CO_2 values if metabolic production remains constant. We call that respiratory acidosis. An increase in alveolar ventilation will cause a drop in CO_2 and promote alkalaemia. We call that respiratory alkalosis. In a healthy individual, brain respiratory centres react swiftly to the changes in hydrogen ion concentration and cause a quick increase in alveolar ventilation.

3. **HCO_3^- (bicarbonate) and BE (base excess) —** The most common way to interpret acid-base disturbances in a blood sample is to analyse bicarbonates as a main buffering mechanism. Their values are regulated in the kidney, and they oppose acids. They act as bases and neutralise hydrogen ions, as shown in the formula below.

$$H^+ + HCO_3^- = H_2CO_3$$

Normal values are 22–26 mmol/L. A decrease in the bicarbonate levels happens when they buffer the extra hydrogen ions. It is a state of metabolic acidosis. An increase in bicarbonate levels causes excess hydrogen ion buffering and a state of metabolic alkalosis.

Metabolic and respiratory components are interdependent, and the derangement of one component typically causes the other to change as well to try to maintain a normal blood pH value. Because the bicarbonate does not directly change blood pH (it only serves as a buffer for the metabolic processes), there are some other things worth noting.

There are three independent variables that affect blood pH:

1. STRONG ION DIFFERENCE (SID) – a difference in the sum of strong cations (e.g., Na^+, K^+) and the sum of strong anions (e.g., Cl^-, lactate).
2. Total concentration of plasma non-volatile buffers (A_{tot}) – mostly determined by phosphates and albumin concentration.
3. $PaCO_2$.

These three variables have a direct influence on blood pH, whereas hydrogen ion and bicarbonate concentrations act as dependent variables. To further explain this, it would take more than the scope of this manual, so the author kindly advises you, should you have additional interest, to seek information online. Standard base excess is a calculated value defined by how the SID should change to bring the pH value back to 7.4 with a steady-state CO_2 value of 5.4 kPa (40 mmHg) at a normal body temperature.

$$SID = ([Na^+] + [K^+] + [Ca_2^+)] + [Mg_2^+)]) - ([Cl^-] + [lactate])$$

Metabolic acidosis is a product of a decreased SID value, which in turn causes chemical reactions that increase hydrogen ion concentration. SID is reduced when the human body loses sodium bicarbonate (e.g., diarrhoea) or in the abundance of exogenous anions (e.g., intoxication). Metabolic alkalosis is a product of an increase in SID, which occurs when there is a drop in strong anion concentration or an abundance of strong cations (e.g., vomiting, diuretic therapy).

SID value is mostly regulated by kidneys, although the gastrointestinal system also contributes. Kidneys are unable to quickly excrete significant amounts of strong ions, and it takes a couple of minutes to hours to have an impact on SID. For every chloride ion that is filtrated and not reabsorbed, a reduction in SID happens. The equilibrium of chloride ion concentration is the main drive for SID changes. In the stomach, chloride ions are pumped into the lumen, which causes an increase in SID. However, it is reabsorbed in the duodenum and returns plasma SID to normal values. Should a person vomit, the chloride ion cannot be reabsorbed. The body loses chloride ions and that causes metabolic alkalosis (contrary to the seemingly logical conclusion that the loss of hydrogen ions is the main reason for alkalosis in such a scenario). Unlike the stomach, the pancreas produces fluid with higher SID levels than plasma (scarce in chloride ions). In the case of a significant loss of that fluid (as may happen with surgical drainage), acidosis will ensue. In the intestine, there is a small number of chloride ions, which are

reabsorbed in the duodenum. We have a high concentration of sodium, potassium, and bicarbonate. During diarrhoea, loss of that fluid causes a reduction in SID and subsequent acidosis.

4. **PaO_2 – normal partial pressure of oxygen in our atmosphere at sea level is 21 kPa (157.5 mmHg). As it passes through the airway, it gets moistened and mixed with carbon dioxide, which causes it to drop to about 13 kPa (97.5 mmHg). PaO_2 is always smaller than PAO_2 (partial pressure of oxygen in the alveoli). The normal value is higher than 11 kPa (82.5 mmHg). In extremely old patients and those with chronic lung disease, a value higher than 8 kPa is tolerated.**

   The fraction of oxygen in the inspired air, F_iO_2, is 0.21.

   An estimated expected PaO_2 value is 10 points lower than the percentage of inhaled oxygen. For example, if a person with no disruption in lung oxygenation breathes an oxygen admixture of 40% (F_iO_2 = 0.4), the predicted value of PaO_2 is 30 kPa.

ABG ANALYSIS (APPROACH ACCORDING TO THE EUROPEAN RESUSCITATION COUNCIL):

1. Step: **Assess the patient!**

2. Step: **Is the patient hypoxic?** (don't ignore the F_iO_2)

3. Step: **Is the blood acidic or alkaline?** > 7.45 — alkalaemia, < 7.35 — acidaemia

4. Step: **What is happening with the $PaCO_2$ level?**

4a. If the pH is < 7.35 (acidaemia):
Step: **Is the $PaCO_2$ level high? > 6.0 kPa (45 mmHg)** – respiratory acidosis (or respiratory compensation of metabolic alkalosis)

4b. If the pH is > 7.45 (alkalaemia):
Step: **Is the $PaCO_2$ level low? < 4.7 kPa (35 mmHg)** – respiratory alkalosis (or respiratory compensation of metabolic acidosis)

5. Step: **What is happening with HCO_3– and BE?**

5a. If the pH is < 7.35 (acidaemia):
Step: **Is the BE lower (more negative) than –2 mmol/L and/or HCO3– lower than 22 mmol/L?** If so, metabolic acidosis is present (or renal compensation of respiratory alkalosis).

5b. If the pH is > 7.45 (alkalaemia):
Step: **Is the BE higher (more positive) than +2 mmol/L and/or HCO_3 – higher than 26 mmol/L?** If so, metabolic alkalosis is present (or renal compensation of respiratory acidosis).

SOURCES

1. Vincent JL, Moore FA, Bellomo R, Marini JJ. Textbook of Critical Care. 8th ed. Elsevier; 2023.
2. Al-Shaikh B, Stacey GS. Essentials of Equipment in Anaesthesia, Critical Care, and Peri-Operative Medicine. 5th ed. Edinburgh: Elsevier; 2019.
3. Soar J, Böttiger BW, Carli P, Couper K, Deakin CD, Djärv T et al. European Resuscitation Council Guidelines 2021: Adult advanced life support. Resuscitation. 2021; 161: 115–51.

3. 4. ANAPHYLAXIS

Anaphylaxis is a severe, life-threatening, generalised, or systemic hypersensitivity reaction to an allergen. It is characterised by the rapid development of life-threatening problems with the airway and/or breathing and/or circulation. Although commonly associated with anaphylaxis, skin and mucosal changes do not have to be present.

3. 4. 1. INITIAL ASSESSMENT

A (AIRWAY):
Assess whether the airway is open. Anaphylaxis is associated with airway obstruction at several different levels:

- Swelling of the throat and tongue (pharyngeal/laryngeal oedema): the patient breathes audibly, has difficulty breathing and swallowing, and feels “their throat is closing”.
- A hoarse voice is a sign of vocal cords oedema, in which case the airway must be managed urgently by endotracheal intubation.
- Inspiratory stridor — high-frequency inspiratory noise caused by upper airway obstruction.

B (BREATHING):

- Swelling of the respiratory tract and bronchoconstriction can lead to hypoxia. The patient usually tries to compensate by

increasing both the frequency and work of breathing, which can lead to the patient becoming tired and unable to breathe adequately due to exhaustion.
- Bronchospasm can be heard on auscultation of the lungs in the form of wheezing, prolonged expiration, and, in severe cases, the so-called silent chest.
- A decrease in oxygen saturation is possible.
- Anaphylaxis can lead to respiratory arrest!

C (CIRCULATION):
- Circulatory derangements (often called anaphylactic shock) can be caused by myocardial depression, vasodilation, and increased capillary permeability, followed by fluid loss from the intravascular space.
- Signs of shock may be present: tachycardia, hypotension, patient complaining of dizziness, and/or pending collapse.
- Anaphylaxis can cause myocardial ischaemia and ECG changes even in patients with normal coronary arteries.
- Anaphylaxis can lead to cardiac arrest!

D (DISABILITY):
- The previously mentioned problems with the airway, breathing, and circulation can lead to reduced brain perfusion and affect the patient's neurological status. Confusion, agitation, and loss of consciousness may be present.

E (EXPOSURE AND ENVIRONMENT):
- In about 80% of anaphylactic reactions, changes occur on the skin and/or mucosa.
- Erythema may be present – spotted or generalised red rash.
- Urticaria may be present (also called hives), which can appear anywhere on the body. Hives can be pale, pink, or red and remind of nettle burns.
- Angioedema is similar to urticaria but includes swelling of deeper tissues, most often the eyelid area and sometimes the mouth and throat.
- Although skin changes are stressful and cause concern for both patients and those treating them, skin changes without life-threatening problems with the airway, breathing, or circulation are not anaphylaxis.

SAMPLE — patients who have already had anaphylaxis are at a higher risk of developing it again. Positive history of exposure to an allergen is also important, as well as previously known allergies.

The most common allergens that cause anaphylaxis in adults are insect bites, food intake (fish, shellfish, nuts), intake of drugs (penicillin and cephalosporin antibiotics, non-steroid antirheumatic drugs), and latex.

3. 4. 2. TREATMENT

Once the ABCDE assessment is made and anaphylaxis is confirmed, an imperative is to call for help. Treatment of anaphylaxis is based on managing respiratory and circulatory problems. Adrenaline will act on several different levels: by constricting the dilated vessels and reducing their permeability and by having the bronchodilating effect in the lungs. Thus, it has a therapeutic effect on the systemic effects of released histamine.

In anaphylaxis, adrenaline is given only intramuscularly (IM) in a dose of 0.5 mg in the anterolateral side of the middle third of the thigh.

Adrenaline is given intravenously only by experienced specialists in a diluted form due to the disastrous consequences it can have. However, it is important to have an **intravenous or intraosseous route open** for administering intravenous fluids to replace the intravascular volume (significant hypovolaemia is present due to increased capillary permeability and vasodilatation). The airway and breathing are **managed by setting up a high flow of oxygen and endotracheal intubation** (if you are not skilled — ask for help!).

Antihistamines and corticosteroids are only the second-line treatment for anaphylaxis and must not delay the administration of adrenaline, administration of oxygen, endotracheal intubation, and intravenous replacement of fluids.

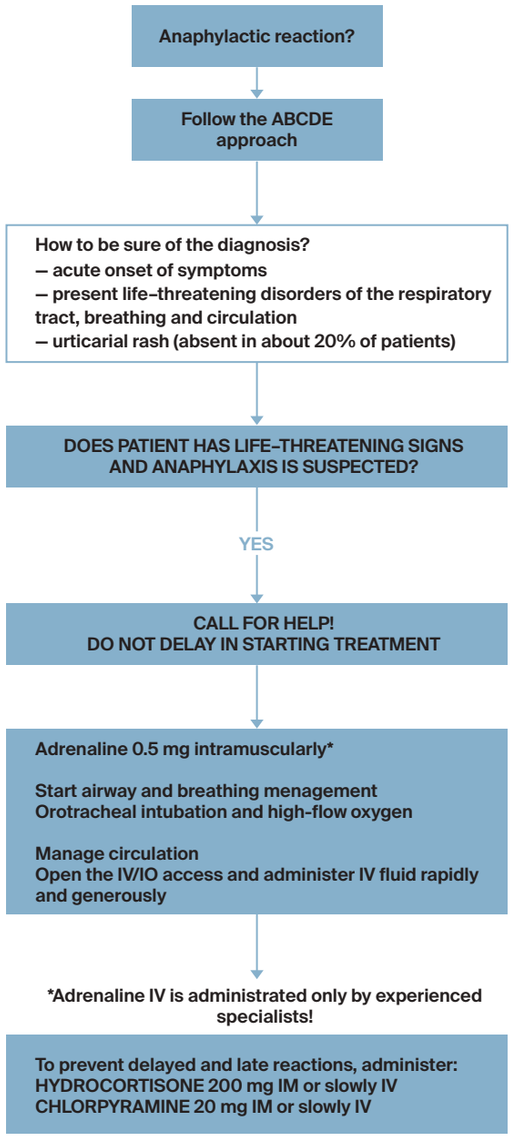

FIGURE 1. *Anaphylaxis Algorithm (adapted from ERC)*

SOURCES

1. Soar J, Böttiger BW, Carli P, Couper K, Deakin CD, Djärv T et al. European Resuscitation Council Guidelines 2021: Adult advanced life support. Resuscitation. 2021; 161: 115–51.
2. Tintinalli JE, O. John Ma, Yealy D, Meckler GD, J. Stephan Stapczynski, Cline DM, Thomas SH. Tintinalli's Emergency Medicine: A Comprehensive Study Guide. 9th ed. New York: McGraw Hill Education; 2020.

3. 5. ACUTE BRONCHOSPASM

In acute respiratory failure, one of the most common physical findings alongside dyspnoea and tachypnoea is wheezing. It occurs with prolonged airflow, through narrowed central and distal airways. It is more pronounced during exhalation, unlike in cases of upper respiratory tract obstruction when stridor is more pronounced during inspiration.

Airway obstruction can occur due to bronchial muscle contraction (bronchospasm), smooth muscle hypertrophy, increased mucus secretion, and/or peribronchial inflammation. Wheezing is usually associated with lower airway diseases such as asthma or other obstructive pulmonary diseases, primarily chronic obstructive pulmonary disease (COPD). Paradoxically, patients with severe airflow obstruction may not have audible wheezing, the so-called trapped lung syndrome. **Approach to patient, diagnosis, and treatment differ depending on the underlying disease and the obstruction/spasm site. The most common diseases that lead to bronchospasm are asthma and COPD.** Wheezing is also very common in heart failure (elderly patients without a history of asthma present with other symptoms and signs of heart failure). Other conditions that can manifest by wheezing and mimic asthma are presented in Table 1.

3.5.1. ASTHMA

Asthma is a chronic inflammatory disorder characterised by increased reversible airway reactivity that can occur in people sensitive

to numerous stimuli. It leads to recurrent episodes of shortness of breath, chest tightness, and coughing, especially at night or early in the morning. A common clinical finding is wheezing in the chest. However, the severity of the wheezing does not correspond to the severity of the obstruction. Attacks often start accompanied by a feeling of tightness in the chest and a cough. As the exacerbation progresses, wheezing becomes audible, the exhalation is prolonged, and the use of auxiliary respiratory muscles can occur. Acute asthma attacks are reversible and regress spontaneously or within minutes to hours of therapy with symptom-free intervals between seizures. Despite this, many patients with asthma develop chronic airway obstruction.

The findings of the physical examination vary widely, from coughing and wheezing at the end of the expiration in patients with mild attacks to respiratory distress in patients with severe asthma attacks. Sometimes wheezing can be heard without a stethoscope. **A change in mental status (e.g., lethargy, exhaustion, agitation, or confusion) is a serious warning about a potential respiratory arrest. The "silent chest" reflects a very serious obstruction in which airflow is insufficient to cause audible wheezing. The finding of hypercapnia is an extremely unfavourable sign in this group of patients.**

TABLE 1. *Differential diagnosis of wheezing (retrieved and adapted from Tintinalli's Emergency Medicine, 9th edition, 2020)*

Upper airway (more likely stridor, may be difficult to distinguish from wheezing)	Angioedema: allergic, ACEi, idiopathic
	Foreign body
	Infection: croup, epiglottitis, tracheitis
Lower airway	Asthma
	Transient airway hyperractivity (caused by infection or irritation)
	Bronchiolitis
	Chronic obstructive pulmonary disease
	Foreigh body
Cardiovascular causes	Cardiogenic pulmonary oedema ("cardiac asthma")
	Pulmonary embolism (rarely)
Non-cardiogenic pulmonary oedema (ARDS)	
Psychogenic	

If patients with acute asthma attacks have $SpO_2 < 92\%$, it is necessary to perform arterial blood gas analysis **due to the risk of hypercapnia**. Specifically, in acute asthma attacks, ventilation is stimulated, usually resulting in a decrease in the partial pressure of carbon dioxide ($PaCO_2$) in arterial blood. **Therefore, normal, or slightly elevated $PaCO_2$ (e.g., > 42 mmHg) indicates extreme airway obstruction and fatigue.**

The aim of the oxygen therapy is to titrate the oxygen flow and concentration to target SpO_2 of 94–98%.

THERAPY:

1. Salbutamol (5 mg in a nebuliser) is the standard of asthma therapy (in acute exacerbations). The dose can be repeated every 15–20 minutes. The oxygen/air flow on the nebuliser must be at least 6 L/min. If the patient does not breathe sufficiently, the nebulised drug will not be delivered adequately.
2. Nebulised anticholinergics (ipratropium bromide 0.5 mg) can produce an additional bronchodilating effect.
3. The use of systemic corticosteroids is also useful.

MAGNESIUM

In severe asthma, intravenous administration of magnesium (2 g in slow infusion) may also be useful. The side effects are nausea, fatigue, headache, and hypotension.

AMINOPHYLLINE

There is no evidence of benefit from aminophylline compared with standard therapy. However, if it is decided to give it (in the presence of an experienced specialist), administer it at a dose of 5 mg/kg for 20–30 min, then with an infusion at a dose of 500–700 µg/kg/h.

FLUID THERAPY

It is used in patients who are dehydrated or have hypovolaemia. **Caution:** beta-2 agonists and corticosteroids may cause hypokalaemia.

PATIENT MONITORING

If the patient does not respond to the initial therapy and develops signs of life-threatening asthma, intensive care is required. Sometimes it is difficult to distinguish between severe life-threatening asthma and anaphylaxis. **If patients have asthma-like symptoms and do not have previously known asthma or COPD, we treat them as if the cause was anaphylaxis.**

TABLE 2. *Severity of asthma exacerbation (retrieved from Adult Advanced Life Support Course Manual published by the European Resuscitation Council, 2015)*

Asthma severity		
Near-fatal asthma	Elevated $PaCO_2$ and/or requiring mechanical ventilation with increased inflation pressures	
Life-threatening asthma	Any of the following in patients with a severe form of the disease:	
	Clinical signs	Measurements
	altered conscious level	PEF <33%
	exhaustion	SpO_2<92%
	arrhythmia	PaO_2<8 kPa
	hypotension	"normal" $PaCO_2$ (4.6-6.0 kPa)
	cyanosis	
	silent chest	
	poor expiratory effort	
Acute severe asthma	Any of the following: – PEF 33–50% – respiratory rate > 25/min – heart rate > 110/min – inability to complete sentences in one breath	
Moderate asthma exacerbation	1. increasing symptoms 2. PEF > 50–70% 3. no signs of acute severe asthma	
Brittle asthma	TYPE 1: wide PEF variability (> 40% diurnal variation for > 50% of the time over a period of > 150 days) despite intense therapy	
	TYPE 2: sudden severe attacks on a background of apparently well-controlled asthma	

3. 5. 2. CHRONIC OBSTRUCTIVE PULMONARY DISEASE

Chronic obstructive pulmonary disease (COPD) is characterised by persistent airflow disturbances that are progressive and associated with an impaired inflammatory response to numerous harmful particles in the inhaled air. COPD has two main forms: chronic bronchitis and emphysema, which, however, often occur together, so the traditional classification is of limited clinical significance.

Chronic bronchitis is defined by the presence of a chronic productive cough for 3 months in two consecutive years, excluding other causes of chronic cough. Emphysema results from the destruction of the bronchioles and alveoli.

Obstruction of the lower respiratory tract causes expiratory wheezing, especially during maximum forced expiration and prolonged expiration. In the early stages, arterial blood analysis reveals mild to moderate hypoxaemia without hypercapnia. As COPD progresses, hypoxaemia becomes more severe, and hypercapnia develops. Arterial oxygenation worsens during acute exacerbations, exercise, and sleep. Clinical signs include swelling of facial blood vessels due to secondary polycythaemia and tremor, somnolence and confusion due to hypercarbia, and right heart failure.

CHRONIC THERAPY

1. Pharmacotherapy: does not affect the progression of the disease, but leads to a decrease in symptoms, reduces the frequency of exacerbations, and improves the quality of life.
   a) Inhaled long-acting β2-agonists and inhaled corticosteroids.
   b) Theophylline: adjunctive therapy, inhibits phosphodiesterase and has an anti-inflammatory effect. Reserved for a limited number of cases.
   c) Mobilisation of secretions: stimulated by abundant oral fluid intake and humidification of the room. Limit the use of antihistamines, antitussives, mucolytics, and decongestants. Expectorants are of no clear benefit.
2. Smoking cessation: the only intervention that can reduce loss of lung function and mortality from respiratory causes.
3. Pulmonary rehabilitation: can improve exercise capacity and quality of life and is recommended for patients with moderate to severe COPD.
4. Vaccinations against pneumococcus and influenza are key to combating acute infections.
5. Oxygen (home oxygen therapy): long-term oxygen therapy reduces COPD mortality, and the goal is to increase the basic arterial oxygen partial pressure (PaO_2) to ≥ 8 kPa or arterial oxygen saturation (SaO_2) to ≥ 90% at rest.

3. 5. 2. 1. ACUTE EXACERBATIONS OF COPD

Acute exacerbations of COPD are characterised by worsening respiratory symptoms beyond normal daily variations and are usually caused by infection (over 75%) or tracheobronchial tree irritation (cold weather, hypoxia, narcotics, sedatives). Acute exacerbations of COPD are primarily due to mismatches of ventilation and perfusion rather than the limitation of expiratory airflow seen in acute asthma attacks. The clinical presentation is dominated by hypoxaemia (tachypnea, tachycardia, systemic hypertension, cyanosis, and changes in mental status) with CO_2 retention and the development of respiratory acidosis. Diseases such as pneumonia, pneumothorax, pulmonary embolism, congestive heart failure (CHF), or acute abdomen can cause exacerbation of COPD.

A patient with acute exacerbation of COPD should have arterial blood gas analysis, venous blood tested for inflammatory markers, blood count with blood differential test, lung ultrasound (pneumothorax), and chest X-ray with further testing depending on the clinical presentation.

3. 5. 2. 2. THERAPY FOR ACUTE EXACERBATIONS

The objectives of treatment are reversible bronchospasm relief, tissue oxygenation correction, and treatment of the underlying cause of exacerbation.

1. **Oxygen +/- mechanical ventilation:** the goal is to correct PaO_2 > 8.0 kPa or SaO_2 88–92% with CAUTION and monitoring of $PaCO_2$ so that hypercapnia does not occur. Oxygen therapy increases the ventilation-perfusion mismatch **(hypoxic pulmonary vasoconstriction)** and leads to reduced haemoglobin affinity for CO_2 **(Haldane effect).** Choosing aids for oxygen delivery and avoiding too high oxygen content are actions crucial for preventing **the development of respiratory acidosis.** If adequate oxygenation is not achieved and/or respiratory acidosis develops, assisted ventilation, primarily non-invasive, is required. In the case of inadequate response and/or further deterioration, invasive mechanical ventilation is required. **Non-invasive ventilation** provides a better outcome in terms of lower intubation rates, lower mortality rates, and shorter duration of hospitalisation (Table 4 and Table 5).
2. **Pharmacotherapy:** short-acting β2-agonists and anticholinergics are the first-line treatment for acute severe COPD exacerbation.
3. **Corticosteroids:** systemic administration for 5–7 days leads to improvement in lung function and hypoxaemia.
4. **Antibiotics:** administered if infection is suspected (change in volume and appearance of sputum). The most common causes are S. pneumoniae, H. influenzae, and M. catarrhalis. Choice of antibiotics: macrolides, tetracyclines, or amoxicillin with/without clavulanic acid.

TABLE 3. *Differential diagnosis of acute exacerbation of COPD (retrieved and adapted from Tintinalli's Emergency Medicine, 9th edition, 2020)*

Diagnosis	Clinical presentation	Caution
Asthma	Earlier onset Different symptoms Family history Reversible airflow	Can coexist with COPD. Many patients diagnosed with asthma actually have COPD or coexisting asthma and COPD.
Heart failure	Presence of orthopnoea and dyspnoea with exertion speaks in favour of CHF Jugular vein distention, Hepatojugular reflux, Bibasilar rales, Chest X-ray: cardiomegaly and/or interstitial oedema BNP < 100 pg/mL not likely to be CHF BNP > 500 pg/mL more likely to be CHF	Can coexist with COPD. Shares some common anamnestic features with COPD. Multiple conditions can falsely elevate or decrease the BNP level.
Pulmonary embolism	Risk factors: older age, recent surgery or trauma, history of thromboembolism, hereditary thrombophilia, malignancy, smoking, estrogen-containing drugs. D-dimer may be useful in ruling out PE in low-risk patients.	20–25% of patients with severe COPD exacerbation without an unclear trigger have PE. Triad of PE (pleuritic chest pain, dyspnea, and tachycardia) plus hypoxaemia is rare.
Acute coronary syndrome	Obtain ECG and troponin in those with chest pain and risk factors for ACS.	Dyspnea may be the leading symptom in patients with ACS.
Pneumothorax	Obtain chest X-ray, US, or CT.	COPD is a risk factor for pneumothorax.
Pneumonia	Obtain chest X-ray.	Frequently coexists with COPD exacerbation

*Abbreviations: ACS – acute coronary syndrome; BNP – B-type natriuretic peptide; CHF – chronic heart failure; CT – computed tomography; ECG – electrocardiogram; COPD – chronic obstructive pulmonary disease; PE – pulmonary thromboembolism; *odds ratio*

TABLE 4. *Indications and relative contraindications for non-invasive ventilation (NIV) (retrieved and adapted from Tintinalli's Emergency Medicine, 9th edition, 2020)*

Selection criteria	Acidosis (pH < 7.36)/hypercapnia ($PaCO_2$ > 50 mmHg)/hypoxia (PaO_2 < 60 mmHg or SaO_2 < 90%) Severe dyspnoea with clinical signs such as muscle fatigue or increased work of breathing
Exclusion criteria (any)	Respiratory arrest Cardiovascular instability (hypotension, arrhythmias, heart attack) Change in mental status; uncooperative patient High aspiration risk Recent facial or gastroesophageal surgery Craniofacial trauma Burns Extreme obesity

Abbreviations: PaO_2 – partial pressure of arterial oxygen; $PaCO_2$ – partial pressure of arterial carbon dioxide; SaO_2 – arterial oxygen saturation.

TABLE 5. *Indications for intubation and mechanical ventilation (retrieved and adapted from Tintinalli's Emergency Medicine, 9th edition, 2020)*

Inability to tolerate NIVa or NIVa failure
Respiratory or cardiac arrest
Decreased consciousness or increased agitation (relative)
Massive aspiration
Persistent inability to remove respiratory secretions
Hypotension
Persistent hypoxaemia despite optimal respiratory treatment
Marked haemodynamic instability

SOURCES

1. Tintinalli JE, O. John Ma, Yealy D, Meckler GD, J. Stephan Stapczynski, Cline DM, Thomas SH. Tintinalli's Emergency Medicine: A Comprehensive Study Guide. 9th ed. New York: McGraw Hill Education; 2020.
2. Soar J, Böttiger BW, Carli P, Couper K, Deakin CD, Djärv T et al. European Resuscitation Council Guidelines 2021: Adult advanced life support. Resuscitation. 2021; 161: 115–51.

3. 6. PULMONARY OEDEMA

It is an emergency condition requiring immediate medical attention. It can be cardiogenic (an increase in hydrostatic pressure) or non-cardiogenic (an increase in microvascular permeability).

If no lung injury is present, fluid flow in the intercellular space of the lung is primarily defined by hydrostatic pressure.

3. 6. 1. INITIAL PATIENT ASSESSMENT: ABCDE + SAMPLE

A (AIRWAY):

- Possible auditory phenomena from the upper airway because of fluid accumulation.

B (BREATHING):

- Impaired gas exchange with subsequent hypoxia increases the work of breathing (WOB). It manifests with accessory respiratory muscle activation, high respiratory rate, and diaphoresis.
- Hypoxia-caused confusion may occur.
- Auscultatory findings: crackles, in the case of pleural effusion – absent lung sounds.

C (CIRCULATION):

- Signs of shock (tachycardia, initial hypertension as a product of sympathetic activation, hypotension develops in later stages)
- Skin is pale and clammy!

- ECG – various findings are possible, often depending on the underlying pathology causing pulmonary oedema.

D (DISABILITY):
- All of the above can lead to impaired cerebral perfusion, which can affect consciousness. Confusion, agitation, and coma may occur.

E (EXPOSURE AND ENVIRONMENT):
- Search for signs of heart failure (e.g., pretibial oedema indicates a probable cardiac cause).

SAMPLE – symptoms: dyspnoea, orthopnoea, weakness, etc. Cardiogenic pulmonary oedema is usually preceded by chronic heart failure (of various causes). Non-cardiogenic oedema is more probable with a history of traffic accidents, infections, fever, etc.

NON-CARDIOGENIC PULMONARY OEDEMA

Encompasses all clinical states promoting increased vascular permeability in the lungs resulting in an inflow of liquid and proteins in the interstitial and alveolar space.

Causes of increased permeability:
- Direct injury (gastric fluid aspiration, pneumonia).
- States promoting acute capillary endothelial injury – systemic infections (sepsis), burns, trauma – they cause ALI (acute lung injury) and ARDS (acute respiratory distress syndrome).

CONSEQUENCES:
- Respiratory failure (primarily type 1, progression to type 1 + type 2 in severe forms).

TREATMENT:
- Treatment of the primary cause.
- Supportive treatment.
- Mechanical ventilation.

CARDIOGENIC PULMONARY OEDEMA

If there is an underlying heart disease and it progresses or is left untreated, pulmonary oedema may occur at some stage (e.g., valvular defects, cardiomyopathies, etc.). Rarely it may happen due to other causes (e.g., volume overload, metabolic diseases, uncontrolled arterial hypertension).

Typically, patients present with progressive dyspnoea. They are seated in the upright position, hoarse crackles are audible, they are tachypnoeic, and peripheral saturation is low with evident use of accessory respiratory muscles. Heart rate is usually rapid, the skin is diaphoretic, cold, and pale. Capillary refill time is prolonged. Blood pressure is usually elevated. The 12-lead ECG may show various pathological findings, such as ischaemia, rhythm abnormalities, and more. In advanced stages, altered and reduced levels of consciousness can be expected.

The purpose of this text is not to describe all possible pathological cardiac states that can lead to heart failure, but to present the most typical presentation and its treatment. The primary cause that leads to this presentation is left ventricle systolic failure. Because of a reduced ejection fraction, the LVEDP (left ventricular end-diastolic pressure) rises and fluid retention occurs. This increases high hydrostatic pressure in blood vessels and fills the alveoli with transudate (large amounts lead to pulmonary effusion).

When the alveoli are filled with fluid, it is difficult for oxygen to reach the capillary bed, and because of reduced systolic function, blood delivery to the organs is compromised.

BASIC PRINCIPLES – TREATMENT OF CARDIOGENIC OEDEMA

A (AIRWAY) AND B (BREATHING):
- **Apply oxygen therapy** if the patient is fully conscious and peripheral saturation is lower than 90%.
- In the case of inadequate response of a fully alert patient to oxygen therapy and peripheral saturation < 90% or PaO_2 < 8 kPa with a respiratory rate of 25/min and over, **non-invasive mechanical ventilation (NIV)** should be considered.
- **Endotracheal intubation and invasive mechanical ventilation** are suitable for patients with life-threatening hypoxia and/or altered level of consciousness.

C (CIRCULATION):
- **High-dose loop diuretics** with monitoring of urine output (caution – Hypokalaemia).
- E.g., intravenous furosemide – the decision about initial dosing depends on the patient's chronic intake and dosage. If a new-onset acute heart failure occurs, the initial dose is a bolus of 40 mg IV, and the subsequent doses are adjusted depending on the diuretic response. In the case of acutely decompensated chronic heart failure (which is more common), patients often take doses as high as 125–250–500 mg daily (furosemide has a high therapeutic range). Repeating the chronic dose does not always result in expected or significant urine output, so repeated doses are necessary. Luckily, we can monitor the response by placing a urinary catheter and measuring urine output. Relief in symptoms is also a good sign of appropriate dosage. Always treat the patient, never the numbers.
- **Nitrates or nitroprusside** (contraindicated if the systolic blood pressure is below 110 mmHg), continuous intravenous infusion is the preferred route of administration (via perfusion pump).
- E.g., nitroglycerine (nitrate) at a dosage of 5–40 µg/min (fast

onset and offset; should be carefully titrated for the desired clinical effect; avoid hypotension). It significantly decreases afterload.

- **Inotropes** in a continuous intravenous infusion (e.g., dobutamine, enoximone, adrenaline, levosimendan) – numerous adverse effects, indicated only when the systolic blood pressure is below 90 mmHg. Do not do it alone – seek expert help!

D (DISABILITY):

- **Morphine** – intravenous administration, anxiolysis, enhances oxygen delivery to the myocardium (indirectly reduces sympathetic activity, tachycardia, and hypertension) and can increase the compliance of a patient with non-invasive ventilation.
- Dosage – initially 2–4 mg IV, followed by further doses of 2 mg IV until the expected clinical response occurs (cooperative, calm). In the case of overdose – administer naloxone!

CARDIOGENIC SHOCK

It is a condition defined as inadequate tissue perfusion following heart failure, and after excluding hypovolaemia as a possible cause. It is the most advanced stage of acutely decompensated heart failure (it happens in less than 2% of all cases).

CHARACTERISTICS:

1. Systolic blood pressure below 90 mmHg (or a drop in MAP greater than 30 mmHg).
2. Heart rate over 60.
3. Oliguria.
4. Signs of organ congestion may or may not be present.

The most common cause is acute decompensation of a previously known heart failure. It occurs rarely in patients with previously normal heart function/unknown heart disease (e.g., massive myocardial infarction).

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3. 7. MECHANICAL VENTILATION SNIPPETS

The first ventilators, so-called iron lungs, were used during the polio pandemic. The patient was placed in a chamber that generated cyclical negative pressure, mimicking the respiratory cycle, and enabling mechanical ventilation. This is physiologically very similar to our natural breathing, where the respiratory muscles create negative pressure and literally suck air into the lungs. Such ventilators had an advantage because of this similarity. However, there were many pitfalls, the first of which was that the patient was trapped in the chamber, making access very difficult and worsening the quality of life. Advances in modern medicine have resulted in positive pressure ventilators used today. The main principle is the generation of positive pressure that "pushes" the air into the lungs, which is opposite to normal breathing.

The aim of mechanical ventilation is to bypass the period in which various reversible clinical states make spontaneous breathing inadequate (e.g., traumatic brain injury requiring deep sedation, severe COVID-19 pneumonia, septic shock, haemorrhagic shock, etc.). Such patients would not survive without mechanical ventilation.

Invasive mechanical ventilation is applied if a patient is endotracheally intubated, and **non-invasive mechanical ventilation** is applied in a fully alert patient and is contraindicated in an altered state of consciousness.

Intubated patient in the early stages of treatment is usually sedated and does not breathe spontaneously. In later phases, when

the favourable clinical effect is present, a clinician reduces the level of mandatory breaths and increases the level of spontaneous (albeit usually still supported) breaths. There is a plethora of ventilation modes to achieve this. Non-invasive ventilation is applied via continuous positive airway pressure (CPAP) masks that fit tightly on the patient's face, providing a seal to prevent air leakage. The patient must be fully awake and cooperative. It is most commonly used in patients with severe respiratory failure after an unsuccessful attempt at oxygen therapy. Sometimes it can be a first-line therapy, e.g., in a COPD exacerbation with acutely elevated CO_2.

Whether we use a transport ventilator or a stationary one, the basic principles and setup are the same. This chapter discusses ONLY the initial setup in emergency settings outside the ICU and does not go into detail.

The basic settings of lung-protective mechanical ventilation are as follows:

1. F_iO_2 — desired oxygen percentage in the inspired air mixture (21–100%, initially can be set up to 100% but only for a short period; it is important to titrate it adequately, e.g., reduce it to target SpO_2 values and avoid hyperoxia, 40–60% is usually sufficient).
2. Tidal volume (Vt) 6–8 mL/kg of ideal body weight (IBW).
3. Set up frequency (f) to ensure adequate CO_2 removal.
4. Positive end-expiratory pressure (PEEP) — setup depends on the disease, 4–15 cmH_2O.
5. Avoid plateau pressure (Pplat) higher than 30 cmH_2O (set the Ppeak limit at 35 cmH_2O).

The above-mentioned recommendations are a result of multiple studies designed to find optimal ventilation settings to reduce complications of mechanical ventilation (which, don't forget, is not a physiological way of breathing). We call that ventilator-induced lung injury, which has different underlying mechanisms.

1. Volutrauma — lung damage following mechanical ventilation with inappropriately high volumes.
2. Atelectrauma — lung damage when an insufficient amount of PEEP is delivered. PEEP-less alveoli are prone to collapse at the end of each expiration and then reopen on inspiration. Cyclical closing/opening results in damaged alveoli, which are further prone to collapse and are unable to reopen, reducing the total alveolar surface needed for effective gas exchange.
3. Barotrauma — lung damage caused by excessive pressures in the lungs.
4. Biotrauma — lung damage caused by the release of proinflammatory mediators, oxygen radicals, etc.

Ventilators are equipped with different ventilation modes, some of which are brand-specific and patent-protected.

The two main modes are:

a) Volume-controlled mode — volume is guaranteed with each breath, ignoring the pressure required to deliver that volume (CAUTION – possible high pressures and barotrauma).
b) Pressure-controlled mode – a breath is delivered at a fixed time and constant pressure (gentler on the lungs; CAUTION – must be closely monitored, high risk of completely inadequate tidal volumes).

For those who want to know more — modern ventilators usually combine those two modes for volume-guaranteed, pressure-regulated breaths. In emergency settings, with a transport ventilator, the mode selection is usually very limited, and we rarely choose pressure modes because of poor conditions for close monitoring.

Non-invasive ventilation is usually performed via the CPAP mode. By setting the PEEP value, this mode delivers a continuous positive pressure in the airway and alveoli, thus preventing collapse. When a patient has inadequately low tidal volume, pressure support (PS) may be added to supplement each breath with additional pressure, which ensures normal tidal volume. Non-invasive ventilation can be used in COPD exacerbation, pulmonary oedema, pneumonia, etc.

Synchronised intermittent mandatory ventilation (SIMV) is the most important invasive ventilation mode on a transport ventilator. It is appropriate for all sedated or comatose patients without spontaneous breathing effort. The advantage of SIMV (compared with others such as CMV) is that it still allows occasional spontaneous

TABLE 1. *Basic settings of mechanical ventilation medical emergencies*

	NON — INVASIVE MECHANICAL VENTILATION	INVASIVE MECHANICAL VENTILATION
Mode	CPAP	SIMV/MMV
Tidal volume (Vt)	Patient-dependent variable, if a spontaneous breath volume is lower than 6–8 mL/kg IBW, set the PS to achieve 6 mL/kg IBW. PS — 2–8 cmH_2O	Vt 6–8 mL/kg IBW PS is also adjustable to provide support in the case of a spontaneous breath. PS — 2–8 cmH_2O
Respiratory rate (RR)	Patient-dependent	Depends on the CO_2 value, initially 12–20/min
Positive end-expiratory pressure (PEEP)	4–8 cmH_2O, if the setting does not suit the patient, they usually complain.	4–15 cmH_2O At least 4, higher if needed (15 cmH_2O max). Suspected intracranial bleeding — never above 4 cmH_2O!
Inspiration/expiration ratio (I:E)	Patient-dependent	I:E 1:2 (in the case of COPD or asthma, 1:2.5 or 1:3).
Pressure limit (Ppeak limit)	Set the Ppeak limit at 35 cmH_2O.	Set the Ppeak limit at 35 cmH_2O.

breaths, thus reducing asynchrony between the patient and the ventilator. Asynchrony can lead to desaturation, elevated CO_2 levels, lung damage, and an increase in intracranial pressure (ICP), which should be avoided at all costs. In emergency settings, it usually happens due to inadequate dosing of sedatives and/or muscle relaxants.

IN SHORT, MECHANICAL VENTILATION IN EMERGENCY SETTINGS IN THE FOLLOWING SITUATIONS:

a) When patients require analgaesia and sedation and have inadequate breathing/respiration due to the effect of the drugs.
b) In respiratory arrest.
c) In severe respiratory failure and increased WOB.

If a patient has low saturation and low PaO_2 (if available), better oxygenation is achieved through optimal PEEP settings. PEEP prevents alveolar collapse and increases the gas exchange area. If CO_2 elimination is deranged and/or in the case of small tidal volumes accompanied by high respiratory rate and accessory respiratory muscle activation, adding the PS (in non-invasive ventilation) or adjusting Vt and I:E ratio (in invasive ventilation) may further help.

SOURCES

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Stick to the ABCDEs!

Acute respiratory failure — happens either as a breathing failure by itself or following other disorders in the body — know the difference.

Oxygen-haemoglobin dissociation curve — not a linear relationship, resulting in a possible significant drop in PaO_2 for a relatively "small" drop in SpO_2.

Oxygen therapy — use only when indicated.
— Low SpO_2/SaO_2
— Pending airway closure (e.g. anaphylaxis)
— Significant blood loss (trauma, hypovolaemic shock)

FiO_2 is the oxygen "dose"!

ABG analysis will rarely give you the cause of the state but will inform you on the severity. Therefore, you should always look at the patient and think clinically!

Use the stepwise approach.

When the metabolic part is the cause of the problem — don't forget about the SID/gap!

In anaphylaxis, give adrenalin 0.5 mg IM as soon as possible, administer oxygen, call for help, secure the airway and open large bore IV line to give fluids FAST!

When treating either acute asthma or COPD exacerbation — administer oxygen, bronchodilators via a nebulizer and give corticosteroids! Be extra careful with oxygen therapy in COPD patients!

Cardiogenic pulmonary oedema requires prompt treatment:
— Oxygen. Consider NIV if the patient is fully alert and in severe distress!
— High-dose loop diuretics! Beware of hypokalaemia.
— Nitrates/nitroprusside iv. continuously if the blood pressure allows.
— SEEK EXPERT HELP before administering inotropes (they sound good in theory but come with a lot of unwanted effects!)
— Morphine — to reduce sympathetic activity (in this case — NOT for pain)

Learn about non-invasive mechanical ventilation — it can help you save lives!

Part 4: Sudden neurological deterioration

125—144

4.1. SUDDEN NEUROLOGICAL DETERIORATION

Sudden neurological deterioration is a common reason for patient admission to emergency medical services. It can be caused by various critical illnesses or injuries. Prompt treatment is essential to limit secondary brain injury and improve patient outcome. Primary injury is the initial brain damage caused by a critical illness or injury. Regardless of the cause, primary injury triggers secondary injury, which can lead to further brain damage. Our management can affect secondary brain damage through interventions that address **hypoxia**, **hypotension**, **and increased intracranial pressure**.

4.1.1. CAUSES OF SUDDEN NEUROLOGICAL DETERIORATION AND POTENTIAL DIFFERENTIAL DIAGNOSIS

NEUROLOGICAL CAUSES

- Ischaemic or haemorrhagic stroke.
- Traumatic brain injury (TBI; the morphological substrate spanning from epidural haematoma, subdural haematoma, brain contusions to diffuse axonal injury, and brain oedema).
- brain tumours, etc.

SYSTEMIC CAUSES

- Infection (septic encephalopathy, central nervous system (CNS) infection in the form of meningitis and/or encephalitis).

- Organ system failure (e.g., shock, renal failure, hepatic failure, hypoxaemia and/or hypercarbia due to respiratory failure).
- Endocrine disorders (hypoglycaemia, hyperglycaemia, thyroid dysfunction, Addison crisis, etc.).
- Electrolyte and acid-base disorders (hypo- or hypernatraemia, hypercalcaemia, hypophophytaemia, etc.).
- Drugs/poisoning (narcotics, benzodiazepines, CO poisoning)
- Hypo- and hyperthermia.

4.1.2. INITIAL PATIENT ASSESSMENT AND MANAGEMENT: ABCDE + SAMPLE

A (AIRWAY):

- Check the airway patency as CNS depression may cause partial or complete loss of airway patency. Noisy breathing/snoring and asynchronous movement of the chest and abdomen are signs of partial obstruction, while complete obstruction is silent with visible accessory respiratory muscles involvement and see-saw phenomenon for a brief period before the patient goes into cardiac arrest.
- In suspected trauma, place the cervical spine stabilisation.
- Aspirate any blood or gastric content from the airway, if present.
- Use simple airway opening manoeuvres (head-tilt chin-lift or jaw-thrust) or oro-/nasopharyngeal airway to secure the airway if it is not patent.
- Tracheal intubation and mechanical ventilation are indicated in patients with GCS ≤ 8, as they may lose protective airway reflexes and develop hypoventilation with resultant hypercapnia, which worsens intracranial hypertension. However, advanced techniques for securing the airway, including the use of anaesthetic and neuromuscular blocking drugs, should only be performed by experienced personnel, as failed attempts of tracheal intubation can worsen secondary brain injury.

B (BREATHING):

- Assess breathing using the “look, listen, and feel” approach to evaluate respiratory rate and effort, breath and added sounds, and chest movement symmetry, as well as breathing patterns associated with different causes of decreased consciousness, such as Cheyne-Stokes and Kussmaul breathing. Look for possible causes leading to hypoxia, such as subcutaneous emphysema, tracheal deviation, and jugular vein distention.
- Monitor oxygen saturation (SpO_2) continuously using a pulse oximeter and end-expiratory CO_2 ($EtCO_2$), if available.
- Administer oxygen if SpO_2 is below 90%.
- **Ensure adequate oxygenation at all times** to prevent secondary brain injury and maintain blood oxygen saturation above 90%.

- If the patient has insufficient minute ventilation and oxygenation, even after opening the airway and administering oxygen therapy, start with manual ventilation using the tight-fitting face mask and self-inflating bag (e.g., Ambu bag) and consider anaesthesia and tracheal intubation if experienced in the procedure. Insufficient minute ventilation can be expected in bradypnoeic patients, for instance, those overdosed with opioids or with imminent brainstem herniation, and patients with respiratory muscle fatigue, such as those with acute respiratory insufficiency.

C (CIRCULATION):

- Assess heart rate, blood pressure, capillary refill time, skin colour, and (hourly) urine output, and monitor these parameters every 3–5 minutes. Look for patterns that might indicate the cause of sudden neurological deterioration, such as Cushing's triad.
- Place continuous ECG monitoring.
- **Hypotension, along with hypoxia, is a major contributor to secondary brain injury** (CPP=MAP–ICP, where CPP is cerebral perfusion pressure, MAP is mean arterial pressure, and ICP is intracranial pressure).
- In the initial management of a patient with neurological deterioration, **aim for systolic blood pressure of above 110 mmHg** (or MAP above 70 mmHg) to achieve adequate brain perfusion.
- If blood pressure is below the target values, consider administering:
  a) **volume resuscitation with 0.9% sodium chloride** (or hypertonic sodium chloride), as this is the crystalloid solution with osmolarity similar to or higher than the human plasma and will not worsen brain oedema.
  b) **vasoconstrictors**, such as noradrenaline, 0.01–0.5 µg/kg/min IV, dose titrated up to target blood pressure with regular blood pressure assessments.
- Once IV access is established, obtain blood samples for testing, including complete blood count (CBC), glucose, sodium, potassium, and coagulation tests; with additional tests such as C-reactive protein (CRP), procalcitonin, urea, creatinine, ammonium, liver enzymes, ABG, etc., added depending on the differential diagnosis.

D (DISABILITY/OTHER ACUTE CONSIDERATIONS):

- Before performing the neurological assessment, correct hypoxia and hypotension as they can cause or contribute to the depth of decreased consciousness.
- Begin by assessing blood glucose level, Glasgow Coma Score (GCS), pupillary size, symmetry, and light response. Check for differences in motor response between the left and right side, and perform a targeted neurological examination to look for signs of increased intracranial pressure, meningeal irritation, and subtle myoclonus such as eyelid fasciculation, as well as evidence of previous seizure: lateral tongue biting, uncontrolled urination, etc.).

- Normal pupils are equal in size (symmetric), reactive to light, and range 2–8 mm in diameter. Any finding outside of these norms, such as asymmetry or decreased responsiveness to light stimuli, is considered pathological in suddenly deteriorated neurological status and can help in making a differential diagnosis.
- **Targeted neurological examination includes the following:**
- GCS: assesses the patient's ability to open their eyes, vocalise, and move their limbs. If the patient is awake, assess their orientation. If the patient does not respond to verbal command, apply a painful stimulus (e.g., by pinching the trapezius muscle, supraorbital ridge, or temporomandibular joint), and assess the best motor response, as well as the difference in motor response between the left and the right side of the body.

TABLE 1. *Glasgow Coma Scale (GCS)*

Behaviour	Response	Score
Eye opening	Spontaneously	4
	To speech	3
	To pain	2
	No response	1
Verbal response	Oriented (in time, space, and towards person)	5
	Confused, disoriented	4
	Inappropriate words	3
	Incomprehensible sounds	2
	No response	1
Motor response	Obeys command	6
	Localises pain	5
	Withdraws to painful stimulus	4
	Abnormal flexion (decortication)	3
	Abnormal extension (decerebration)	2
	No response	1

- Assess breathing pattern, which can help with diagnosing the depth and cause of comatose state (e.g., Cheyne-Stokes breathing, bradypnoea, irregular breathing, Kussmaul breathing).
- Assess plantar and deep tendon reflexes – useful when there is an asymmetry in motor response between the left and the right side of the body.
- Assess brain stem reflexes, e.g., pupillary, corneal, oculovestibular (doll's eye movement); suction from the trachea (useful if brain death is suspected).
- HYPOGLYCAEMIA: hypoglycaemia is a common and easily treatable cause of a comatose state. If present, administering glucose is an absolute priority, ideally in high concentration (e.g., 40% glucose), titrated up to recovery of consciousness and adequate blood glucose levels. If rapid glucose testing is not available in a patient with sudden neurological deterioration, then 50 mL of 40% glucose should be administered, as the potential benefit of treating hypoglycaemia outweighs the potential risk associated with hyperglycaemia.
- SEIZURES: repeated or prolonged seizures add substantially to secondary brain injury and should be stopped immediately. Administer IV benzodiazepines.
- SIGNS OF ELEVATED INTRACRANIAL PRESSURE: For example, decreased consciousness with present anisocoria or dilated unresponsive pupils, Cushing's triad, etc. Administer the following:

1. mannitol 0.5–1 g/kg IV +/- furosemide 20 mg IV (before applying this therapy, place a urinary catheter)
2. consider hyperventilation (shortly)
3. elevation of the head of the bed to 15–30° (increases venous drainage and decreases ICP)
4. if adequately trained for the administration of anaesthetics, give analgaesia and sedation and proceed with tracheal intubation and mechanical ventilation of the patient.

- DRUG OVERDOSE: if available, administer an antidote (e.g., naloxone in opiate overdose, flumazenil in benzodiazepine overdose, etc.).

SAMPLE:

S (Signs/symptoms): Obtain information from any witness present. How did the acute deterioration develop – over several minutes, hours? Did the patient complain of any other symptoms before the acute neurological deterioration, such as headache, vomiting, fever, etc.?

A (Allergies): Does the patient have any known allergies or hypersensitivity to medications or other substances?

M (Medications): Inquire about any medications the patient is taking, with special emphasis on antiplatelet and anticoagulation therapy, psychiatric therapy, insulin, oral antidiabetic medications, and antiepileptic therapy. This information can guide

treatment decisions and help prevent adverse effects.

P (Past illnesses): Ask about recent surgeries, infections, and other comorbidities. Is there a family history of psychiatric disorders or other relevant conditions?

L (Last oral intake): If less than six hours have passed since the last meal, they are at a higher risk of regurgitation and aspiration of gastric contents during procedures such as analgosedation and tracheal intubation. Follow the rapid sequence intubation protocol if these procedures are planned.

E (Events leading up to present illness/injury): Look for any empty medication containers or bottles near the patient. Are there any signs that this is a suicide attempt or intentional overdose? Gathering this information can help guide treatment decisions and prevent future occurrences.

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4. 2. SEIZURES

4. 2. 1. INTRODUCTION

Seizures are caused by abnormal electrical impulses in the brain that disrupt normal brain function, resulting in acute neurological deterioration. This can manifest as decreased consciousness and focal or generalised involuntary movements. Diagnosis is usually based on clinical presentation, laboratory findings, electroencephalogram (EEG), and imaging studies such as CT or MRI of the brain. Treatment involves identifying and addressing the underlying cause, administering anticonvulsant drugs, and in some cases, surgical intervention.

Overall, understanding the causes and types of seizures is crucial for the effective diagnosis and management of patients with seizure disorders. As medical professionals, it is important to have a thorough understanding of the conditions to provide optimal care for our patients.

Seizures can be classified into different types, including:

- Epilepsy (epileptic convulsive disorder) – a chronic brain disorder characterised by recurrent seizures that are not caused by a reversible factor.
- Symptomatic epilepsy – seizures due to a known reason, such as brain tumour, TBI, stroke, etc.
- Non-epileptic seizures – seizures caused by a temporary systemic disorder that interferes with normal brain function, e.g., electrolyte disorders, hyperthermia, etc.

The patient's age at the time of the first seizure can provide clues to the underlying cause. For example, febrile states and hereditary neurological or metabolic disorders are common causes of seizures in patients younger than 2. Idiopathic convulsive disorders are the most common cause of seizures in patients between the ages of 2 and 14. In adulthood, seizures most often occur due to brain tumours, TBI, or stroke.

To properly classify seizure disorders, it is important to recognise whether the seizure is focal or generalised, as well as whether a motor component is present. Generalised convulsive disorders involve electrical impulses that affect both hemispheres of the brain, often leading to impaired consciousness. Within this category, seizures can present with a motor response such as tonic-clonic, exclusively tonic, or exclusively clonic. In contrast, seizure disorders with a focal response are characterised by impulses originating from one hemisphere, usually from subcortical structures. These types of seizures often present with simpler manifestations, such as automatisms, muscle atony, or clonic muscle spasms, with or without loss of consciousness. Focal seizures can also appear without the motor component and may involve dysfunction of the autonomic nervous system (gastrointestinal sensations, feelings of heat or cold) or cognitive and/or emotional impairment. It is important to note that a focal seizure can progress to a generalised seizure at any moment!

Status epilepticus is a life-threatening condition caused by a continuous burst of impulses in the central nervous system. It can manifest in two forms: convulsive and non-convulsive. Convulsive status epilepticus is characterised by tonic-clonic activity lasting longer than **5 minutes or two consecutive seizures without regaining consciousness**. Non-convulsive status epilepticus is typically characterised by changes in mental status and can be more challenging to diagnose.

4. 2. 2. TREATMENT

Effective management of seizures requires attention to both general measures and drug therapy. One of the most critical aspects of management is securing the airway to ensure adequate airflow. For example, in patients with generalised tonic-clonic seizures, it is important to clear the area around the neck to prevent compression from clothing. Attempting to remove the patient's tongue should be avoided, as this can result in injury to the rescuer's fingers, as well as injury to the patient. Patients should be turned to their left side to decrease the risk of aspiration in the case of vomiting. If a generalised motor seizure occurs, immediate administration of medication is indicated.

There are three lines of drug treatment for seizures. The first-line drugs include **benzodiazepines** such as midazolam, lorazepam, and diazepam. In prehospital settings, midazolam can be administered intramuscularly (0.2 mg/kg to a maximum of 10 mg), intranasally (0.2 mg/kg to a maximum of 5 mg), or buccally (0.5 mg/kg to a maximum of 10 mg). Buccal administration of lorazepam (0.1 mg/kg to a maximum of 4 mg) or rectal administration of diazepam (0.5 mg/kg to a maximum of 20 mg) are also options. In the hospital setting, the drugs of choice for convulsive status are also midazolam (0.1 mg/kg to a maximum of 5 mg), lorazepam (0.1 mg/kg to a maximum of 4 mg), or diazepam (0.3 mg/kg to a maximum of 10 mg), administered intravenously. Prehospital medication administration should be considered when choosing the next line and dose of treatment.

If the patient does not respond to the first-line treatment or has received the medication twice, second-line drugs such as phenytoin (20 mg/kg to a maximum of 1000 mg), phenobarbital (20 mg/kg to a maximum of 1000 mg), and levetiracetam (60 mg/kg to a maximum of 3000 mg) can be used. If the patient remains in a convulsive state after the first administration of the second-line drugs, another second-line drug can be given. If symptoms persist, third-line drugs such as propofol (2.5 mg/kg initially, continuous infusion 3 mg/kg/h), thiopental (2–5 mg/kg initially, continuous infusion 1–4 mg/kg/h), or midazolam (continuous infusion 1 µg/kg/h) may be used in the intensive care setting. In such cases, securing the airway with endotracheal intubation and mechanical ventilation is important, along with other intensive care measures.

4. 2. 3. INITIAL PATIENT CARE: ABCDE + SAMPLE

Proper initial care is essential when managing a patient with seizures, The following steps can help ensure the patient's safety:

A (AIRWAY):

- Maintain an open airway by tilting the head and using supraglottic aids such as the oropharyngeal or nasopharyngeal airway.
- Endotracheal intubation may be necessary if hypnotics are used.

B (BREATHING):

- Administer 100% oxygen via face mask with reservoir.
- Continuously measure SpO_2 values.
- Regularly assess the respiratory rate and the need for mechanical ventilation.

C (CIRCULATION):

- Continuously monitor vital signs.
- Check pulse and blood pressure, particularly for hypotension when using hypnotics.
- Place two wide venous lines as soon as possible.
- Perform standard laboratory tests for electrolyte disorders.

D (DISABILITY):

- **Check glucose levels** as hypoglycaemia is a common cause of seizures. Administer 40% glucose in the case of hypoglycaemia.
- Assess the patient's state of consciousness using the AVPU/GCS scale.

E (exposure):

- Remove the patient's clothing but avoid tying the patient to prevent injuries from seizures.
- Remove clothing or jewellery around the neck to prevent suffocation.
- Check for injuries such as tongue biting and sphincter relaxation.

In addition to the ABCDE approach, obtaining a SAMPLE or AMPLE history is important:

S (Signs/symptoms): Examine the symptoms, including any presence of an aura before the seizure (headache, dizziness, sensory sensations).

A (Allergies): Check for drug and food allergies, particularly those related to the use of antiepileptic drugs.

M (Medications): Determine if the patient is already taking antiepileptic drugs or other medications.

P (Past illnesses): Check if this is the first manifestation of epilepsy or if the patient has a previous diagnosis of seizure disorder.

L (Last oral intake): Determine when the patient last ate and whether there is a risk of stomach content aspiration.

E (Events leading up to present illness/injury): Identify any precipitating factors for the onset of seizures, such as loud sounds, bright lights, insomnia, etc.

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4. 3. ACUTE POISONING

4. 3. 1. INTRODUCTION

Poisoning is a clinical condition characterised by a sudden or gradual disease onset due to contact with the poison.

The symptoms that develop are closely related to the type of poison. Its absorption into the bloodstream is essential for the systemic action of the poison. Acute poisoning requires prompt diagnosis and urgent treatment. As these are sometimes substances we do not encounter daily, it is important to have constant access to the data and possible therapeutic interventions that should be done in the event of poisoning (usually a contact number for a poison control centre). Any substance taken in the wrong dose can be toxic.

Acute poisoning can be accidental, suicidal, criminal, environmental, or industrial. They are usually accidental. Suicidal poisoning is caused by the deliberate intake of poison to cause death. Environmental poisonings are usually caused by food, but they can also be caused by plants or animals, such as venomous snakes or spiders.

4. 3. 2. INITIAL PATIENT ASSESSMENT: ABCDE + SAMPLE

A (AIRWAY):

- Patency primarily depends on the effect of the poison on the state of consciousness.
- In the case of poison ingestion, damaged airways are a possibility (e.g., strong acids).
- Gastric content in the mouth (in the case of vomiting).
- Some poisons may cause seizures, and bite marks on the tongue can be visible.
- Airway manoeuvres depend on the patient's state.

B (BREATHING):

- Respiratory rate may be affected (e.g., opiate poisoning — low respiratory rate).
- Abnormal auscultatory findings may help in the suspicion of aspiration of gastric content in a patient with an altered level of consciousness.
- If the airway is not patent and breathing is inadequate, the saturation (SpO_2) will decrease — if so, start oxygen therapy and secure the airway properly.

C (CIRCULATION):

- Possible abnormalities (depending on the poison — fast or slow heart rate, arrhythmias) are diverse.
- Blood pressure can be affected (depending on the dose and poison).
- Skin appearance and CRT may indicate shock.
- Bleeding can also happen because of poisoning (e.g., unintentional anticoagulant poisoning).
- IV access and standard monitoring are obligatory.
- Some poisons have specific antidotes that should be applied in a life-threatening condition.

D (DISABILITY):

- Level of consciousness can be affected due to the direct effect of the poison on the central nervous system (CNS) or indirectly by affecting other vital functions.
- In the case of a coma, advanced airway management and mechanical ventilation are indicated.
- Pupils may help differentiate poisons (e.g., myosis in opiate poisoning, mydriasis in sympathomimetic drug poisoning) and whether the poisoning caused severe CNS damage (for example, diffuse brain oedema — unreactive dilated pupils).
- Glucose levels can be high/normal/low (persons with type 1 diabetes may attempt suicide with insulin).

E (EXPOSURE):

- Very important! Check the patient and their surroundings, which can provide critical clues about the type and dose of poison and potential injuries.
- Suicide attempt — the means of the attempt are usually in the person's immediate environment (i.e., empty pill blisters).
- SAMPLE — be thorough; it varies depending on the type of poisoning.

Specific measures that can be taken include decontamination, stimulation of excretion, and administration of specific antidotes. These are taken immediately, as soon as we identify the poison and assess the patient. Since we are surrounded by an infinite number of poisons, it is impossible to know specific measures for all of them — hence the establishment of poison control centres available 24/7.

Decontamination is directly dependent on the type of contact between the poison and the body. Usually, this means decontamination of the digestive system by performing gastric lavage. Rinsing the eyes that came into contact with a chemical is also a method of decontamination. There is no evidence of the usefulness of gastric lavage and/or activated charcoal.

Stimulation of excretion can be done via the gastrointestinal system or urinary tract. Administering activated charcoal is one way of stimulating excretion. If a poison can be excreted in the urine or filtered by dialysis, renal replacement therapy may be commenced in a critically ill patient. Some poisons suitable for this are barbiturates, lithium, metformin, salicylates, and methanol.

Antidotes are drugs with an effect opposite to the poison. Airway management must be done prior to antidote administration. Also, it is important to note that some antidotes have a shorter time of duration than the poison. Therefore, it is crucial to monitor the patient and repeat the doses if needed. One such antidote is naloxone.

TABLE 1. *Toxic agents and corresponding antidotes and dosages*

POISON	ANTIDOTE	DOSE
Paracetamol	Acetylcysteine	200 mg/mL in 200 mL of 5% glucose adjusted for body weight
Benzodiazepines	Flumazenil	Initial dose 0.2 mg IV, repeated dose 0.5 mg every minute (or until effect) up to 3 mg max.
Opioids	Naloxone	0.4–2 mg IV Repeated dose every 2–3 min up to 10 mg max. CAN BE ADMINISTERED INTRAMUSCULARLY. Additional doses are necessary every 20–60 minutes!
Calcium channel blockers	Calcium chloride 10%	IV bolus 20 mL every 2–5 min
Beta-blockers	Glucagon; insulin, glucose, lipid emulsion, phosphodiesterase inhibitors	Glucagon: 50–150 µg/kg
Local anaesthetics	Intralipid 20%	1.5 mL/kg for 1 min, followed by 15 mL/kg/h
Carbon monoxide	Oxygen	FiO_2 = 1 (100%)

SOURCES

1. Šustić A, Sotošek V. Handbook of Anesthesiology, Reanimatology, and Intensive Care Medicine for students of medicine and dental medicine. 1st ed. Zagreb: Medicinska naklada; 2021.

Keypoints

Stick to the ABCDEs! Do the GCS when appropriate.

Not all causes of sudden neurological deterioration are strictly neurological (e.g, hypoxia and hypotension).

If there are signs of elevated intracranial pressure (ICP):
— Administer mannitol
— Elevate the head by 30°
— If trained, sedate the patient
— Consider short term hyperventilation

In case of seizures, secure the airway and stop the seizure (e.g. by administering a benzodiazepine via the easiest available route) as soon as possible; while promptly identifying and treating the underlying cause (e.g. hypoglycaemia).

When poisoning is suspected, apply basic decontamination, administer specific antidotes if available, consider contacting the toxicology centre for additional tips and guidance, and don't forget to provide supportive care to vital functions until the condition resolves.

Part 5: Sepsis

145–152

5.1. SEPSIS AND SEPTIC SHOCK

5.1.1. INTRODUCTION

Sepsis is a severe and potentially life-threatening condition that occurs when the body's immune system responds inadequately to infection, leading to widespread inflammation and tissue damage. It is important to understand basic pathophysiology to diagnose and treat it effectively.

The mortality rate of sepsis is high, ranging from 20–50%, depending on the underlying pathology and the timing of initial treatment. The excessive release of inflammatory mediators during sepsis leads to generalised vasodilation and increased permeability of the blood vessels, causing relative hypovolaemia and hypoperfusion with damage to tissues and organs throughout the body.

To diagnose sepsis and initiate prompt treatment, two criteria should be met:

1. **the presence of an infection** (suspected or documented: e.g., justified suspicion is sufficient, such as patient history suggesting urinary infection, physical examination involving audible rales over part of the lungs, etc.)
2. **change in Sequential Organ Failure Assessment (SOFA) score of at least 2 points** (Table 2).

However, the SOFA score requires specific laboratory findings for interpretation, which takes time. To overcome this limitation, a **quick SOFA (qSOFA) score** was developed, which consists of three clinical findings available through physical examination, making it

easy to use and repeat as often as necessary. Timely recognition and treatment of sepsis are essential determinants of patient outcome.

SEPTIC SHOCK

Septic shock is a severe and life-threatening condition that can occur in patients with sepsis. It is characterised by abnormal circulatory physiology and cellular metabolism, resulting in increased mortality. To diagnose septic shock, the patient must meet the following criteria, in addition to the diagnosis of sepsis:

1. hypotension with mean arterial pressure below 65 mmHg, despite fluid resuscitation with a 30 mL/kg of balanced crystalloid solution (this is needed to rule out hypovolaemia as the cause of low blood pressure),
2. serum lactate levels higher than 2 mmol/L.

Early recognition and prompt treatment of septic shock are critical for reducing mortality rates. Treatment includes early antibiotic administration, fluid resuscitation, and vasoactive medications to maintain adequate blood pressure and organ perfusion.

TABLE 1. *Criteria for diagnosing sepsis with the most common sources of infection and qSOFA components*

INFECTION (suspected or documented)	INCREASE IN SOFA score ≥ 2
Possible sources: 1. Respiratory tract 2. Abdomen 3. Urinary tract 4. Indwelling catheters 5. Skin and soft tissues	qSOFA (≥ 2) – increased respiratory rate (≥ 22/min) – hypotension, systolic BP < 100 mmHg – altered mental status, GCS < 15

5.1.2. INITIAL MANAGEMENT: ABCDE + SAMPLE

The initial management of sepsis involves the following:

A (AIRWAY):

- Assess the patient's airway to ensure it is open. In cases of obstruction due to the collapse of soft tissues, open the airway by tilting the head, lifting the mandible, and placing an oropharyngeal or nasopharyngeal airway if necessary.
- It is essential to note that patients with sepsis, particularly the elderly, may have reduced consciousness that can lead to airway obstruction, often audible as snoring.

B (BREATHING):

- Check the symmetry of chest expansion on inhalation and respiratory rate, and perform lung auscultation and percussions.
- Place a pulse oximeter to monitor oxygen saturation.
- Provide high-flow oxygen.

TABLE 2. *SOFA score*

Organ system/ parameter	0 points	1 points	2 points	3 points	4 points
Respiratory system/PaO_2/ FiO_2, mmHg	≥ 400	< 400	< 300	< 200	< 100
Coagulation/ platelets, µmol/L	> 150	< 150	< 100	< 50	< 20
Cardiovascular system catecholamine doses/µg/kg/ min	MAP ≥ 70 mmHg	MAP < 70 mmHg	Dopamine < 5 or dobutamine any dose	Dopamine 5.1–15 or adrenaline ≤ 0.1 or noradrenalin e ≤ 0.1	Dopamine > 15 or adrenaline > 0.1 or noradrenaline > 0.1
CNS/GCS score	15	13–14	10–12	6–9	< 6
Kidneys/creatinine (µmol/L), urine output (mL/h)	< 110	110–170	171–299	300–440 < 500	> 440 < 200
Liver/bilirubin (µmol/L)	20	20–32	33–101	102–204	> 204

- Patients with sepsis may experience respiratory failure due to several reasons. For instance, sepsis often originates from pneumonia, leading to tachypnoea as the body compensates for the reduced diffusion of gases in the lungs due to the inflammatory infiltrate. Another possibility is sepsis caused by an infection of a different origin, leading to metabolic acidosis, which the body compensates for by increasing the respiratory rate (by reducing p_aCO_2, the body returns the pH to physiological levels). Patients with sepsis often have a respiratory rate greater than 22/min. If a patient with sepsis has a respiratory rate < 20/min, uses auxiliary respiratory muscles, and has a small inspiratory volume, this indicates respiratory fatigue, which needs to be treated with non-invasive or invasive mechanical ventilation.

C (CIRCULATION):

- Check the patient's heart rate, blood pressure, capillary refill time, and central and peripheral pulses. Maintain ECG monitoring.
- Note that in patients with sepsis, the pulse rate may or may not be increased. Additionally, elderly patients taking chronic medications, such as beta-blockers, may not be able to compensate for changes in circulation by increasing their heart rate.
- Hypotension is common in patients with sepsis due to generalised vasodilation and increased endothelial permeability caused by inflammatory mediators. It is important to assess

hypovolaemia and treat it with a bolus of balanced crystalloid solution. If hypotension persists after the bolus, a vasopressor, such as noradrenaline (0.01–0.5 µg/kg/min), should be initiated.

- Prolonged capillary refill time may be indicative of tissue and organ hypoperfusion; the more pronounced the hypoperfusion, the longer the capillary refill time will be.
- ECG monitoring is necessary as sepsis can cause new arrhythmias or worsen existing ones, and may lead to cardiac arrest due to hypoxia and/or (relative) hypovolaemia.
- If hypotension is marked, the pulse may only be palpated in the central blood vessels, with a systolic blood pressure greater than approximately 80 mmHg when palpated on the radial artery.
- Establish at least one large-bore intravenous line (≤ 18 G) to enable the rapid administration of balanced crystalloid intravenous fluids at a dose of 30 mL/kg of body weight.
- Once an intravenous line is established, perform laboratory tests such as complete blood count, C-reactive protein, procalcitonin, sodium, potassium, creatinine, and liver function test, and obtain blood and microbial samples from the presumed origin of infection (such as urine culture). Additionally, arterial blood gases should be measured to calculate the SOFA score.
- Place a urinary catheter to monitor hourly urine output, check for pyuria, and send urine for analysis.
- The administration of intravenous antibiotics as soon as possible is an ABSOLUTE PRIORITY in patients with suspected sepsis. Start with a broad-spectrum antibiotic, and adapt it to the microorganisms most likely causing the infection at the presumed site of origin.
- The first-hour bundle indicates procedures that should be initiated within one hour of the patient's admission to the hospital.

D (DISABILITY):

- Perform a neurological assessment of the patient, including AVPU or GCS scale and pupil examination. Check the patient's blood glucose to rule out hypo-/hyperglycaemia as the cause of altered consciousness.
- Patients with diabetes may experience difficulty regulating their blood glucose levels, which could be the first sign of an infection.

E (EXPOSURE AND ENVIRONMENT):

- If the source of sepsis has a surgical cause, such as perforated appendicitis, it should be surgically managed as soon as possible to achieve source control.
- Measure the patient's body temperature: a patient with sepsis may or may not be febrile. Often, elderly patients with multiple comorbidities do not develop a fever in sepsis: normal body temperature does not exclude the patient's septic condition.
- Examine the patient thoroughly, checking for visible sources of infection, such as gangrenous foot or abdominal distension.
- Keep the patient warm.

Patients with sepsis require hospitalisation, and those with respiratory insufficiency or septic shock require intensive care. Sepsis is a life-threatening condition with a mortality rate of up to 50% and is a significant cause of death worldwide. Timely recognition and appropriate management are essential to improve patient outcomes. Therefore, it is crucial to recognise the signs and symptoms of sepsis and initiate appropriate treatment promptly.

SOURCES

1. Vincent JL, Moore FA, Bellomo R, Marini JJ. Textbook of Critical Care. 8th ed. Elsevier; 2023.
2. Evans L, Rhodes A, Alhazzani W, Antonelli M, Coopersmith CM, French C, et al. Surviving Sepsis Campaign: International Guidelines for Management of Sepsis and Septic Shock 2021. Crit Care Med. 2021; 49(11): 1063–e143.

Stick to the ABCDEs!

To diagnose sepsis a patient should have a documented or suspected infection with a change in SOFA score of at least 2 points, which is a time consuming and somewhat impractical scoring system in an acute setting. That is why we use qSOFA as a first step.

qSOFA:
- Respiratory rate ≥ 22/min,
- Systolic blood pressure < 100 mmHg,
- Altered mental status with < GCS 15.

If sepsis is suspected:
- Obtain cultures.
- Administer broad spectrum antibiotics i.v.
- Search for signs of shock (not only, but most importantly hypotension and lactate levels).
- Administer fluids (30 mL/kg).
- Apply vasopressors if the patient remains hypotensive after 30 mL/kg of fluid administration.

Do all of this within one hour of suspected sepsis!

Part 6: Blood, fluids and electrolytes

153–180

6.1. INTRAVENOUS FLUIDS

6.1.1. INTRODUCTION

Intravenous (IV) fluid is a drug administered when a person is, due to the nature of a disease, unable to sustain their own basal metabolic needs. This happens in a couple of situations. Human metabolic needs are met with water and food intake. If the level of consciousness is altered, that becomes impossible. Also, when there is a severe gastrointestinal abnormality, even if the patient is theoretically able to swallow food and drink water, absorption is impaired, and deficit develops. There are some states in which IV fluids are administered to compensate for the losses (blood transfusion product in the case of bleeding). To understand indications, contraindications, and for proper choice of an IV fluid, some basic physiological facts need to be stated. Approximately 60% of the human body weight is water. It varies with age, sex, and body composition.

Total body water is located intracellularly and extracellularly. Extracellular fluid can be divided into the following compartments:

- interstitial fluid (lymph, protein-poor fluid between the cells),
- intravascular fluid (plasma volume),
- transcellular fluid (gastrointestinal fluid, bile, urine, vitreous body, joint fluid, pleural, peritoneal, and pericardial fluid),
- fluid in bones and dense connective tissue (anatomically significant portion, but does not form functional extracellular fluid because of the slow flow between this and other compartments).

6.1.2. BASAL METABOLISM

Urine, gastrointestinal fluid production, perspiration, and invisible losses (from skin and lungs) are responsible for basal metabolic needs. If a patient is unable to maintain their basal metabolic requirements through oral intake, fluid and electrolyte disorders will occur. The 4-2-1 formula is used to calculate hourly basal metabolic needs. In some conditions, the retrograde deficit should also be calculated into the deficit (e.g., a patient with a prolonged fasting period) and calculated extra losses should be added. Some losses are impossible to determine precisely (vomiting, diarrhoea). As always, clinical presentation and ABCDE assessment findings guide us to optimise fluid administration (for example, but not limited only to, heart rate, blood pressure, CRT, and urine output).

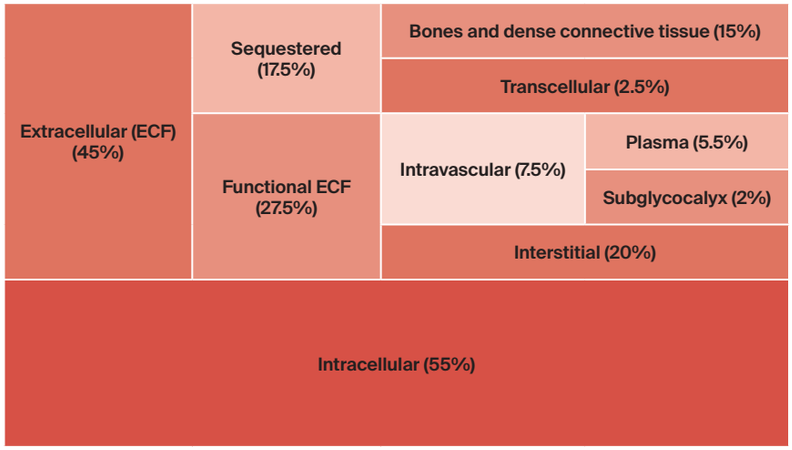

FIGURE 1. *Total body water distribution*

TABLE 1. *Basal metabolic requirements (if a person weights 70 kg, first 10 kg = 40 mL/h, next 10 kg = 20 mL/h, and for the remaining 50 kg, add 50 mL/h – in sum 110 mL/h) (retrieved from Morgan & Mikhail's Clinical Anesthesiology, 6th edition, 2018)*

Ideal body weight	Requirements
first 10 kg	4 mL/kg/h
next 10 kg	add 2 mL/kg/h
each kg above 20 kg	add 1 mL/kg/h

6.1.3. TYPE AND SELECTION OF INTRAVENOUS FLUID

The Table 2 presents IV fluids sorted by type. The basic classification distinguishes crystalloid and colloid IV fluids. The main difference between them is in the plasma half-life. Crystalloids have a significantly shorter half-life (approx. 20–40 min), while the colloid half-life is around 4–6 hours. Indications, contraindications, and side effects vary depending on the different properties of each fluid.

6.1.3.1. CRYSTALLOIDS

UNBALANCED SOLUTIONS

Saline (0.9% NaCl) has a higher chloride ion (strong ion difference — the sum of strong cation concentrations (Na, Mg, K) minus the sum of strong anion concentrations [Cl, lactate]). SID of extracellular fluid is approx. 40 mEq/L and SID of saline is 0, meaning that infusing a 0.9% NaCl solution reduces SID and causes metabolic acidosis. It has a clinically significant effect at doses higher than 1,000 mL when it causes hyperchloraemic metabolic acidosis.

BALANCED SOLUTIONS

They have chloride concentrations similar to human plasma concentrations. They have added anions such as acetate, malate, and gluconate. They behave as buffers and create bicarbonates. Acetate is metabolised throughout the body, is not dependent on liver metabolism, and metabolises faster than lactates. It does not interfere with glucose or insulin levels. Lactate is metabolised in the liver and can become glucose through gluconeogenesis. In severe shock, there is a disturbance in liver metabolic function, and acetate metabolism remains unaffected. Gluconate metabolism is not fully explained. Besides additional buffers, these solutions contain added different cation concentrations (Na, K, Ca, Mg).

TABLE 2. *Classification of IV fluids*

IV fluid	Example	Characteristics
Crystalloids		
Unbalanced solutions	0.9% NaCl (saline)	Isotonic, contains higher Cl– concentration than plasma
	Hypertonic saline (1.8%, 3%, 7.5%)	For correction of hypoosmolar hyponatraemia and elevated ICP
Balanced solutions	Ringer lactate	Hypotonic, contains buffer (lactate)
	Plasma-Lyte 148	Isotonic, contains buffers (acetate and gluconate)
Colloids		
Hydroxyethyl starch (HES)	HES 6%	Plasma volume expansion, numerous adverse effects
Plasma derivatives	Human albumin 20%	
	Freshly frozen plasma	

6.1.3.2. COLLOIDS

Colloids have higher colloid-osmotic pressure and reduce transcapillary filtration rate in the event of low capillary hydrostatic pressure. Therefore, they expand the plasma volume. Their half-life is longer compared with crystalloids. However, they tend to have a nasty side-effect profile, which limits their indications and allowed volumes to be infused. In emergency settings, the most commonly used colloid is HES 6% (hydroxyethyl starch), which is notorious for coagulopathy, anaphylactoid reactions (< 0.6%), and renal toxicity, making it contraindicated in patients with renal failure. Nevertheless, it is indicated in severe haemorrhagic shock when blood product availability is limited.

TABLE 3. *Characteristics of commonly used IV fluids compared to human plasma*

	Plasma	0.9% NaCl	Ringer lactate	Plasma-Lyte 148
Sodium (mmol/L)	136–145	154	130	140
Potassium (mmol/L)	3.5–5.0		4	5
Magnesium (mmol/L)	0.8–1.0			1,5
Calcium (mmol/L)	2.2–2.6		3	
Chloride (mmol/L)	98–106	154	109	98
Acetate (mmol/L)				27
Gluconate (mmol/L)				23
Lactate (mmol/L)			28	
eSID (mEq/L)	42		28	50
Osmolarity	291	308	273	295
Osmolality	287	286	256	271
pH	7.35–7.45	4.5–7.0	5.0–7.0	4–8

SOURCES

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3. Butterword JF, Mackey DC, Wasnick JD. Morgan & Mikhails Clinical Anesthesiology, 6th ed. Philadelphia: McGraw-Hill Education; 2018.

6.2. MANAGEMENT OF ELECTROLYTE DISORDERS

Electrolytes are electrically charged substances commonly found in blood and other bodily fluids. They are essential for various physiological processes, such as the maintenance of proper membrane potential and the generation and propagation of the action potential in nerve and muscle cells. Sodium, potassium, and chloride are the most significant among them, along with magnesium, calcium, and phosphates. Electrolyte disorders are most commonly classified according to the change in their (usual) plasma concentration and can have increased (hyper-) and decreased (hypo-) values. They can lead to potentially life-threatening conditions, so it is important to recognise and treat them as soon as possible.

6.2.1. HYPOKALAEMIA AND HYPERKALAEMIA

Potassium is an electrolyte vital for maintaining normal body functions. Changes in its concentration can cause the sudden onset of potentially life-threatening conditions.

Potassium is primarily an intracellular positively charged ion (cation). The sodium-potassium pump (Na^+/K^+-ATPase, Na^+/K^+ pump) found in the membrane of all cells is responsible for maintaining potassium homeostasis in the body. It maintains the normal resting membrane potential through the activity of the enzyme Na/K-ATPase, which transports three sodium ions out of the cell in exchange for two potassium ions, thus removing one positive charge carrier from inside the cell. The normal intracellular concentration of potassium when

fully balanced is 130–150 mmol/L, and its plasma concentration varies between 3.5 and 5.0 mmol/L (values can vary depending on laboratory parameters).

Kidneys are responsible for the removal of roughly 90% of potassium, the remainder being secreted through the digestive tract and other bodily fluids like sweat and saliva. In kidneys, the reabsorption of potassium happens in the proximal convoluted tubule and the ascending limb of the loop of Henle, while its secretion happens in the distal convoluted tubule.

Changes in serum pH, insulin, various medications (e.g., diuretics), as well as various hormones, can cause changes in extracellular potassium concentration, either hyperkalaemia or hypokalaemia. What makes potassium especially dangerous is its effect on myocardial cells, where it can cause the development of cardiac arrhythmias that can progress to malignant arrhythmias and cause cardiac arrest. Therefore, it is vital to recognise potassium disorders and their clinical presentation, even before laboratory tests are completed.

HYPERKALAEMIA

Hyperkalaemia is defined as the elevation of potassium concentration above its upper limit, usually above 5.0 to 5.5 mmol/L. It can be classified as mild, moderate, or severe, depending on the plasma potassium concentration.

Mild hyperkalaemia is defined as a potassium concentration of 5.5 to 5.9 mmol/L. It doesn't usually cause any symptoms and is typically detected accidentally while performing routine laboratory tests. This condition can also be chronic, usually in patients with a certain degree of decreased kidney function, also often without a clinical manifestation. The need for treatment depends on the presence of symptoms and is usually done by stimulating the elimination of potassium through either oral or rectal suspensions, as well as modifying the diuretic therapy, if present.

Moderate hyperkalaemia is defined as a potassium concentration of 6.0 to 6.4 mmol/L. Symptoms can be present, and more worrying, changes in heart rhythm can also occur.

In severe hyperkalaemia (potassium concentration of 6.5 mmol/L or higher), these changes are almost always present, making this a potentially life-threatening condition requiring emergency treatment.

These patients should be examined using the ABCDE approach and their history should be taken according to the SAMPLE acronym. It is of utmost importance to monitor these patients' vital signs using a **standard 3-lead ECG monitor, regularly measure blood pressure, and continuously monitor SpO_2.** Serial ECG readings should be recorded to monitor changes in the ECG pattern or to gauge the effectiveness of therapy.

Symptoms are non-specific and depend on the overall level of potassium. They vary from diarrhoea, nausea, vomiting, thirst, muscle weakness, muscle pain or numbness, and heart palpitations with or without chest pain.

The expected ECG changes include first-degree AV block, the loss of P waves, broadened QRS complex, ST denivelation, the fusion of S and T waves, formation of the so-called sine-wave pattern, and tall T waves (tenting T waves). Patients can present with bradycardia and ventricular tachycardia, and if left untreated, it can progress to cardiac arrest.

Treatment of moderate or severe hyperkalaemia depends on the clinical presentation (see Figure 1). The three basic steps of treatment are to protect the heart muscle, stimulate the potassium flux into the cells, and remove excess potassium from the body.

If ECG changes are present, start the treatment with the application of intravenous calcium in the form of calcium gluconate or calcium chloride in a slow IV infusion. This protects the myocardium from the deleterious effects of high potassium concentrations. However, this does not lower the actual potassium concertation.

Therefore, treatment is continued by trying to move the excess potassium inside the cells by administering intravenous glucose solution with short-acting insulin and nebulised salbutamol. **In cases of severe hyperkalaemia or in cases of moderate therapy-resistant hyperkalaemia with pronounced ECG changes and/or cardiac arrhythmias, it is important to seek immediate expert help from a nephrology specialist and consider emergency dialysis.** Furthermore, it is important to recognise the cause of hyperkalaemia.

HYPERKALAEMIA CAN BE CAUSED BY:

- acidaemia, i.e., lowering of serum pH – this leads to the removal of potassium from the cells in exchange for the influx of hydrogen ions
- diabetes — the insulin-regulated transport of glucose inside the cells is impaired (co-transport with potassium)
- diuretics — potassium-sparing diuretics: aldosterone-receptor antagonists: spironolactone — Aldactone, eplerenone – Inspra®, Diuron®), thiazide diuretics (indapamide, hydrochlorthiazide)
- chronic kidney failure or acute kidney injury
- various other medications — ACE inhibitors, beta-blockers, non-steroid anti-inflammatory drugs (NSAIDs)
- tissue breakdown — rhabdomyolysis, tumour necrosis, haemolysis
- potassium-rich diet
- **pseudohyperkalaemia** — caused by haemolysis due to inadequate blood sampling or storage of blood samples.
- The management of hyperkalaemia is illustrated in Figure 1.

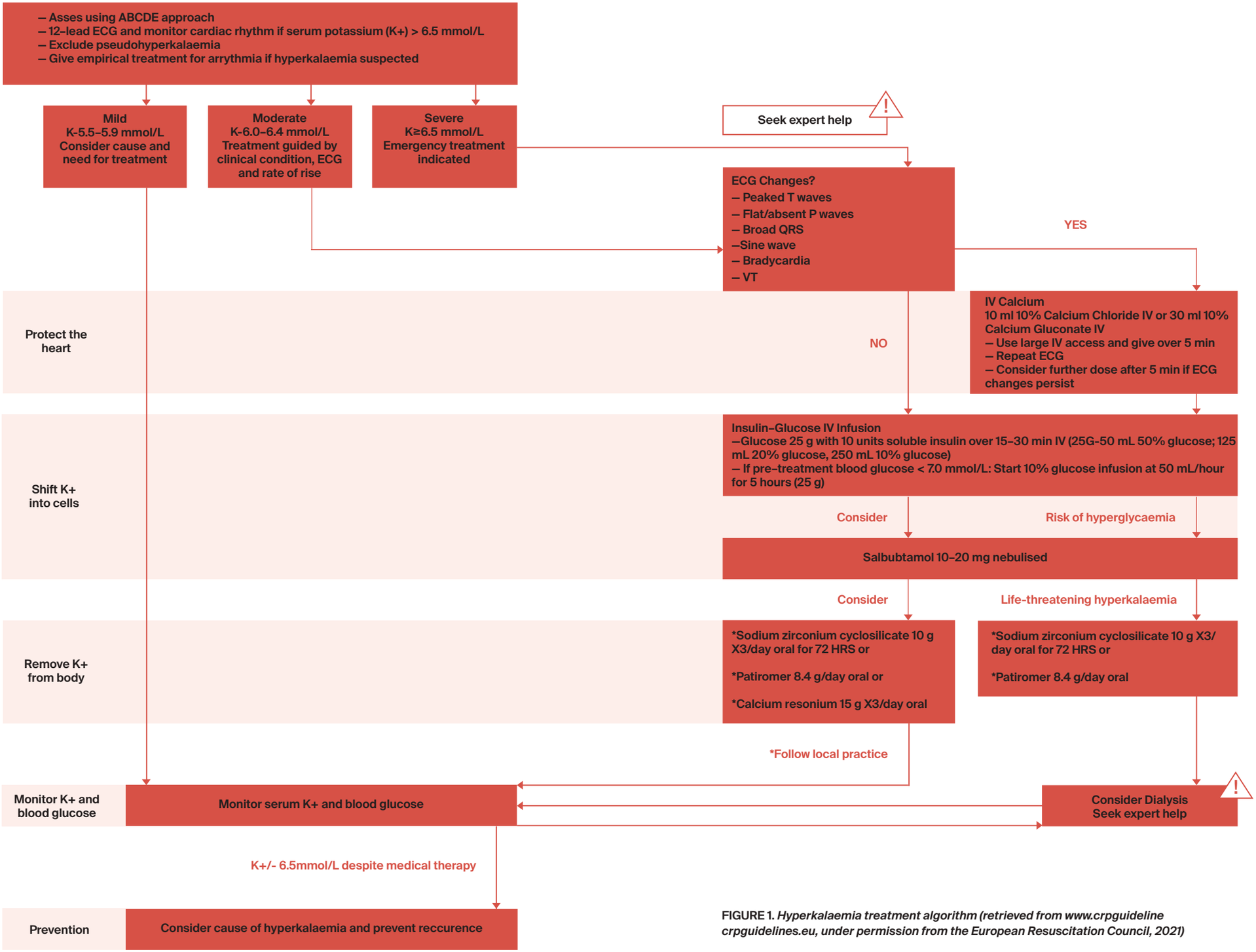

FIGURE 1. *Hyperkalaemia treatment algorithm (retrieved from www.crpguideline crpguidelines.eu, under permission from the European Resuscitation Council, 2021)*

HYPOKALAEMIA

Hypokalaemia is one of the most frequent electrolyte disorders recorded in everyday practice. It is more common than hyperkalaemia and usually presents with milder symptoms. It is defined as a potassium concentration below 3.5 mmol/L and can be classified as mild (3.0 to 3.4 mmol/L), moderate (2.5 to 2.9 mmol/L), or severe (< 2.4 mmol/L). It is very frequent in inpatients, with cardiac, nephrological, and paediatric patients being the most susceptible.

THE MOST COMMON CAUSES OF HYPOKALAEMIA ARE:

- Reduced potassium dietary intake — viewed in isolation, it rarely causes hypokalaemia due to the kidney's ability to compensate by reducing the secretion of potassium. However, it can still cause symptomatic hypokalaemia in severe malnutrition combined with certain medications.
- Increased intracellular influx of potassium — can be caused by the elevation of serum pH (alkalaemia), hyperinsulinaemia, aldosterone, or salbutamol.
- Increased potassium loss via kidneys or the digestive tract — losses through kidneys are usually the result of primary hyperaldosteronism, Cushing syndrome, treatment with the loop or osmotic diuretics, as well as in hypomagnesaemia. Hypokalaemia can also be the result of laxative use or in the case of severe diarrhoea.

Hypokalaemia can usually be discerned from an adequately taken patient history even before laboratory results are in. It can manifest in the musculoskeletal system and, more dangerously, in the cardiovascular system. Symptoms can vary due to the patient's response to the lowering of potassium levels. Mild forms of hypokalaemia are characterised by the development of muscle cramps. Symptoms progress further with the development of paralytic ileus or nausea and vomiting, possible rhabdomyolysis, and cardiac arrhythmias. Patients can also present with various ECG changes like the lowering of T waves, ST depression, and prominent U waves (especially in precordial V4–C6 leads). The most common cardiac manifestation in practice is the sudden onset of atrial fibrillation.

Treatment depends on the severity of the presentation. It is most often treated by administering oral potassium supplements like potassium citrate in a dilute aqueous solution (Kalinor®), which is usually effective within 1–2 hours. In cases of inability to administer potassium via the oral route or in severe cases, potassium can be supplemented via a peripheral vein. Application of intravenous potassium is limited to 20 mmol/h, with 10 mmol/h being the recommended dose. In very severe cases, potassium can be supplemented with 10 mmol/10 minutes and an additional 10 mmol/5–10 minutes, with mandatory ECG monitoring.

HYPONATRAEMIA AND HYPERNATRAEMIA

Sodium is the main extracellular positively charged ion, i.e., cation. Its plasma concentration varies from 135 to 145 mmol/L and usually stays within this range despite daily variations in salt and fluid intake.

Along with chloride and bicarbonate anions, it maintains roughly 90% of extracellular osmolarity.

Sodium concentration is primarily dependent on the overall volume of extracellular fluid, i.e., fluid depletion or overload that results in hyper- or hyponatraemia. Mild disorders are usually asymptomatic and self-limiting, while more severe disorders can lead to death.

TABLE 1. *Sodium disorders: causes, clinical presentation, and treatment options*

Disorder	Causes, according to the overall extracellular fluid volume	Clinical presentation	Treatment
Hypernatraemia	LOW: diuretics, vomiting, diarrhoea, thirst disorders	The leading symptom is thirst. More severe forms can lead to impaired functioning of the central nervous system – disorientation, irritability, muscle cramps, altered mental status, and changes in the level of consciousness leading up to and including coma.	Treatment usually consists of fluid resuscitation, except in conditions with high volumes of extracellular fluids where treatment starts with increasing urine output and then careful fluid resuscitation.
	NORMAL: diabetes insipidus		
	HIGH: loss of central regulatory mechanisms (e.g., after traumatic brain injury), overuse of hypertonic IV solutions (rarely)		
Hyponatraemia	LOW: increased fluid loss via the digestive tract (pancreatitis, peritonitis), via the urinary tract with diuretic overuse, in various nephropathies, or through burns	Leading symptoms are caused by the dysfunction of the central nervous system — disorientation, inertia, with more severe cases presenting with stupor, convulsions, coma and death.	Careful sodium replenishment with hypertonic sodium chloride (NaCl) solutions and with restrictions in fluid intake depending on the overall extracellular fluid volume with the optional addition of a diuretic agent.
	NORMAL: increased release of antidiuretic hormone (ADH) (SIADH — syndrome of inappropriate antidiuretic hormone secretion)		
	HIGH: liver cirrhosis, cardiac decompensation, kidney failure (acute or chronic), nephrotic syndrome		Caution is needed when replenishing sodium because of possible central pontine demyelination.

HYPOCALCAEMIA AND HYPERCALCAEMIA

Calcium is an electrolyte largely found in bones (roughly 99% of total body calcium), with the remaining 1% found in plasma where it circulates in three forms — protein-bound (albumin fraction), chelated,

and in an ionised, active form. Total calcium and ionised calcium are usually measured in laboratories and their concentrations vary from 2.2 to 2.6 mmol/L (total calcium) and 1.15 to 1.25 mmol/L (ionised fraction of calcium). Calcium is essential in coagulation processes, neuromuscular conduction, and muscle contractions. Increased calcium concentration is called hypercalcaemia, whereas reduced concentration is called hypocalcaemia.

TABLE 2. *Calcium disorders: causes, clinical presentation, and treatment options*

Disorder	Causes	Clinical presentation	Treatment
Hypercalcaemia	hyperparathyroidism, malignant diseases, granulomatous diseases (sarcoidosis, TBC)	nausea, vomiting, constipation, ileus, hypotension, QT interval shortening, arrhythmias, polyuria, altered mental status – most common with ionised calcium levels above 3.5 mmol/L	IV fluid replenishment, diuretics, calcitonin, and bisphosphonate to block the osteoclast activation, corticosteroids to reduce intestinal calcium absorption
Hypocalcaemia	hypoparathyroidism, vitamin D deficiency, kidney disease, magnesium depletion, sepsis, blood transfusion	tetany, hyperreflexia, paraesthesia, epileptic seizures, heart rhythm disorders	Calcium replenishment using either 10% calcium chloride or 10% calcium gluconate, magnesium or vitamin D replenishment as needed

HYPOMAGNESAEMIA AND HYPERMAGNESAEMIA

Magnesium is a cation important in maintaining various physiological processes. It takes part in the functioning of the Na+/K+ pump and regulates the flow of calcium into muscle cells, a vital process for the maintenance of normal cardiac contractility and vascular tone.

TABLE 3. *Magnesium disorders: causes, clinical presentation, and treatment options*

Disorder	Causes	Clinical presentation	Treatment
Hypermagnesaemia	kidney failure	typically, cardiac manifestations present because of the antagonising effect on calcium: first-degree AV block, third-degree AV block, and cardiac arrest (with levels above 13 mmol/L).	emergency dialysis, calcium supplementation (temporary cardiac stabilisation), diuretics
Hypomagnesaemia	medications (furosemide, aminoglycosides, digitalis), alcoholism and malnutrition, increased loss via the digestive tract, diabetes with glycosuria	dysfunction of the central nervous system with attention deficits, tetany, fasciculations, generalised tonic-clonic seizures, cardiac manifestations (arrhythmias, ischaemia, digitalis poisoning)	oral or parenteral magnesium supplementation

As with calcium, its active form is found ionised in plasma. Laboratories usually measure total magnesium with concentrations varying between 0.7 and 1.0 mmol/L. Disorders are classified as hypermagnesaemia (concentrations above 1.0 mmol/L) and hypomagnesaemia (concentrations below 0.7 mmol/L).

6. 2. 3. INITIAL PATIENT ASSESSMENT

A (AIRWAY):

- Loss of consciousness due to CNS-related symptoms or the progression of cardiac presentations can result in **sound phenomena** like snoring due to upper airway soft tissue obstruction.
- In the case of seizures, **fluid contents can be found in the oral cavity** (foam, vomit, etc.) with resulting sound phenomena like gargling.

B (BREATHING):

- **Respiratory rate** can be increased due to irritability or psychomotor agitation in milder electrolyte disorders or as compensation for metabolic acidosis. In more severe cases, the respiratory rate can be reduced resulting in bradypnea or apnoea as symptoms progress or cardiac arrest occurs.
- It is important to consider possible injuries that can lead to pathological **chest wall motions** (e.g., pneumothorax).
- Auscultatory phenomena, e.g., rhonchi or rales due to secretions in the respiratory tract or aspiration of gastric contents.
- **With SpO_2 < 94% (< 88% with COPD) – apply oxygen therapy.**

C (CIRCULATION):

- **Due to possible life-threatening cardiac manifestations, it is important to monitor the patient: 3-lead ECG cardiac rhythm monitoring, SpO_2, blood pressure monitoring, and regular standard 12-lead ECG recording. Also, wide-diameter peripheral venous access is important.**
- **Pulse** can be altered in line with various arrhythmias connected with electrolyte disorders (bradycardic, tachycardic, regular, or irregular pulse, loss of pulse in cardiac arrest!).
- **Hypotension** can be linked to various arrhythmias.
- In the case of arrhythmias, hypotension, or altered level of consciousness, **a state of shock can occur with pale, cold, dry skin and prolonged capillary refill time.**
- Urinary catheter should be placed to monitor **urine output** (consider the prior history of diuretic therapy when suspecting hypokalaemia) or response to diuretic therapy.

D (DISABILITY):

- Assess the level of consciousness according to the **AVPU scale**

or GCS.

- Repeat assessment as disorders of consciousness can continue to progress.
- Measure **blood sugar level** to exclude possible hyperglycaemia or hypoglycaemia.
- **Pupils** — monitor possible changes with loss or change in the level of consciousness.

E (EXPOSURE AND ENVIRONMENT):

- Possible signs of liver cirrhosis, cardiac decompensation, pancreatitis, malnutrition, AV fistulas (kidney replacement therapy).

6. 2. 4. PATIENT HISTORY ACCORDING TO THE SAMPLE APPROACH

S (Symptoms): diarrhoea, nausea, vomiting, thirst, muscle cramping or spasms, palpitations with or without chest pain, in crease in urine output with or without diuretics, insufficient urine output, CNS-related symptoms — excitability or depression of CNS.

A (Allergies): ask for known allergies to food or medications.

M (Medications): DIURETICS (potassium-sparing diuretics/aldosterone receptors antagonists) or mannitol with inpatients, ACE inhibitors, beta-blockers, non-steroid anti-inflammatory drugs (NSAIDs), beta-2 agonists, insulin, arrhythmic drugs (e.g., digitalis).

P (Past or Pertinent medical history): chronic heart failure, chronic kidney disease, dialysis treatment, diabetes, endocrine disorders (Cushing syndrome, SIADH, primary hyperaldosteronism, diabetes insipidus, hyper- or hypoparathyroidism, hypovitaminosis D), alcoholism, malnutrition.

L (Last oral intake): potential food intoxication with resultant vomiting or diarrhoea.

E (Events leading up to present illness/injury): previous dialysis history, food intake history, drug intake (deviations from regular therapy schedule).

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6. 3. HAEMORRHAGE AND HAEMORRHAGIC SHOCK

Haemorrhage **results from blood loss** from the intravascular space, which leads to a decrease in oxygen delivery to the cells. When oxygen levels decrease to levels insufficient to maintain aerobic metabolism, a state of shock occurs, irrespective of the initial cause. If the state of generalised ischaemia persists, cell death with organ damage and the development of multiorgan failure occurs. Ultimately, this leads to the death of the organism.

Oxygen delivery to cells (DO_2) depends on two factors:

1. **cardiac output (CO),** which is the product of stroke volume and heart rate,
2. **oxygen content in the arterial blood**, which is dependent on the haemoglobin concentration (Hb) in the blood, saturation of the arterial blood with oxygen (SaO_2), and oxygen partial pressure in the arterial blood (PaO_2).

The aforementioned can be summarised in the following formula:

$$\text{oxygen content in arterial blood} = (1.38 \times Hb \times SaO_2) + (0.0031 \times PaO_2)$$

We do not expect you to memorise the formula: the important thing to note is that the delivery of the primary substrate (oxygen) for the work

of each cell in the body, along with CO, mostly **depends on the concentration of haemoglobin – which is lost due to bleeding!**

In summary, the **absolute priority in this type of shock is to stop the bleeding as quickly as possible and to compensate for blood loss.** The former prevents further loss of haemoglobin – the main carrier of oxygen, and the latter allows the heart to achieve sufficient cardiac output.

NORMAL BLOOD VOLUME:

- ADULTS: 70 mL/kg ideal body weight
- CHILDREN: 80 mL/kg ideal body weight

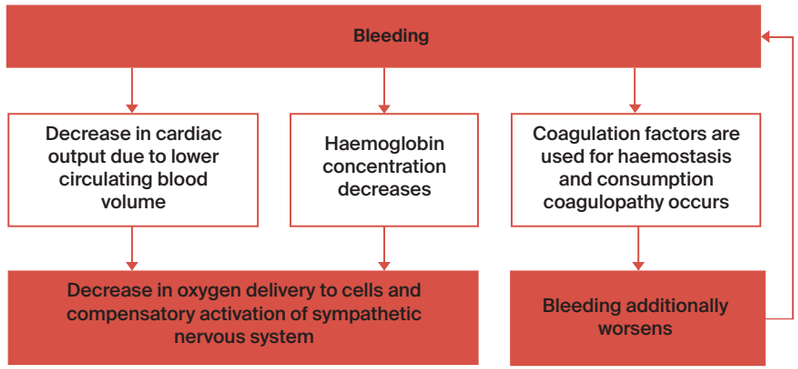

FIGURE 1. *Overview of the pathophysiological course and the vicious circle that occurs in bleeding*

6.3.1. CLASSIFICATION OF BLEEDING

1. STAGE: < 15% blood volume loss: usually without significant changes in vital signs, IV fluid resuscitation is generally not required.
2. STAGE: 15–30% blood volume loss: the patient becomes tachycardic, and the difference between systolic and diastolic pressures decreases. Compensatory mechanisms are activated, peripheral vasoconstriction occurs, and the skin and mucosa are pale and cold to the touch. The patient may become agitated. Resuscitation of circulating volume with **crystalloids and colloids** is required. In patients with cardiovascular diseases, consider blood products.
3. STAGE: 30–40% blood volume loss: blood pressure drops (systolic pressure < 90 mmHg), heart rate increases (> 110/min), visible signs of peripheral hypoperfusion/shock – capillary refill time is prolonged, urine output decreases, and consciousness is impaired as cerebral blood flow becomes insufficient. **Transfusion of blood products is required.**
   *Haematocrit and haemoglobin findings are not objective and possibly false due to haemoconcentration. **The decision on the transfusion of blood products is based on clinical assessment.**
4. STAGE: > 40% blood volume loss; compensatory mechanisms are reached.

BLOOD VOLUME LOSS IS DIFFICULT TO ASSESS IN THE FOLLOWING PATIENTS:

1. ELDERLY PATIENTS have a reduced ability to compensate in the case of acute bleeding due to their reduced cardiorespiratory reserve. Their blood pressure will decrease more rapidly even in the case of minor bleeding compared with the general population. Similar can be seen in patients who have a so-called low fixed cardiac output, e.g., patients with aortic stenosis that prevents an increase in cardiac output. Their heart is already using compensatory mechanisms for their steady state and further compensation is not possible.
2. PATIENTS TAKING CHRONIC THERAPY AND PATIENTS WITH A PACEMAKER. In patients taking beta-blockers, bleeding will not lead to an increase in heart rate and will also cause a sooner drop in blood pressure. Assessing bleeding based on heart rate in these patients will lead to a significant underestimation of blood loss (no compensatory tachycardia), and the same applies to patients whose heart rate is defined by their pacemakers.
3. PREGNANT WOMEN under normal circumstances have a higher heart rate, increased circulating blood volume, and increased respiratory rate, which may complicate the assessment of blood loss.
4. ATHLETES under normal circumstances and at rest may have a heart rate of < 50/min. In cases of bleeding, significant relative tachycardia in this group of patients may already be present well under 100/min.

6.3.2. CAUSES OF SEVERE BLEEDING

- traumatic injury (e.g., car accident, fall from a great height, stab or gunshot wound, etc.)
- bleeding from the gastrointestinal tract (e.g., bleeding gastric or duodenal ulcer, intestinal obstruction with necrosis, bleeding from haemorrhoids, etc.)
- massive bleeding in gynaecology and obstetrics (e.g., placental abruption, placenta previa, extrauterine pregnancy, etc.)
- bleeding from large blood vessels (e.g., aortic dissection, ruptured thoracic/abdominal aortic aneurysm)
- tumour infiltrating large blood vessels (e.g., thoracic aorta infiltration by a lung or oesophageal cancer).

6.3.3. INITIAL PATIENT ASSESSMENT: ABCDE + SAMPLE

GOAL:

- Recognise and **stop the bleeding** as soon as possible (damage control resuscitation)!
- Check breathing and pulse.
- If severe bleeding is present, try to limit/stop bleeding by applying pressure/bandages.

CLINICAL PRESENTATION OF HAEMORRHAGIC SHOCK:
— increase in heart rate (due to catecholamine effect on the heart)
— tachypnoea
— flat jugular veins
— circulation centralisation — vasoconstriction in the "less important tissues":
  — cold, mottled skin
  — decreased urine output
  — consciousness impairment

FIGURE 2. *Clinical presentation of haemorrhagic shock*

A (AIRWAY) / B (BREATHING):

— Check if the patient is able to keep the airway open. In the case of obstruction, **open the airway** while **stabilising the cervical spine** if you suspect an injury.
— Check for symmetrical **chest movement, respiratory rate, auscultation, and lung percussion**. Use a **pulse oximeter**.
— Set high-flow oxygen (15 L/min): $SpO_2 \geq 94\%$.
— If the patient has acute respiratory insufficiency and/or impaired state of consciousness (GCS ≤ 8), secure the airway with an endotracheal tube.

C (CIRCULATION):

— Connect the patient to **ECG monitoring**, measure the **heart rate** simultaneously on the central and peripheral arteries, **blood pressure** (repeat the measurement every 3–5 minutes), and **capillary refill time**. Place a urinary catheter, and **monitor hourly urine output**.
— Obtain at least **two wide-bore intravenous lines (≤ 18 G)** → alternative vascular access if the above is not possible (intraosseous access, central venous access).
— Try to assess the source of bleeding and blood loss:
  — Do not forget about possible "hidden" bleeding in the thorax, abdomen, pelvis, retroperitoneal space, and thigh muscles.
  — If bleeding is > 15% of total blood volume, administer **crystalloids** (saline, Ringer's lactate, Plasma-Lyte, etc.) and **colloids** (6% HES, human albumin).
  — Start targeted fluid resuscitation only in the case of hypotension until blood products are ready. Target pressures of controlled/targeted hypotension are presented in more detail below (damage control resuscitation).
  — Whatever intravenous fluid you choose, make sure it is warmed to body temperature (hypothermia further worsens coagulopathy).
— Aggressive resuscitation with large volumes of intravenous fluids will lead to an increase in blood pressure, which carries the risk of displacement/damage of an already-formed thrombus at the bleeding site and can cause dilutional coagulopathy. This leads to further bleeding, which creates the need for even greater volume resuscitation – thus creating a **vicious cycle**. The priority is to provide the patient with **emergency surgical/radiological/endovascular/other haemostasis.** During transport to the hospital, prior to haemostasis, systolic blood pressure should be maintained at a minimum level sufficient for the short-term maintenance of organ viability, however low enough not to stimulate bleeding further.
  According to the latest recommendations, the target pressure values for traumatic bleeding in patients with stab wounds without suspected brain or spinal cord injury and no other risk factors are MAP > 65 mmHg or **systolic pressure 80–90 mmHg**. In patients with blunt injury and/or brain injury, spinal cord injury, or other risk factors, the pressure values are MAP > 85–90 mmHg or **systolic pressure 110–120 mmHg.** Adjust the volume of intravenous fluids administered according to the target pressure values, i.e., do not administer intravenous fluids until blood pressure drops **below** target. Typically, intravenous fluid boluses of 500–1000 mL are administered until the target pressure is reached.
— As soon as we are able to give the patient blood products and significant bleeding is present, thromboelastography should be monitored and targeted therapy should be applied according to the guidelines. In case the above is not possible, the massive transfusion protocol is 1:1:1 in the form of erythrocyte concentrate, fresh frozen plasma, and platelets with additional clotting factors as needed.
— Tranexamic acid is an antifibrinolytic agent indicated in massive bleeding that preserves the fibrin plaque at the bleeding site. It is indicated to administer it within three hours of the onset of bleeding, with the most commonly used scheme of 1 g in 100 mL of saline for 10 minutes and continuous administration of 1 g in 500 mL of saline during the next 8 hours. No significant side effects have been reported, but caution should be exercised in patients with existing thromboembolic events, epilepsy, and visual disturbances.
— Consider commercially available clotting factors:
  — recombinant factor VII, prothrombin complex (coagulation factors II, VII, IX, X), antithrombin III, and factor XIII
  — cryoprecipitate
— Vasoconstrictors:
  — administer in the case of traumatic brain injury and spinal cord injury to maintain MAP in addition to fluids.

VITAL SIGNS TO BE REPEATED EVERY FEW MINUTES IN PATIENTS WITH HAEMORRHAGIC SHOCK:

– respiratory rate and SpO_2
– heart rate, arterial pressure, pulse strenght (central and peripheral)
– ECG (rhythm, rate, and shape of QRS)
– capillary refill time.

D (DISABILITY):

– Examine the patient's state of consciousness (GCS), pupils, and blood glucose.
– Compensatory mechanisms in bleeding are aimed at maintaining sufficient cerebral blood flow. When bleeding causes consciousness impairment, it usually means > 30% of circulating blood volume is lost and compensatory mechanisms become insufficient.

E (EXPOSURE):

– Measure the patient's body temperature and warm them.
– Examine the body in detail for the source of bleeding:
– In traumatic bleeding, look for direct and indirect signs of long bone or pelvic fractures, rib fractures, or abdominal distension.
– In women of childbearing age with abdominal pain, do not forget about the possibility of extrauterine pregnancy. Significant bleeding in pregnant women in the later stages of pregnancy is one of the most common causes of maternal mortality.
– In the case of suspected bleeding from the digestive system, do not forget to do a digital rectal examination (a trace of black stool indicates bleeding from the oesophagus or stomach, and a trace of fresh blood may be the result of bleeding from any part of the digestive system).

SAMPLE HISTORY:

S (Signs / symptoms): adjust the questions to the potential cause and source of bleeding.
A (Allergies): check if the patient has allergies.
M (Medications): examine the patient for chronic medications; check especially for antiplatelets (acetylsalicylic acid, clopidogrel, prasugrel), anticoagulant drugs (warfarin, NOACs, and beta-blockers).
P (Past illnesses): check if the patient has significant comorbidities and whether they have had surgery in the past.
L (Last oral intake / last menstrual cycle): check when the patient last ate (possibility of aspiration of gastric contents); in women of childbearing age, check the date of the last menstruation.
E (Events leading up to present illness): check for significant events that preceded the current condition.

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Stick to the ABCDEs!

Humans are watery creatures, not all of it is within the blood vessels.

Use the 4-2-1 formula to assess the basal metabolic needs. This does not account for the extra loses such as vomiting, diarrhoea and bleeding, which are often time impossible to measure exactly and then you must approximate.

First line treatment – balanced crystalloids. Use the one you have available. Although 0,9% NaCl is not physiological and is not balanced, there are some situations where one might still use it (such as brain injury, hyperkalaemia or loss of chloride ions).

When suspecting an electrolyte disorder, try to find previous labs from a patient to see the trend.

Hypokalaemia is the most common electrolyte disorder. Luckily, it is easier to supply what is missing then to remove that what is in a surplus.

Hyperkalaemia happens often with chronic kidney failure (and less often from other causes). It can be life-threatening, proceed carefully. Look at the ECG changes, protect the heart with calcium if needed, administer insulin (with glucose, of course) and salbutamol to shift the potassium into the cells. You've just bought the patient enough time to seek expert help and consider/organize dialysis.

Significant bleeding is a big problem:
— decrease in cardiac output and haemoglobin concentrations decreases oxygen delivery to the tissues,
— losing blood results in losing coagulation factors as well — promotes further bleeding.

Haemorrhagic shock: the patient is pale, cold, mottled, can be disoriented, agitated, unconscious, usually tachypnoeic and tachycardic.

Be extra careful with paediatric and geriatric patients. The former (once they are no longer babies) have a big reserve, they will compensate vigorously but once they stop compensating – cardiac arrest is imminent. On the other side, geriatric patients tend to decompensate rapidly. Both age groups require extra caution. When out of the hospital and the suspected blood loss is higher than 15% total blood volume, try to stop the bleeding, if possible, open two wide-bore IV lines, administer crystalloids and colloids (6% HES 500-1000 ml max.). If available, administer other specific blood products per local protocol.

When managing suspected blood loss exceeding 15% of total blood volume outside the hospital, first attempt to control the bleeding if possible. Establish two wide-bore IV lines, administer crystalloids (adjust the volume to achieve, but not exceed, the target blood pressure), and consider adding colloids—specifically 6% HES, up to a maximum of 500–1000 mL.

Follow the targeted MAP values for traumatic bleeding while performing fluid resuscitation!

Part 7: Injuries

180—212

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7.1. TRAUMA ASSESSMENT AND MANAGMENT

7.1.1. INTRODUCTION

Polytrauma, along with other conditions such as acute coronary syndrome and stroke, is a “time-dependent disease”, meaning that the recognition of life-threatening injuries, initial stabilisation, and definitive care are directly connected with the survival rate, which is referred to in the literature as the “golden period” (previously known as the “golden hour”). The specified time starts from the moment of injury to the moment of definitive treatment in the operating room. When talking about the pre-hospital management of polytrauma as part of the golden period, the aim is to spend as less time as possible at the scene of the accident, ideally 10 minutes, to examine the patient, identify and manage life-threatening conditions, immobilise the patient, and transport them to the appropriate institution – trauma centre.

When talking about the approach to the polytrauma patient, there are several different associations that hold courses and provide guidelines for patient care. The most common approaches used in Europe are the International Trauma Life Support (ITLS) approach, which is adapted to outpatient conditions, and the European Trauma Course (ETC) approach as primary hospital trauma care, but the trauma examination remains the same.

7.1.2. MANAGEMENT OF TRAUMA PATIENTS

We use the internationally recognised ITLS algorithm for the proper survey of injured patients in outpatient emergency medicine (Figure 1):

- ITLS primary survey – scene size-up, initial assessment, and rapid trauma survey or focused survey
- ITLS ongoing exam – monitoring of changes in the patient's condition
- ITLS secondary survey – complete examination and identification of all injuries.

SCENE SIZE-UP

Assessment of the scene involves taking precautions to identify and remove all hazards from the scene of the accident before approaching the patient. If it is determined that the place of the accident is not safe (e.g., risk of electric shock, risk of explosion or fire, risk of building collapse, unsecured road, etc.), it must be made safe by other emergency services or the victim must be removed from the scene without jeopardising your own safety or the safety of your team. It is always necessary to use your own protective equipment due to possible contact with blood or other potentially infectious material. You should also protect the patient from contact with other people's body fluids by changing protective gloves when examining multiple injured persons.

When assessing the scene, we must also assess the mechanism of injury (MOI), the number of victims, the equipment necessary for treating the injured, and the need for additional medical assistance. If it is determined that there are more injured people at the scene, reinforcements should be requested. As a rule, one vehicle and one ambulance team are needed for each severely injured person.

When approaching an injured person, we must assess the mechanism of injury – the way the patient was injured, which is especially important in considering possible injuries.

7.1.3. MECHANISM OF INJURY

The mechanism of injury can be generalised when large forces act on the whole body and life-threatening injuries can be expected, or focused, when forces act on an isolated region (e.g., a stone falling on the foot), although these injuries can also be life-threatening (e.g., a penetrating wound with a sharp object). A significant mechanism of injury includes falling out of a vehicle, falling from a height of more than four and a half meters, overturning a vehicle, colliding with a car at high speeds, colliding with a pedestrian, and having a motorcycle accident. When we examine a patient with any of the significant mechanisms of injury, we should always expect more severe injuries that may not be visible at first glance.

A motor vehicle collision implies three consecutive collisions: a vehicle collision, a body collision, and an organ collision. Additional causes of injuries can be secondary and additional collisions – when

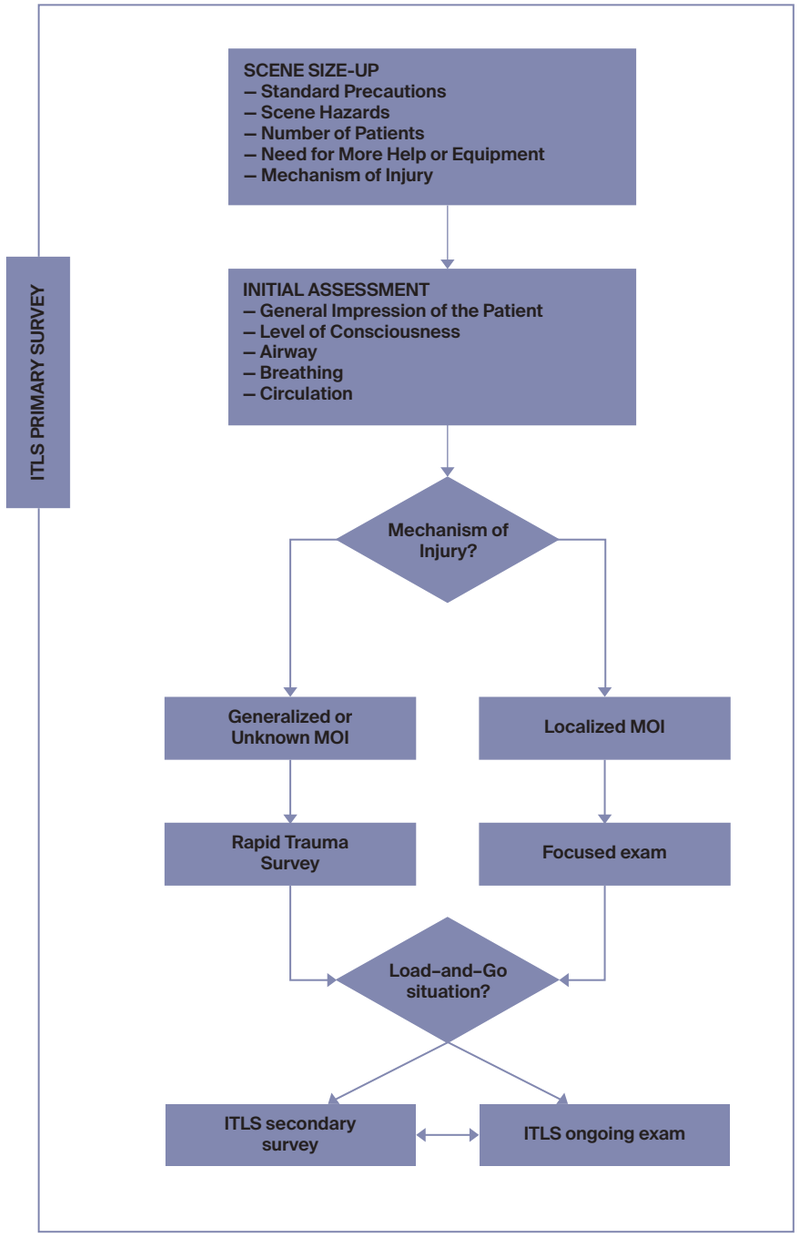

FIGURE 1. *ITLS trauma patient assessment (retrieved and adapted from International Trauma*

unsecured objects move like projectiles at the speed of the vehicle or the vehicle hits the nearest object or other vehicles collide with it. Likewise, vehicle safety systems can cause additional injuries to the abdomen, cervical and lumbar spine, shoulders, and collarbone. By knowing the principle of injury formation in a particular mechanism, it is possible to predict the injury to certain organ systems (Table 2).

In the event of a fall, injuries depend on the height of the fall, the part of the body hit, and the surface on which the body hit.

The severity of a stab wound with a knife depends on the part of the body that was stabbed (a stab below the fourth intercostal space can also affect the abdomen), the length of the blade, and the angle of the stab.

TABLE 1. *Basic mechanisms of injury*

The basic mechanisms of injury caused by movement	
Blunt injuries	Penetrating injuries
– sudden forward deceleration (collision) – sudden vertical deceleration (fall) – energy transfer from blunt objects (impact)	– missiles – knives – falls on a fixed object

TABLE 2. *Potential injuries depending on the trauma mechanism*

MECHANISM OF INJURY	POTENTIAL INJURY
Frontal impact: – Steering wheel deformity – Dashboard cracking – Windscreen deformity	– Cervical spine fracture – Flail chest – Myocardial contusion – Pneumothorax – Aortic disruption – Spleen or liver laceration – Posterior hip dislocation – Knee dislocation
Lateral impact	– Cervical spine fracture – Flail chest – Pneumothorax – Aortic disruption – Diaphragmatic rupture – Laceration of spleen, liver, kidney – Pelvic fracture
Rear impact	– Cervical spine injury
Pedestrian vs. car	– Head injury – Abdominal visceral injuries – Fracture of lower extremities and pelvis – Spine fracture

7.1.4. INITIAL ASSESSMENT

The team leader performs the primary ITLS assessment and does not stop it until it is fully completed, except in cases of cardiopulmonary arrest, airway obstruction, external bleeding that cannot be stopped, or new danger at the scene of the accident. When dealing with people who are in critical condition, the goal is to complete the primary examination within two minutes and stay on the scene for less than ten minutes. To perform procedures and interventions during an ITLS assessment, the team leader delegates team members using the “fix it” tactic. For example, after the doctor checks the airway and cervical spine, the “fix it” procedure may be to place a Schantz collar and an oxygen mask. One member of the team is positioned above the injured person’s head and manually limits the mobility of the cervical spine until the final immobilisation on a long backboard with a strap and lateral head restraints or in a vacuum mattress.

Before approaching the injured person, the team leader assesses the general impression. The team leader notices major deformities and obvious breathing disorders and determines the risk group (elderly, children, pregnant women). Impaired state of consciousness, compromised airway, respiratory or circulatory disorder, and penetrating injuries to the head, neck, body, or extremities proximal to the elbow or knee indicate a critical patient. If the team leader notices major external bleeding after stabilising the head and neck with their hands, they delegate to a team member to stop the bleeding, which changes the usual priority approach from ABC to C-ABC.

The initial assessment is performed by the team leader using the ABCDE algorithm. The team leader approaches the injured person and addresses them only when the team leader has stabilised with their hands the injured person’s head and neck. The team leader introduces themselves and asks what happened. Depending on the patient’s answer, the level of consciousness is quickly assessed using the AVPU scale.

“A” (alert) means the patient is awake, oriented, and obeys commands. “V” (verbal) means the patient responds to verbal stimuli, is awake but confused, or unconscious but responds in some way to verbal stimuli. “P” (pain) means unconscious but responds in some way to touch or painful stimuli. “U” (unresponsive) means that the patient does not respond to any stimuli.

If the level of consciousness is “P” or “U”, there is a high risk of airway compromise.

The stabilisation of the head and neck is taken over by a team member, and the team leader continues with the examination.

A (AIRWAY):

If the victim responds, it means that the airway is open. If consciousness is impaired, we listen for sound phenomena: snoring indicates a partial obstruction and requires interventions to ensure airway patency (initial modified mandibular ejection grip or placement of

an oropharyngeal or nasopharyngeal tube), gurgling indicates liquid contents in the oral cavity and requires aspiration. The see-saw phenomenon (a non-synchronised movement of the chest and abdomen during inhalation) during inspiration and vice versa during expiration) is a sign of complete airway obstruction.

B (BREATHING):
It consists of calculating the respiratory rate, observing the symmetry of chest elevation, using the auxiliary respiratory muscles, and attaching a pulse oximeter. High-flow oxygen (12–15 L/min) should be delivered via a mask with a tank to each critically injured person. If the patient is not breathing, it is necessary to provide bag-mask ventilation using the self-expanding balloon with a container (e.g., Ambu bag). Provide assisted ventilation if the patient breathes inadequately (with too low frequency or insufficient volumes).

C (CIRCULATION):
It consists of pulse assessment (rate, rhythm, strenght), skin (temperature, colour, dryness/sweating), and capillary refill time. In the early stage of shock, the skin is pale, cold, and sweaty, the pulse is poorly palpable and rapid, and the capillary refill time is prolonged. We check again to see if there is any visible external bleeding. On top of that, we try to obtain IV access. However, establishing IV access must not delay critical interventions and emergency transport of a vitally endangered injured person.

D (DISABILITY):
We assess the patient's state of consciousness as part of the initial assessment using the AVPU scale. A more detailed neurological examination, with the assessment of the state of consciousness using the Glasgow Coma Scale and estimation of pupil size, symmetry, and reactivity to light, is performed at the end of the primary ITLS assessment before entering the vehicle if the victim has a disturbed state of consciousness. If the victim is fully conscious, we can postpone and perform the detailed neurological examination during the secondary survey. If the injured person has an impaired state of consciousness or diabetes, it is necessary to measure their blood glucose level.

E (EXPOSURE):
By evaluating exposure to environmental factors, we identify objects found in the immediate environment (needles in intravenous drug addicts, pills in suspected overdoses) or factors related to the victim (stab wounds, body temperature, etc.).

SAMPLE
Since it is not uncommon for an injured person to lose consciousness before arriving at the hospital, the emergency medical service team is the only one that has the opportunity to obtain anamnestic data. If the injured person is already unconscious during the primary examination, it is necessary to take a heteroanamnesis from eyewitnesses or family members.

S symptoms
A allergies
M medications
P past medical history
L last oral intake
E events preceding the incident

7. 1. 5. RAPID TRAUMA SURVEY

Rapid trauma survey is a systematic examination of the injured person by body regions to determine the resulting injuries and their extent (Figure 2).

It begins after the initial ABCDE assessment is completed. It is performed in patients with a generalised mechanism of injury or if any part of the initial assessment is abnormal. In patients with isolated or minor trauma, instead of a quick trauma survey, only a targeted physical examination of the injured body part is required.

A rapid trauma survey is performed by assessing region by region (head, neck, chest, abdomen, pelvis, extremities, and finally the back). We first assess each region by inspection and then by palpation (we also perform auscultation of the chest and, if necessary, percussion), and only then do we move on to the next region ("from head to toe").

Inspection enables us to detect visible injuries, hematomas, bleeding, and deformities. The neck is specially inspected for jugular venous distention (if present, it may indicate tension pneumothorax or pericardial tamponade). In chest examination, we should notice open wounds (suspected open pneumothorax) and unstable segments (flail chest), whereas the mark of a seat belt on the abdomen indicates a high probability of injury to the abdominal organs and the possibility of developing shock.

By examining the extremities, we can notice open fractures. We never remove foreign bodies (except in the case when they cause airway obstruction or are located in the cheek area), but we stabilise them in the found position. Do not return abdominal protrusions to the abdominal cavity but cover them with sterile gauze abundantly soaked in isotonic saline solution and covered with an impermeable foil. The amputated parts are placed in a dry bag, which is then immersed in another bag containing ice and water. The amputated part must not come into contact with water or into direct contact with ice. Palpation reveals sensitivity, instability, and crepitation which indicate a possible fracture. After palpation of the posterior parts of the body inaccessible to inspection, it is necessary to determine the presence of blood on the gloves. During the palpation of the cervical

spine (and later when examining the entire spine), it is necessary to identify pain or deformities in the spinous processes, as well as paravertebral muscle tension.

We palpate the chest with the “click-clack” technique. Spread the palms horizontally above and below the breast, first on one side of the chest (e.g., left hemithorax). Press with one hand and then with the other and ask the injured person if they are in pain and palpate for crepitations or instability. Repeat the same with the other side of the chest.

Palpate the abdomen in four quadrants. If the patient is in pain, we consider that there is an injury to the intra-abdominal organs with the threatening risk of shock. Palpate the pelvis by pressing on the symphysis and with the movement of closing the book we embrace both iliac cristae with our palms and squeeze inwards. If the pelvis is already painful when pressing on the symphysis or we feel the instability of the pelvic ring, we do not examine the pelvis further. In the case of an unstable pelvis, it is necessary to install a pelvic splint.

When palpating the extremities, we examine individual segments. First one leg (upper leg, lower leg, sensation, and motor function), then the other leg, then one arm (upper arm, forearm, sensation, motor function), and finally the other arm. In each segment, we first determine instability (which is a definite sign of fracture) if there are no obvious deformities (which is also a definite sign of fracture). We fixate the joint against the base by pressing from above (knee or elbow), and with the other hand we reach under the upper end of the femur or humerus and by lifting that part, we determine whether the bone is unstable. We also examine the distal parts of the extremities (with one hand we squeeze the knee or elbow against the ground, and with the other hand we lift the distal part of the lower leg or forearm). After assessing the instability, we put our hand under a certain segment and pass along the entire length by palpating possible crepitations. We then ask the injured person if they are in pain, and look at the glove to see if there are any traces of blood. If there are signs of limb fracture, we must assess the presence of pulsations below the fracture site.

The assessment of a possible spinal injury includes the assessment of motor function and sensation in the hands and feet. If the injured person has an altered state of consciousness but still reacts to painful stimuli, then during palpation we observe the injured person’s face and determine the pain level according to their grimace.

Among other methods of examination, we auscultate the chest at two points in the midaxillary line on both sides of the chest and listen to breathing sounds. If breath sounds are unequal, by percussion in four places we identify if the chest is hyperresonant (pneumothorax) or dull (haemothorax). We also auscultate the heart sounds (are they still normal or have they become muffled).

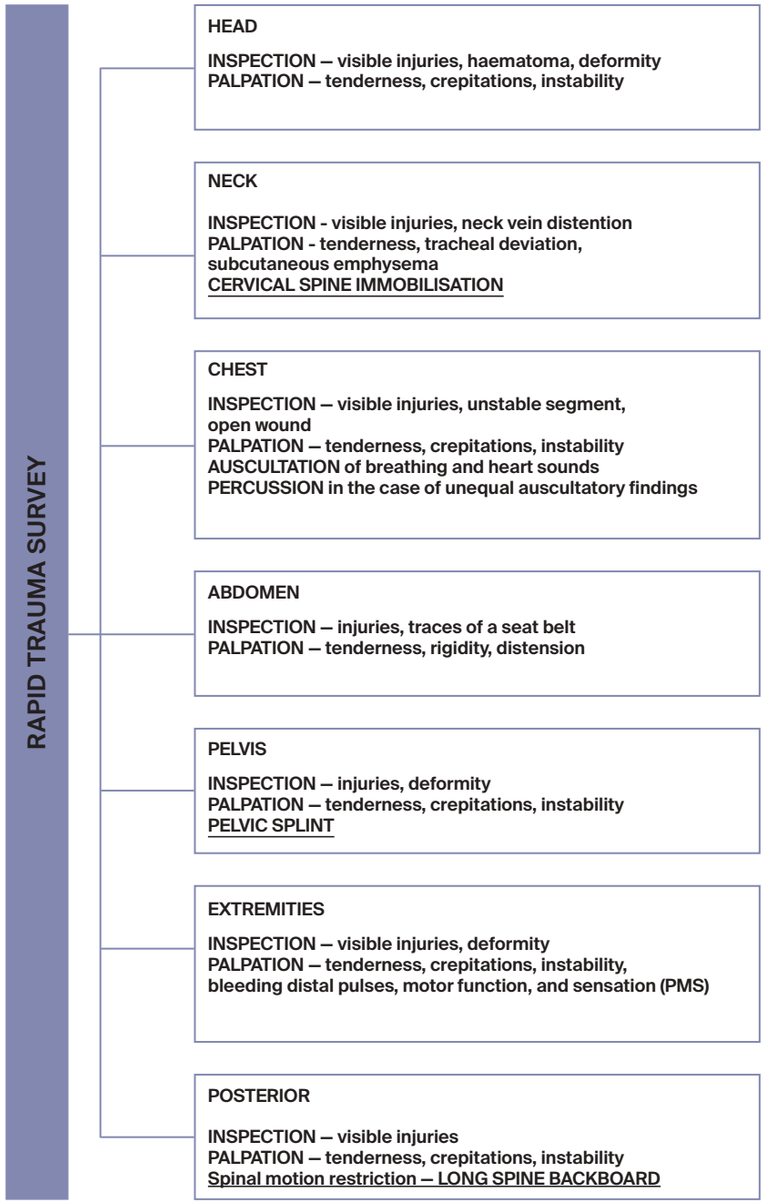

FIGURE 2. *Rapid trauma survey*

We examine the injured person's back after turning them on the side using the so-called log roll technique. In the case of contraindications for turning on the side (unstable pelvis, bilateral femoral fracture, foreign body in the abdomen or chest, or clear signs of a spinal injury – abnormality of sensation or motor function of the extremities, deformity and pain of the cervical spine), the back is examined after lifting the patient with a **scoop stretcher** (Figure 3).

The assessment should be carried out as described, without skipping steps, so that injuries can be detected and addressed with a systematic approach. In patients at risk of developing shock or with signs of developed shock, it is necessary to obtain two large-bore IV lines and start with fluid replacement. In the case of suspected internal bleeding (intra-abdominal, haemothorax), the goal is to achieve a systolic blood pressure of 90 mmHg.

Excessive fluid replacement can increase blood pressure and thus worsen bleeding (the formed clot dislodges, and the pressure at the bleeding site is higher), dilution of clotting factors occurs, and cold fluid additionally contributes to coagulation disorder. When possible, remove the patient's clothes for proper examination. It should always be assumed that an injured patient with a significant mechanism of injury may also have a spinal injury.

Contraindications for turning on the side are unstable pelvis, bilateral femoral fracture, foreign body in the abdomen or chest, or clear signs of a spinal injury.

7.1.6. IMMOBILISATION

7.1.6.1. CERVICAL SPINE IMMOBILISATION WITH A CERVICAL COLLAR

Immobilisation of the spine begins with manual stabilisation by placing the patient's head between the rescuer's two hands and continues with the placement of a cervical collar and other immobilisation equipment.

The cervical collar supports the head and neck and maintains a neutral position of the cervical spine. However, spinal immobilisation is not complete until the patient is secured on a long backboard with lateral head stabilisers.

Before installing the collar, it is necessary to examine the entire neck because after its installation, inspection and palpation of that area are difficult. It is necessary to determine the correct size of the collar for cervical spine immobilisation by measuring the distance between the point where the trapezius muscle touches the side of the neck and the point where the imaginary line starting from the chin (perpendicular to the floor for a lying patient, parallel for a sitting patient) intersects the side of the neck. The second point is usually located one to two centimetres below the angle of the mandible. The measurement

FIGURE 3. *Scoop stretcher*

is expressed by the number of fingers that fit between these two points, and the measured distance is transferred to the plastic part of the collar after which the width is adjusted. With the victim lying down, the first step is to place the back of the collar under the head and neck, and then the collar is placed on the front of the neck under the chin. If the injured person is sitting, we first place the collar on the front of the neck.

7. 1. 6. 2. SPINAL IMMOBILISATION WITH A LONG BACKBOARD – LOG ROLL

To place the patient on a long backboard and completely immobilise the spine, we need the following: a cervical collar, a long backboard, lateral stabilisers for the head and neck, two adhesive headbands, straps, and three or four people (Figure 5).

Manual stabilisation of the head and neck is the fastest way to protect the spine until an appropriate immobilisation device is selected. If the patient is conscious, we should explain the spinal immobilisation procedure to them and ask for their cooperation. By placing a cervical spine immobilisation collar, we keep the injured person's head and neck in a neutral position while the spine is being immobilised. We still use our hands to stabilise the head and neck. The long backboard should be positioned parallel to the patient. While one team member maintains in-line manual stabilisation, another supports the shoulder and hips, and the third supports the hips and lower extremities. A team member on the patient's head directs the other team members to log roll the patient as a unit onto the side toward the emergency care providers. While the injured person is on their side, the back and buttocks are examined for possible injuries.

The team member holding the injured person's shoulder must not move it, as this would endanger the stability of the spine, and performs the back examination with their other ("lower") hand. The fourth person, if there is one, places a long backboard for immobilisation parallel to the patient's body. The patient is then log rolled on the long backboard back to the supine position.

The injured person needs to be secured on the immobilisation board using straps. First, place the upper horizontal strap under the arms, above the breasts. Then place the second horizontal strap across the pelvis, the third strap above the knees (in the case of a femur fracture, place it below the knees), and with the fourth strap, form an "8" around the feet. Finally, we install the head motion-restriction device – lateral stabilisers for the head and neck. It is important to always place the straps and lateral stabilisers in this order, otherwise, in the event of vomiting and turning the patient on the long backboard onto the side, the patient could slide off the board and the spine would become unstable. Another option for immobilisation is a vacuum mattress, which is increasingly recommended by professional associations, especially for longer transports.

FIGURE 4. *Semi-rigid Schanz collar (down) and collar placed for cervical spine immobilisation (up)*

FIGURE 5. *Immobilisation equipment*

7. 1. 7. ETC APPROACH TO TRAUMA PATIENTS

The first European Trauma Course was held in 2002. and was developed through a collaboration of the main European medical associations such as anaesthesiology, traumatology, and emergency medicine associations. It was developed in response to the need for a multidisciplinary team focused on trauma approach and initial care in the hospital emergency department.

Upon the patient's arrival at the emergency hospital department begin with a quick assessment, the so-called **5-second round**:

- Are there signs of airway obstruction?
- Is there massive external bleeding visible?
- Are there signs of arrest?

After the exclusion of the above, the senior doctor, the team leader, gathers information about the patient's status and the event. A rapid trauma survey and an ultrasound examination following the FAST protocol are performed, and, depending on the clinical status, a decision is made on the necessary laboratory and radiological tests and removal of the immobilisation equipment.

7. 1. 8. SPECIFIC INJURIES

7. 1. 8. 1. CHEST INJURIES

Thoracic bones protect most of the vital organs in the chest: lungs, heart, large blood vessels, liver, spleen, transverse colon, and spinal cord. Trauma can cause significant disruptions in each of these systems that can be life-threatening if not quickly identified and treated. According to the mechanism of injury chest injuries are divided into blunt or closed and open or penetrating. Closed ones are usually caused by traffic accidents, falls from heights, being buried, or a blunt blow to the chest. Open injuries are most often caused by stab wounds or gunshot wounds. A foreign body should not be removed from the wound, but it should be stabilised.

Life-threatening injuries should be identified during the primary ITLS survey. Difficulty breathing and chest pain are the main signs of chest injuries.

Tenderness, instability, and crepitations can be determined by palpation. Auscultation checks whether breath sounds are present and symmetrical. Multiple rib fractures with or without the formation of an unstable segment can lead to hypoxia due to disorders of respiratory mechanics and lung contusion. It is important to constantly monitor the patient and use pulse oximetry and capnography to detect hypoxia or hypoventilation in time. If necessary, apply oxygen therapy, constantly monitor the heart rhythm, and establish intravenous access for possible therapy.

Secondary examination and inspection will provide information on other possible life-threatening injuries, such as pulmonary

contusion, myocardial contusion, traumatic diaphragmatic rupture, and tracheobronchial tree disorders. A sternal fracture can be suspected by palpation of the sternum, and it is caused by blunt trauma to the anterior chest wall. A simple rib fracture is one of the most common chest injuries.

TENSION PNEUMOTHORAX

Pneumothorax is the accumulation of air in the potential space between the visceral and parietal pleura, resulting in complete lung collapse. **Tension pneumothorax** is a progressive accumulation of air in the pleural space that continuously accumulates without exiting, which results in the "one-way valve" phenomenon and further accumulation of air. Intra-thoracic pressure on the affected side increases, displacing the heart and trachea to the opposite side and collapsing the superior and inferior vena cava, thus occluding venous return to the heart. This leads to a drop in cardiac output and obstructive shock. If rapid intervention is not performed, respiratory and cardiac arrests occur within a few minutes.

AIRWAY OBSTRUCTION
TENSION PREUMOTHORAX
OPEN PREUMOTHORAX
MASSIVE HAEMOTHORAX
FAIL CHEST
CARDIAC TAMPONADE

FIGURE 6. *Life-threatening conditions in chest injuries that are important to recognize during primary survey*

Classic signs of tension pneumothorax include dyspnoea, tachypnoea, possibly tracheal deviation, distended jugular veins (caution during hypovolaemic shock), overexpansion of that part of the chest with minimal respiratory movements, auscultatory absent or decreased breath sounds on the affected side, and hyperresonant sound during percussion (on the affected side). Needle decompression is a temporary but life-saving method.

Needle thoracocentesis is performed by inserting a needle with the widest possible calibre (14 or 16 Fr, or < 10 G) with a catheter into the pleural space through the second intercostal space in the midclavicular line along the upper edge of the lower rib. The sound of hissing air coming out under high pressure from the pleural space confirms the diagnosis. A possible approach is the fourth and fifth intercostal space in the front or midaxillary line, where the thoracic wall is thinner, and, according to some research, the percentage of success is, therefore, higher than with the classic approach.

Definitive therapy for tension pneumothorax includes placement of thoracic drainage in hospital conditions.

OPEN PNEUMOTHORAX

Open pneumothorax is an accumulation of air in the potential space between the visceral and parietal pleura due to an open chest wound. It occurs as a result of penetrating chest trauma or an open rib fracture. If the chest wound is greater than two-thirds the diameter of the trachea, air will dominantly enter through the wound during inspiration.

Open pneumothorax must be covered with an occlusive dressing that is attached to the skin on three sides with adhesive tape so that one corner remains free. This forms a "flutter valve", which enables air to be expelled through the wound during exhalation but prevents "suction" during inhalation.

MASSIVE HAEMOTHORAX

Massive haemothorax can result from blunt or penetrating trauma, and the source of bleeding can be in the lungs, major vessels, intercostal vessels, or even the heart. Massive haemothorax is defined by the presence of at least 1,500 mL of blood loss into the pleural space of the thoracic cavity. Clinical signs of shock may be apparent. The indications for thoracotomy are the following:

— blood loss > 1,500 mL or 1/3 blood volume
— blood loss > 200 mL/h (3 mL/kg/h) over 2–4 hours.

FLAIL CHEST

Flail chest is an emergency condition resulting from a fracture of three or more adjacent ribs in two or more sites, causing chest wall instability and paradoxical movement of the "flail segment" in a spontaneously breathing patient. A flail segment no longer moves in continuity with the rest of the chest and retracts during inspiration and protrudes during expiration.

It occurs as a result of blunt thoracic trauma in traffic accidents, violence, and falls. The most significant complication and accompanying pathophysiology is lung contusion, respiratory insufficiency, lung atelectasis, and pneumothorax.

Management includes patient stabilisation, so adequate analgaesia, endotracheal intubation, and assisted ventilation are often required.

CARDIAC TAMPONADE

Pericardial tamponade often occurs as a result of penetrating chest injuries. If blood accumulates quickly, even small amounts can cause tamponade because the pericardium cannot stretch fast enough to accommodate it. The ventricles get compressed, making the heart less able to refill, and cardiac output decreases. This leads to the

development of obstructive shock. Cardiac tamponade is characterised by Beck's triad, which includes hypotension, systemic venous stasis (jugular vein distention), and muffled tones during heart auscultation. Accelerated heart rate, tachypnoea, and paradoxical pulse may also appear. It should be emphasised that without the help of ultrasound, it is very difficult to clinically distinguish between tension pneumothorax and cardiac tamponade during the primary survey.

Treatment of tamponade includes emergency pericardiocentesis with drain age, usually performed by an experienced person.

DIAPHRAGMATIC RUPTURE

Diaphragmatic rupture caused by blunt trauma is not a common injury, but about 5% of polytrauma patients have this injury. Traffic accidents are the main cause of traumatic diaphragmatic ruptures. In many cases, the diaphragmatic rupture is initially not recognised due to severe multiple injuries. Most often, the diaphragm ruptures on the left side. Diaphragmatic injuries can occur as a result of blunt or penetrating trauma (especially on the left side). Blunt injury causes lacerations that can lead to herniation of abdominal structures into the thoracic cavity. Penetrating injuries can cause small perforations that may present as herniations years later. Diaphragm injuries should be suspected in severe blunt chest trauma, especially if compressive or rapid deceleration forces are involved.

7.1.8.2. ABDOMINAL INJURIES

Abdominal trauma is life-threatening due to the high possibility of haemorrhagic shock. Abdominal trauma is very difficult to assess even in hospital conditions, therefore, it is very important to obtain as much information as possible on the mechanism of injury in pre-hospital conditions and then consider an intra-abdominal injury. Minimising the time spent in the field and fast transport increases the chances of survival. According to the mechanism of occurrence, abdominal trauma can be divided into blunt or penetrating injuries. Signs and symptoms include abdominal pain, tenderness, abdominal wall tension, and bruising of the outer abdominal wall. In the case of a blunt abdominal injury, it is very important to look for external signs of injury such as skin abrasions, hematomas, and seat belt marks. A horizontal belt track is a clinical sign that carries a 30% risk of mesenteric and intestinal injury.

The abdomen is palpated in four quadrants, and the pelvis and lower ribs should be examined for signs of fracture. In many patients subsequently diagnosed with an intra-abdominal injury, there are no clear signs of such an injury on the initial ITLS examination, so it is very important to obtain information about the mechanism of injury. For larger open injuries, it is extremely important not to attempt to return the organs to the abdominal cavity. Instead, cover them with warm gauze moistened with saline or water. Patient management includes obtaining several venous accesses, administering prescribed analgaesics, constantly monitoring the patient, and observing their condition until arrival at the hospital.

BLUNT ABDOMINAL INJURIES

Blunt abdominal injuries are caused by the blunt action of an external force and are a diagnostic challenge due to the very frequent absence of visible signs on the body surface. They are usually caused by traffic accidents, falls from a height, bicycle handlebar injuries, sports injuries, and intentional injuries.

Intra-abdominal bleeding, hollow organ injury, and retroperitoneal injury may occur in the case of a blunt abdominal injury.

The most common causes of intra-abdominal bleeding are ruptured spleen, liver, and, less often, mesenteric blood vessels. The symptoms are the same as in hypovolaemic shock, with pain sensitivity and distension. Rupture of the spleen is most often associated with a fracture of the lower ribs.

The consequence of rupture of the initial jejunum or terminal ileum is peritonitis. Rupture occurs due to midabdominal compression between the external force and the spine. The main symptoms are pain, nausea, and hard stomach. Vomiting and abdominal distention occur in the case of intestinal paresis. In the case of abdominal contusion, rupture most often occurs in the less mobile part of the intestines, such as the jejunum, and terminal ileum. Duodenal rupture usually occurs if the abdominal muscles are relaxed at the time of abdominal contusion. The spleen is the most frequently injured organ of the abdominal region during trauma and the second most frequently injured organ in children.

Retroperitoneal haematoma is most often the result of a kidney rupture, which can be combined on the left side with a ruptured spleen. In the retroperitoneal space, the pancreas and duodenum can be injured as a result of compression between the external force and the spine. Pancreatic injury can manifest as traumatic pancreatitis, which is the most common form of pancreatic injury.

PENETRATING ABDOMINAL INJURIES

Penetrating injuries are mostly caused by stabbing or firearms. These injuries cause damage to the blood vessels of the abdominal wall, but injuries to the internal abdominal organs (liver, spleen, stomach, intestines, and mesentery) can also occur. Sometimes a part of the intestinal coil protrudes through the opening on the abdominal wall, which is a definite sign of a penetrating injury. If the wound is caused by a projectile, then a penetrating abdominal injury could result from an entry wound at almost any part of the body. In the case of larger open injuries, it is extremely important not to attempt to return the organs to the abdominal cavity. Instead, cover them with warm gauze moistened with saline or water.

7.1.9. PELVIC INJURIES

Pelvic injuries are complex because they are mostly caused by trauma with high energy transfer (traffic accidents, falls from a greater height...) and can be accompanied by profuse bleeding and rapid development of haemorrhagic shock. In most cases, a pelvic fracture appears associated with injuries to other body regions (intra-thoracic and intra-abdominal injuries). In the case of a pelvic fracture, we can expect profuse bleeding and the rapid development of circulatory shock. Stabilisation of the pelvis is performed with a pelvic splint.

Pelvic stabilisation is an important and simple intervention in the treatment of severe pelvic trauma. It has four main goals:

- preventing further tissue damage caused by possible pathological movement of the fractured pelvic bones
- reducing the volume of the pelvis
- minimising bleeding from injured bony structures and soft tissue reducing the patient's pain.

When do we need to use a pelvic splint?
- in a haemodynamically unstable patient with a mechanically unstable pelvis
- in a haemodynamically unstable patient with a suspected pelvic fracture
- in a haemodynamically stable patient with a mechanically un stable pelvis.

FIGURE 5. *Illustration of the placement of the pelvic splint. It is important to check the 4 Ps before and after immobilisation: pain, pockets, penis, and pulse. The splint is placed at the level of the greater trochanters. The buckle ensures the proper clamping force of the splint, but it must not be too tight. Splint is X-ray transparent, can be used in MRI, and is intended for multiple uses.*

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7. 2. BURNS

7. 2. 1. INTRODUCTION

Most physicians who work in emergency medical services will encounter a patient with large burns, a challenging and life-threatening injury. **The initial management of these patients will make a difference between good and poor outcomes.** Epidemiology data suggest that burns are more common in men than women, with flame and scald as the most common causes. Moreover, children under 5 years of age are five times more likely to be injured than adults. Approximately 10% of burned patients will need hospital admission, and 10–15% of hospitalised patients will need to be treated in burn centres. The incidence of severe burns in Europe is 0.2–2.9/10,000.

RISK FACTORS FOR DEVELOPING SEVERE COMPLICATIONS AND POOR OUTCOMES INCLUDE:

- second-degree burns covering more than 10% of total body surface area (TBSA) in patients under 10 and over 50 years of age
- second-degree burns covering more than 20% TBSA
- third-degree burns covering more than 5% TBSA
- burns that involve the face, hands, feet, genitalia, perineum, and/or major joints
- chemical burns (management includes contacting the Poison Control Centre, for Croatia available at +38512348342)
- electrical burns (including lightening injuries)
- burns associated with other trauma injuries
- patient's preexisting medical condition that has implications on wound healing and/or burn management (e.g., chronic heart failure, diabetes, immunosuppression, pregnancy).

Patients with one or more of the mentioned criteria will benefit from being managed in specialised burn units. The mortality rate is directly affected by burned TBSA, while the functional outcome depends on the depth and location of the burn injury.

7.2.2. INITIAL PATIENT ASSESSMENT AND MANAGEMENT: ABCDE + SAMPLE

- Assess the safety of the patient and the rescuer(s), e.g., consider potential hazards due to electricity, gas, traffic, etc.
- If ongoing, stop the burning process.
- Every burned patient should also be managed as a trauma patient due to common concomitant injuries such as head trauma due to suffocation and loss of consciousness, escaping from fire by jumping through a window, etc.
- Check if the patient responds to stimuli:
- If not, call for help, open the airway, and check for breathing sounds and pulse for 10 seconds (follow the advanced life support algorithm).
- If the patient is alive, proceed with **ABCDEF**:

A (AIRWAY) / B (BREATHING) / CERVICAL SPINE CONTROL:

- Check the airway patency and open the airway in the case of partial or complete obstruction (apply the cervical spine control in suspected trauma).
- Evaluate breathing ("look, listen, and feel" approach: respiratory rate and effort, breath and added sounds, symmetry of chest movement, check for subcutaneous emphysema, tracheal deviation, jugular vein dilation, cyanosis), and use pulse oximetry.
- Apply high-flow oxygen therapy through a non-rebreathing reservoir mask.
- If the patient has deep burns to the lower face and/or neck, burned nasal hair, blisters, or oedema of the lip and/or tongue, and if the fire was in a closed room, a direct and thorough inspection of the oropharynx is needed, preferably by a physician experienced in airway management.
- Hoarseness, carbonaceous sputum, hypoxia, stridor, wheezing, signs of respiratory distress, and/or use of accessory muscles **indicate inhalation injury of the airway and/or lungs**, and most experts advocate for emergency tracheal intubation with a cuffed endotracheal tube (ETT). Prepare a selection of ETT sizes as smaller size might be needed due to airway oedema and secure the ETT position thoroughly as the intubation might be impossible once the oedema fully develops.
- Mind that signs and symptoms of airway oedema might become apparent only after the fluid resuscitation is started, e.g., during the transport of the patient to the hospital.
- Check for signs of carbon monoxide or cyanide poisoning.

C (CIRCULATION):

Every burn injury larger than 10% of TBSA in children or 15% of TBSA in adults requires meticulous intravenous fluid resuscitation to prevent the development of hypovolaemic shock!

- Check for heart rate, blood pressure, capillary refill time, skin colour, and (hourly) urine output. Monitor the dynamics every 3–5 minutes.
- Place two large bore intravenous cannulas (≤ 18 G, "green, grey, or orange"), preferably at sites that are not burned, and secure them well.
- If this is not possible, establish intraosseous access.
- Development of hypovolaemic shock takes time: if the burned patient has low blood pressure immediately after the injury, search for other causes of hypotension (such as bleeding).
- Place continuous ECG monitoring, especially in the case of electrical injury or hypothermia.
- Check for peripheral pulses on all extremities due to the risk of developing compartment syndrome, especially in the case of circumferential burns.
- Place a urinary catheter to monitor hourly urine output (target value: > 1 mL/kg/h in adults, > 2 mL/kg/h in children).
- After finishing the initial assessment, apply analgaesia: e.g., morphine 4–6 mg IV, add 1–2 mg IV every 10 minutes until an adequate level of analgaesia is achieved (pain activates the sympathetic nervous system and releases catecholamines, which may intensify skin hypoperfusion and depth of burns).

D (DISABILITY):

- Check for blood glucose level, AVPU/GCS, reactivity, and symmetry of the pupils. Put special emphasis on toxicological examinations in the case of suspected carbon monoxide or cyanide poisoning.

E (EXPOSURE/ENVIRONMENTAL CONTROL):

- Consider cooling the burned sites with clean and cold water, at an approximate temperature of 12–15 °C (not colder than that), for 2–3 minutes for the theoretical possibility of limiting the thermal injury, reducing oedema, and relieving pain. Note that cooling may cause hypothermia.
- Irrigate chemical injuries.
- Wet and dirty clothes should be removed, but do not pull burnt clothing that is stuck to the skin. Remove all jewellery.
- Estimate the percentage of the burned TBSA using the "**rule of nines**" (Figure 1).
- If the burns are patchy, the TBSA can be estimated by approximating the area of the patient's hand, in which case the patient's hand represents approximately 1% of their TBSA.
- Burned sites should be covered with sterile gauze soaked in

sterile saline or commercially available materials for burn coverage. Do not use ointments, powders, or creams.

- **Watch for hypothermia** (with a core body temperature < 35 °C): although it is rarely present on initial examination, its incidence increases to 35% when burned TBSA is > 70% and can lead to increased fluid requirements, coagulopathy, depressed cardiac function, arrhythmia, respiratory depression, impaired oxygen delivery, acidosis, and increased mortality. Hence, once the patient is fully assessed, cover them with a clean blanket, use warmed intravenous fluids, and warm the vehicle for transport.
- Check when the patient last received tetanus immunisation.

F (EARLY FLUID THERAPY):

- In adult patients with burns covering more than 20% of TBSA, start with 500 mL of warmed intravenous fluid during the first hour after the injury.
- In children with burns covering more than 10% of TBSA, start with 50–100 mL of warmed intravenous fluid, depending on the child's body weight.
- Mark the vials/packages so that you are aware of the exact volume of applied intravenous fluids (excessive fluid resuscitation can exaggerate the extracellular space oedema, worsening the airway oedema and/or decreasing skin perfusion with the resultant delay of burn wound healing).
- Intravenous fluid: balanced crystalloid solution, ideally Ringer lactate or Ringer acetate, prewarmed at body temperature.
- Once hospitalised, burns should be assessed and managed by the surgeon, and intravenous fluid balance during the first 24 hours should be calculated using the Parkland formula

2–4 mL × the percentage of the burned TBSA × body weight

The first half of the calculated volume should be administered during the first 8 hours, and the other half over the next 16 hours.

- Inform the hospital that you are arriving with a burned patient, as these injuries are quite specific, high in resource demands, and require the expertise of multiple specialists.

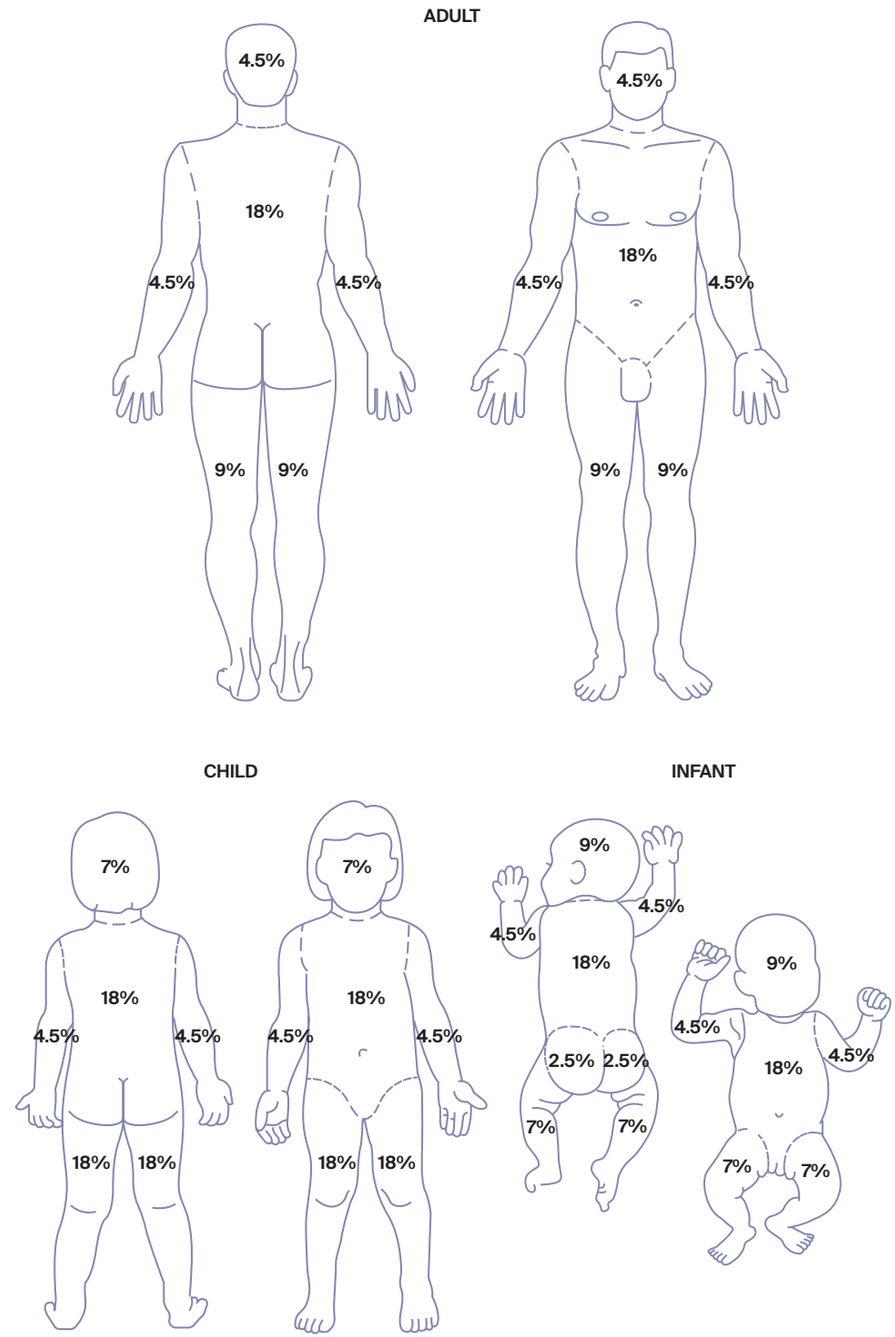

FIGURE 1.

Estimation of the percentage of the burned TBSA (retrieved from https://uscrow.org/2014/07/26/burn-treatment-shtf/burn-treatment-rule-of-nines/)

SOURCES

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2. Tintinalli JE, O. John Ma, Yealy D, Meckler GD, J. Stephan Stapczynski, Cline DM, Thomas SH. Tintinalli's Emergency Medicine: A Comprehensive Study Guide. 9th ed. New York: McGraw Hill Education; 2020.
3. Guttormsen AB, Berger MM, Sjoberg F, Heisterkamp H. Burns Injury. 1st ed. European Society of Intensive Care Medicine – Patient-Centered Acute Care Training; 2012.
4. Vincent JL, Moore FA, Bellomo R, Marini JJ. Textbook of Critical Care. 8th ed. Elsevier; 2023.

Trauma patients — protocol is modified to enhance effective management. Learn the algorithm, do additional courses to improve knowledge and performance.

Do the ABCs.

Look at the scene, it might be dangerous and/or give hints to possible injuries. If you see significant bleeding – try to stop it as soon as possible (it usually means to apply pressure to the wound).

Do the rapid trauma survey whenever possible.

Immobilize the patient properly, do not take lightly this part of the management.

If a patient is burned, make sure to know the mechanism. Underlying injuries must be considered. Also, not all burns carry the same risks (anatomically, chemically...).

Airway/lung burns – might result in complete airway closure and/or pulmonary oedema. Intubation often necessary as soon as possible, if not trained, administer oxygen, and get to the trained personnel.

Calculate TBSA, calculate the Parkland formula if you have time. Open an IV line (preferably on an area of untouched skin), begin with fluid resuscitation, and administer adequate analgesia.